



**Generation Interconnection
System Impact Study Report
for
Queue Project AF2-010
UNION CITY-TITUSVILLE 115 KV
46 MW Capacity / 77 MW Energy**

February 2021

Table of Contents

1	Introduction.....	4
2	Preface.....	4
3	General	5
4	Point of Interconnection.....	6
4.1	Primary Point of Interconnection	6
5	Cost Summary	6
6	Transmission Owner Scope of Work	8
7	Schedule.....	9
8	Transmission Owner Analysis.....	9
9	Interconnection Customer Requirements.....	9
9.1	System Protection.....	9
9.2	Compliance Issues and Interconnection Customer Requirements	9
9.3	Power Factor Requirements.....	10
10	Revenue Metering and SCADA Requirements	10
10.1	PJM Requirements	10
10.2	Meteorological Data Reporting Requirements	10
10.3	Interconnected Transmission Owner Requirements.....	11
11	Summer Peak Analysis	12
11.1	Generation Deliverability	12
11.2	Multiple Facility Contingency	12
11.3	Contribution to Previously Identified Overloads.....	12
11.4	Steady-State Voltage Requirements	14
11.5	Potential Congestion due to Local Energy Deliverability.....	14
11.6	System Reinforcements.....	18
11.7	Flow Gate Details.....	31
11.7.1	Index 1	32
11.7.2	Index 2	37
11.7.3	Index 3	39
11.7.4	Index 4	40
11.7.5	Index 5	46
11.7.6	Index 6	50

11.7.7	Index 7	54
11.7.8	Index 8	60
11.7.9	Index 9	62
11.7.10	Index 10.....	65
11.7.11	Index 11.....	68
11.7.12	Index 12.....	70
11.7.13	Index 13.....	72
11.7.14	Index 14.....	76
11.7.15	Index 15.....	80
11.8	Queue Dependencies	81
11.9	Contingency Descriptions.....	84
12	Light Load Analysis.....	88
13	Short Circuit Analysis.....	88
13.1	System Reinforcements - Short Circuit.....	88
14	Stability and Reactive Power	89
15	Affected Systems	90
15.1	NYISO	90
16	Attachment 1: One Line Diagram	91

1 Introduction

This System Impact Study has been prepared in accordance with the PJM Open Access Transmission Tariff, 205, as well as the System Impact Study Agreement between the Interconnection Customer (IC), and PJM Interconnection, LLC (PJM), Transmission Provider (TP). The Interconnected Transmission Owner (ITO) is Mid-Atlantic Interstate Transmission, LLC (MAIT, PENELEC zone).

2 Preface

The intent of the System Impact Study is to determine a plan, with approximate cost and construction time estimates, to connect the subject generation interconnection project to the PJM network at a location specified by the Interconnection Customer. As a requirement for interconnection, the Interconnection Customer may be responsible for the cost of constructing: Network Upgrades, which are facility additions, or upgrades to existing facilities, that are needed to maintain the reliability of the PJM system. All facilities required for interconnection of a generation interconnection project must be designed to meet the technical specifications (on PJM web site) for the appropriate transmission owner.

In some instances an Interconnection Customer may not be responsible for 100% of the identified network upgrade cost because other transmission network uses, e.g. another generation interconnection or merchant transmission upgrade, may also contribute to the need for the same network reinforcement. The possibility of sharing the reinforcement costs with other projects may be identified in the Feasibility Study, but the actual allocation will be deferred until the System Impact Study is performed.

The System Impact Study estimates do not include the feasibility, cost, or time required to obtain property rights and permits for construction of the required facilities. The project developer is responsible for the right of way, real estate, and construction permit issues. For properties currently owned by Transmission Owners, the costs may be included in the study.

The Interconnection Customer seeking to interconnect a wind or solar generation facility shall maintain meteorological data facilities as well as provide that meteorological data which is required per Schedule H to the Interconnection Service Agreement and Section 8 of Manual 14D.

3 General

The Interconnection Customer (IC), has proposed a Solar generating facility located in Erie County, Pennsylvania. The installed facilities will have a total capability of 77 MW with 46 MW of this output being recognized by PJM as Capacity. The proposed in-service date for this project is December 15, 2022. This study does not imply a TO commitment to this in-service date.

Queue Number	AF2-010
Project Name	UNION CITY-TITUSVILLE 115 KV
State	Pennsylvania
County	Erie
Transmission Owner	MAIT – PENELEC Zone
MFO	77
MWE	77
MWC	46
Fuel	Solar
Basecase Study Year	2023

Any new service customers who can feasibly be commercially operable prior to June 1st of the basecase study year are required to request interim deliverability analysis.

4 Point of Interconnection

4.1 Primary Point of Interconnection

The interconnection of the project at the Primary POI will be accomplished by constructing a new 115 kV three (3) breaker ring bus substation and looping the Titusville – Union City 115 kV Line into the new station. The new substation will be located approximately 1.27 miles from Union City Substation. The IC will be responsible for acquiring all easements, properties, and permits that may be required to construct both the new interconnection switching station and the associated facilities. The IC will also be responsible for the rough grade of the property and an access road to the proposed three-breaker ring bus site. The project will also require upgrades at Union City Substation and Titusville Substation.

Attachment 1 shows a one-line diagram of the proposed primary interconnection facilities for the AF2-010 generation project to connect to the FirstEnergy (“FE”) Transmission System. The IC will be responsible for constructing the facilities on its side of the POI, including the Attachment Facilities which connect the generator to the FE Transmission System’s interconnection facilities.

5 Cost Summary

The AF2-010 project will be responsible for the following costs:

Description	Total Cost
Total Physical Interconnection Costs	\$7,656,600
Total System Network Upgrade Costs (Summer Peak)	\$62,191,000
Total System Network Upgrade Costs(Light Load)	\$0
Total System Network Upgrade Costs (TO Identified)	\$0
Total Costs	\$69,847,600

*As your project progresses through the study process and other projects modify their request or withdraw, then your cost allocation could change.

This cost excludes a Federal Income Tax Gross Up charges. This tax may or may not be charged based on whether this project meets the eligibility requirements of IRS Notice 88-129. If at a future date it is determined that the Federal Income Tax Gross charge is required, the Transmission Owner shall be reimbursed by the Interconnection Customer for such taxes.

Note 1: PJM Open Access Transmission Tariff (OATT) section 217.3A outline cost allocation rules. The rules are further clarified in PJM Manual 14A Attachment B. The allocation of costs for a network upgrade will start

with the first Queue project to cause the need for the upgrade. Later queue projects will receive cost allocation contingent on their contribution to the violation and are allocated to the queues that have not closed less than 5 years following the execution of the first Interconnection Service Agreement which identifies the need for this upgrade.

Note 2: For customers with System Reinforcements listed: If your present cost allocation to a System Reinforcement indicates \$0, then please be aware that as changes to the interconnection process occur, such as prior queued projects withdrawing from the queue, reducing in size, etc, the cost responsibilities can change and a cost allocation may be assigned to your project. In addition, although your present cost allocation to a System Reinforcement is presently \$0, your project may need this system reinforcement completed to be deliverable to the PJM system. If your project comes into service prior to completion of the system reinforcement, an interim deliverability study for your project will be required.

6 Transmission Owner Scope of Work

The interconnection of the project at the Primary POI will be accomplished by constructing a new 115 kV three (3) breaker ring bus substation and looping the Titusville – Union City 115 kV line into the new station. The new substation will be located approximately 1.27 miles from Union City Substation. The IC will be responsible for acquiring all easements, properties, and permits that may be required to construct both the new interconnection switching station and the associated facilities. The IC will also be responsible for the rough grade of the property and an access road to the proposed three-breaker ring bus site. The project will also require upgrades at Union City Substation and Titusville Substation.

The total physical interconnection costs is given in the table below:

Description	Total Cost
Construct new 115 kV three-breaker ring bus substation	\$5,513,000
Loop the Titusville – Union City 115 kV Line into the new AF2-010 interconnection substation	\$559,500
Install SCADA	\$263,200
Install fiber	\$238,200
Project Management, Environmental, Forestry, Real Estate, and Right of Way	\$638,400
Review drawings and provide equipment nameplates for customer's substation	\$68,500
Replace line relaying at Union City	\$191,500
Modify relay settings at Titusville	\$90,900
Modify relay settings at AF1-304	\$88,800
Install revenue metering at customer facility	\$4,600
Total Physical Interconnection Costs	\$7,656,600

7 Schedule

Based on the scope of work for the interconnection facilities, it is expected to take a minimum of 31 months after the signing of an Interconnection Construction Service Agreement and construction kickoff call to complete the installation of the physical connection work. This assumes that there will be no environmental issues with any of the new properties associated with this project, that there will be no delays in acquiring the necessary permits for implementing the defined interconnection work, and that all system outages will be allowed when requested.

The schedule for any required Network Impact Reinforcements will be more clearly identified in future study phases. The estimated time to complete each of the required reinforcements is identified in the "System Reinforcements" section of the report.

8 Transmission Owner Analysis

FE performed an analysis of its underlying transmission <100 kV system. The AF2-010 project did not contribute to any overloads on the FE Transmission <100 kV System.

9 Interconnection Customer Requirements

9.1 System Protection

The IC must design its Customer Facilities in accordance with all applicable standards, including the standards in FE's "Requirements for Transmission Connected Facilities" document located at: <http://www.pjm.com/planning/design-engineering/to-tech-standards/private-firstenergy.aspx>. Preliminary Protection requirements will be provided as part of the Facilities Study. Detailed Protection Requirements will be provided once the project enters the construction phase.

9.2 Compliance Issues and Interconnection Customer Requirements

The proposed Customer Facilities must be designed in accordance with FE's "Requirements for Transmission Connected Facilities" document located at: <http://www.pjm.com/planning/design-engineering/to-tech-standards/private-firstenergy.aspx>. In particular, the IC is responsible for the following:

1. The purchase and installation of a fully rated 115 kV circuit breaker to protect the AF2-010 generator lead line. A single circuit breaker must be used to protect this line; if the project has several GSU transformers, the individual GSU transformer breakers cannot be used to protect this line.
2. The purchase and installation of the minimum required FE generation interconnection relaying and control facilities. This includes over/under voltage protection, over/under frequency protection, and zero sequence voltage protection relays.

3. The purchase and installation of supervisory control and data acquisition (“SCADA”) equipment to provide information in a compatible format to the FE Transmission System Control Center.
4. Compliance with the FE and PJM generator power factor and voltage control requirements.
5. The execution of a back-up service agreement to serve the customer load supplied from the AF2-010 generation project metering point when the units are out-of-service. This assumes the intent of the IC is to net the generation with the load.

The IC will also be required to meet all PJM, ReliabilityFirst, and NERC reliability criteria and operating procedures for standards compliance. For example, the IC will need to properly locate and report the over and under voltage and over and under frequency system protection elements for its units as well as the submission of the generator model and protection data required to satisfy the PJM and ReliabilityFirst audits. Failure to comply with these requirements may result in a disconnection of service if the violation is found to compromise the reliability of the FE Transmission System.

9.3 Power Factor Requirements

The IC shall design its non-synchronous Customer Facility with the ability to maintain a power factor of at least 0.95 leading (absorbing VARs) to 0.95 lagging (supplying VARs) measured at the high-side of the facility substation transformer(s) connected to the FE Transmission System.

10 Revenue Metering and SCADA Requirements

10.1 PJM Requirements

The Interconnection Customer will be required to install equipment necessary to provide Revenue Metering (KWH, KVARH) and real time data (KW, KVAR) for IC's generating Resource. See PJM Manuals M-01 and M-14D, and PJM Tariff Section 8 of Attachment O.

10.2 Meteorological Data Reporting Requirements

The solar generation facility shall provide the Transmission Provider with site-specific meteorological data including:

- Back Panel temperature (Fahrenheit) - (Required for plants with Maximum Facility Output of 3 MW or higher)
- Irradiance (Watts/meter²) - (Required for plants with Maximum Facility Output of 3 MW or higher)
- Ambient air temperature (Fahrenheit) - (Accepted, not required)
- Wind speed (meters/second) - (Accepted, not required)
- Wind direction (decimal degrees from true north) - (Accepted, not required)

10.3 Interconnected Transmission Owner Requirements

The IC will be required to comply with all Interconnected Transmission Owner's revenue metering requirements for generation interconnection customers located at the following link:

<http://www.pjm.com/planning/design-engineering/to-tech-standards/>

11 Summer Peak Analysis

The Queue Project AF2-010 was evaluated as a 77.0 MW (Capacity 46.0 MW) injection tapping the Union City to Titusville 115 kV line in the PENELEC area. Project AF2-010 was evaluated for compliance with applicable reliability planning criteria (PJM, NERC, NERC Regional Reliability Councils, and Transmission Owners). Project AF2-010 was studied with a commercial probability of 100.0 %. Potential network impacts were as follows:

11.1 Generation Deliverability

(Single or N-1 contingencies for the Capacity portion only of the interconnection)

ID	FROM BUS#	FROM BUS	kV	FROM BUS AREA	TO BUS#	TO BUS	kV	TO BUS AREA	CK T ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADIN G %	POST PROJECT LOADIN G %	AC/D C	MW IMPACT
100432694	235204	01KITTA N	138.0	AP	235139	01AL&D6 T	138.0	AP	1	ATSI-P1-2-CEI-345-700T	single	151.0	99.72	101.55	AC	2.91
100432695	235204	01KITTA N	138.0	AP	235139	01AL&D6 T	138.0	AP	1	PN-P1-2-PN-345-107T	single	151.0	99.72	101.55	AC	2.91
100886088	957160	AF2-010 TAP	115.0	PENELEC	200571	26UNION CY	115.0	PENELEC	1	PN-P1-2-PN-345-001	single	120.0	82.03	106.82	AC	32.41

11.2 Multiple Facility Contingency

(Double Circuit Tower Line, Fault with a Stuck Breaker, and Bus Fault contingencies for the full energy output)

None

11.3 Contribution to Previously Identified Overloads

(This project contributes to the following contingency overloads, i.e. "Network Impacts", identified for earlier generation or transmission interconnection projects in the PJM Queue)

ID	FROM BUS#	FROM BUS	kV	FROM BUS AREA	TO BUS#	TO BUS	kV	TO BUS AREA	CK T ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADIN G %	POST PROJECT LOADIN G %	AC/D C	MW IMPACT
98608978	200599	26ERIE W	345.0	PENELEC	238547	02AT	345.0	ATSI	1	PN-P2-2-PN-230-007T	bus	1900.0	107.34	109.07	AC	33.29
98609361	200599	26ERIE W	345.0	PENELEC	238547	02AT	345.0	ATSI	1	Base Case	single	1560.0	100.29	101.53	AC	19.5
98995959	200767	26HOMER CT	230.0	PENELEC	200795	26SHELOCT A	230.0	PENELEC	1	PN_P4-500-002A	breaker	917.0	158.04	158.91	AC	9.6
98995960	200767	26HOMER CT	230.0	PENELEC	200795	26SHELOCT A	230.0	PENELEC	1	PN_P4-500-002F	breaker	917.0	158.04	158.92	AC	9.6

ID	FROM BUS#	FROM BUS	kV	FROM BUS AREA	TO BUS#	TO BUS	kV	TO BUS AREA	CK T ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADIN G %	POST PROJECT LOADIN G %	AC D C	MW IMPAC T
98995735	200769	26HOMER CY	345.0	PENELE C	999391	STAR602	1.0	PENELE C	S	PN-P2-2-PN-230-001T	bus	824.0	123.99	125.0	AC	11.21
98995979	200769	26HOMER CY	345.0	PENELE C	999392	STAR601	1.0	PENELE C	N	PN-P2-3-PN-230-9E	breaker	807.0	143.09	144.12	AC	11.21
98996049	200769	26HOMER CY	345.0	PENELE C	999391	STAR602	1.0	PENELE C	S	PN-P2-3-PN-230-9I	breaker	824.0	123.99	125.0	AC	11.21
98995950	200795	26SHELOCT A	230.0	PENELE C	200810	26KEYSTON E	230.0	PENELE C	1	PN_P4-500-002F	breaker	917.0	153.48	154.49	AC	10.68
98995951	200795	26SHELOCT A	230.0	PENELE C	200810	26KEYSTON E	230.0	PENELE C	1	PN_P4-500-002A	breaker	917.0	153.48	154.49	AC	10.68
100432688	235139	01AL&D6T	138.0	AP	235138	01AL 4J	138.0	AP	1	PN-P1-2-PN-345-107T	single	151.0	103.68	105.51	AC	2.91
100432689	235139	01AL&D6T	138.0	AP	235138	01AL 4J	138.0	AP	1	ATSI-P1-2-CEI-345-700T	single	151.0	103.68	105.51	AC	2.91
100432286	235197	01KARNSC	138.0	AP	235152	01BUTLER	138.0	AP	1	ATSI-P2-3-CEI-345-004D	breaker	179.0	199.56	202.88	AC	6.99
100432568	235197	01KARNSC	138.0	AP	235152	01BUTLER	138.0	AP	1	PN-P1-2-PN-345-107T	single	179.0	154.64	156.96	AC	4.39
100432569	235197	01KARNSC	138.0	AP	235152	01BUTLER	138.0	AP	1	ATSI-P1-2-CEI-345-700T	single	179.0	154.64	156.96	AC	4.39
100432605	235203	01KISSNG	138.0	AP	235197	01KARNSC	138.0	AP	1	ATSI-P1-2-CEI-345-700T	single	268.0	121.78	123.45	AC	4.63
100432606	235203	01KISSNG	138.0	AP	235197	01KARNSC	138.0	AP	1	PN-P1-2-PN-345-107T	single	268.0	121.78	123.45	AC	4.63
100432304	235240	01COLMBG PN	138.0	AP	235202	01KISKIV	138.0	AP	1	ATSI-P2-3-CEI-345-004D	breaker	151.0	162.34	165.34	AC	5.34
100432612	235240	01COLMBG PN	138.0	AP	235202	01KISKIV	138.0	AP	1	PN-P1-2-PN-345-107T	single	151.0	134.25	136.36	AC	3.43

ID	FROM BUS#	FROM BUS	kV	FROM BUS AREA	TO BUS#	TO BUS	kV	TO BUS AREA	CK T ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADIN G %	POST PROJECT LOADIN G %	AC/D C	MW IMPACT
100432613	235240	01COLMBGPN	138.0	AP	235202	01KISKIV	138.0	AP	1	ATSI-P1-2-CEI-345-700T	single	151.0	134.25	136.36	AC	3.43
100432296	235282	01GAR RN	138.0	AP	235240	01COLMBGPN	138.0	AP	1	ATSI-P2-3-CEI-345-004D	breaker	151.0	169.88	172.88	AC	5.34
100432597	235282	01GAR RN	138.0	AP	235240	01COLMBGPN	138.0	AP	1	PN-P1-2-PN-345-107T	single	151.0	141.21	143.31	AC	3.43
100432598	235282	01GAR RN	138.0	AP	235240	01COLMBGPN	138.0	AP	1	ATSI-P1-2-CEI-345-700T	single	151.0	141.21	143.31	AC	3.43
98609743	239036	02PERRY	345.0	ATSI	239334	02L.CENTER	345.0	ATSI	1	ATSI-P7-1-CEI-345-012	tower	1667.0	129.98	131.33	AC	22.1
98609745	239036	02PERRY	345.0	ATSI	238684	02EASTLK	345.0	ATSI	1	ATSI-P7-1-CEI-345-016	tower	1667.0	121.97	123.26	AC	21.37
111556533	943150	AE2-344 TAP	115.0	PENELE C	200573	26VENGO JT	115.0	PENELE C	1	ATSI-P2-3-CEI-345-004D	breaker	179.0	122.92	124.48	AC	4.03

11.4 Steady-State Voltage Requirements

To be determined in the Facilities Study phase.

11.5 Potential Congestion due to Local Energy Deliverability

PJM also studied the delivery of the energy portion of this interconnection request. Any problems identified below are likely to result in operational restrictions to the project under study. The developer can proceed with network upgrades to eliminate the operational restriction at their discretion by submitting a Merchant Transmission Interconnection request.

Note: Only the most severely overloaded conditions are listed below. There is no guarantee of full delivery of energy for this project by fixing only the conditions listed in this section. With a Transmission Interconnection Request, a subsequent analysis will be performed which shall study all overload conditions associated with the overloaded element(s) identified.

ID	FROM BUS#	FROM BUS	kV	FROM BUS AREA	TO BUS#	TO BUS	kV	TO BUS AREA	CK T ID	CON T NAME	Type	Ratin g MVA	PRE PROJEC T LOADIN G %	POST PROJEC T LOADIN G %	AC D C	MW IMPAC T
98996764	200568	26ERIE SO.	230.0	PENELE C	200600	26ERIE SO	345.0	PENELE C	5	PN-P1-3-PN-230-006	operati on	776.0	94.2	97.46	AC	25.31
100886361	200576	26UTICA JT	115.0	PENELE C	200598	26WAYNE	115.0	PENELE C	1	PN-P1-2-PN-345-003	operati on	277.0	89.79	91.54	AC	5.69
100886334	200586	26ECLIPSE	115.0	PENELE C	200576	26UTICA JT	115.0	PENELE C	1	PN-P1-2-PN-345-003	operati on	277.0	92.82	94.57	AC	5.69
98609356	200599	26ERIE W	345.0	PENELE C	238547	02AT	345.0	ATSI	1	Base Case	operati on	1560.0	116.69	118.75	AC	32.64
98609357	200599	26ERIE W	345.0	PENELE C	238547	02AT	345.0	ATSI	1	ATSI-P1-1-CEI-345-711	operati on	1900.0	108.51	110.21	AC	32.64
98996377	200767	26HOMER CT	230.0	PENELE C	200795	26SHELOCT A	230.0	PENELE C	1	Base Case	operati on	731.0	152.92	153.7	AC	7.31
98996378	200767	26HOMER CT	230.0	PENELE C	200795	26SHELOCT A	230.0	PENELE C	1	PN-P1-3-PN-230-001T	operati on	917.0	153.82	154.69	AC	9.62
98996379	200767	26HOMER CT	230.0	PENELE C	200795	26SHELOCT A	230.0	PENELE C	1	PN-P1-2-PN-230-025	operati on	917.0	153.82	154.69	AC	9.62
98996514	200769	26HOMER CY	345.0	PENELE C	999392	STAR601	1.0	PENELE C	N	PN-P1-3-PN-230-004A	operati on	807.0	126.33	127.36	AC	11.28
98996549	200769	26HOMER CY	345.0	PENELE C	999391	STAR602	1.0	PENELE C	S	PN-P1-3-PN-230-003A	operati on	824.0	123.03	124.04	AC	11.22
98996392	200795	26SHELOCT A	230.0	PENELE C	200810	26KEYSTON E	230.0	PENELE C	1	PN-P1-2-PN-230-025	operati on	917.0	148.42	149.43	AC	10.7
98996393	200795	26SHELOCT A	230.0	PENELE C	200810	26KEYSTON E	230.0	PENELE C	1	PN-P1-3-PN-230-001T	operati on	917.0	148.42	149.43	AC	10.7
98996394	200795	26SHELOCT A	230.0	PENELE C	200810	26KEYSTON E	230.0	PENELE C	1	Base Case	operati on	731.0	146.77	147.75	AC	8.27
100432706	235121	01ARMSTR	138.0	AP	235204	01KITTAN	138.0	AP	1	PN-P1-2-PN-345-107T	operati on	228.0	106.59	108.58	AC	5.33
100432707	235121	01ARMSTR	138.0	AP	235204	01KITTAN	138.0	AP	1	ATSI-P1-2-CEI-345-700T	operati on	228.0	106.59	108.58	AC	5.33

ID	FROM BUS#	FROM BUS	kV	FROM BUS AREA	TO BUS#	TO BUS	kV	TO BUS AREA	CK T ID	CON T NAME	Type	Ratin g MVA	PRE PROJEC T LOADIN G %	POST PROJEC T LOADIN G %	AC D C	MW IMPAC T
100432686	235139	01AL&D6T	138.0	AP	235138	01AL 4J	138.0	AP	1	PN-P1-2-PN-345-107T	operati on	151.0	117.75	120.49	AC	4.87
100432687	235139	01AL&D6T	138.0	AP	235138	01AL 4J	138.0	AP	1	ATSI-P1-2-CEI-345-700T	operati on	151.0	117.75	120.49	AC	4.87
100432566	235197	01KARNSC	138.0	AP	235152	01BUTLER	138.0	AP	1	ATSI-P1-2-CEI-345-700T	operati on	179.0	197.89	201.38	AC	7.35
100432567	235197	01KARNSC	138.0	AP	235152	01BUTLER	138.0	AP	1	PN-P1-2-PN-345-107T	operati on	179.0	197.89	201.38	AC	7.35
100432603	235203	01KISSNG	138.0	AP	235197	01KARNSC	138.0	AP	1	ATSI-P1-2-CEI-345-700T	operati on	268.0	155.6	158.47	AC	7.75
100432604	235203	01KISSNG	138.0	AP	235197	01KARNSC	138.0	AP	1	PN-P1-2-PN-345-107T	operati on	268.0	155.6	158.47	AC	7.75
100432692	235204	01KITTAN	138.0	AP	235139	01AL&D6T	138.0	AP	1	PN-P1-2-PN-345-107T	operati on	151.0	113.68	116.42	AC	4.87
100432693	235204	01KITTAN	138.0	AP	235139	01AL&D6T	138.0	AP	1	ATSI-P1-2-CEI-345-700T	operati on	151.0	113.68	116.42	AC	4.87
100432610	235240	01COLMBG PN	138.0	AP	235202	01KISKIV	138.0	AP	1	ATSI-P1-2-CEI-345-700T	operati on	151.0	157.93	161.16	AC	5.73
100432611	235240	01COLMBG PN	138.0	AP	235202	01KISKIV	138.0	AP	1	PN-P1-2-PN-345-107T	operati on	151.0	157.93	161.16	AC	5.73
100432595	235282	01GAR RN	138.0	AP	235240	01COLMBG PN	138.0	AP	1	PN-P1-2-PN-345-107T	operati on	151.0	165.42	168.65	AC	5.73
100432596	235282	01GAR RN	138.0	AP	235240	01COLMBG PN	138.0	AP	1	ATSI-P1-2-CEI-345-700T	operati on	151.0	165.42	168.65	AC	5.73
98609450	238547	02AT	345.0	ATSI	239036	02PERRY	345.0	ATSI	1	ATSI-P1-3-SYS-345-722	operati on	1891.0	91.81	93.46	AC	31.69

ID	FROM BUS#	FROM BUS	kV	FROM BUS AREA	TO BUS#	TO BUS	kV	TO BUS AREA	CK T ID	CON T NAME	Type	Ratin g MVA	PRE PROJE C T LOADIN G %	POST PROJE C T LOADIN G %	AC D C	MW IMPAC T
160679663	946400	AF1-304 TAP	115.0	PENELE C	200584	26GRANDV W	115.0	PENELE C	1	PN-P1-2-PN-115-021-B	operati on	149.0	77.01	125.79	AC	77.0
100886084	957160	AF2-010 TAP	115.0	PENELE C	200571	26UNION CY	115.0	PENELE C	1	PN-P1-2-PN-345-001	operati on	120.0	95.7	137.75	AC	54.25
100886087	957160	AF2-010 TAP	115.0	PENELE C	200571	26UNION CY	115.0	PENELE C	1	Base Case	operati on	120.0	59.46	102.91	AC	54.46

11.6 System Reinforcements

ID	Idx	Facility	Upgrade Description	Cost	Cost Allocated to AF2-010	Upgrade Number																																
100432296,100432598,100432597	12	01GAR RN 138.0 kV - 01COLMBGPN 138.0 kV Ckt 1	<u>APS</u> ProjectId : n6183 (WP-AG1-F-0017) Description : Reconductor 10.4 miles of 4/0 CU from Columbia Big Pine - Garretts Run Jct 138 kV (99 spans). Type : FAC Total Cost : \$27,092,520 Time Estimate : 48.0 Months Ratings : 261.0/311.0/311.0 <table><tr><th>Queue</th><th>MW</th><th>Cost %</th><th>Cost \$</th></tr><tr><td>AE2-344</td><td>5.26</td><td>10.69%</td><td>\$2,897,066</td></tr><tr><td>AF1-170</td><td>5.57</td><td>11.32%</td><td>\$3,067,805</td></tr><tr><td>AF1-276</td><td>5.18</td><td>10.53%</td><td>\$2,853,004</td></tr><tr><td>AF1-277</td><td>5.18</td><td>10.53%</td><td>\$2,853,004</td></tr><tr><td>AF1-304</td><td>8.47</td><td>17.22%</td><td>\$4,665,047</td></tr><tr><td>AF1-306</td><td>14.19</td><td>28.85%</td><td>\$7,815,468</td></tr><tr><td>AF2-010</td><td>5.34</td><td>10.86%</td><td>\$2,941,127</td></tr></table>	Queue	MW	Cost %	Cost \$	AE2-344	5.26	10.69%	\$2,897,066	AF1-170	5.57	11.32%	\$3,067,805	AF1-276	5.18	10.53%	\$2,853,004	AF1-277	5.18	10.53%	\$2,853,004	AF1-304	8.47	17.22%	\$4,665,047	AF1-306	14.19	28.85%	\$7,815,468	AF2-010	5.34	10.86%	\$2,941,127	\$27,092,520	\$2,941,127	n6183
Queue	MW	Cost %	Cost \$																																			
AE2-344	5.26	10.69%	\$2,897,066																																			
AF1-170	5.57	11.32%	\$3,067,805																																			
AF1-276	5.18	10.53%	\$2,853,004																																			
AF1-277	5.18	10.53%	\$2,853,004																																			
AF1-304	8.47	17.22%	\$4,665,047																																			
AF1-306	14.19	28.85%	\$7,815,468																																			
AF2-010	5.34	10.86%	\$2,941,127																																			
100886088	3	AF2-010 TAP 115.0 kV - 26UNION CY 115.0 kV Ckt 1	<u>PENELEC</u> Not a Violation. PENELEC applicable emergency rating on this facility is B: 172 MVA.	\$0	\$0	-																																
98608978,98609362,98609361	1	26ERIE W 345.0 kV - 02AT 345.0 kV Ckt 1	<u>ATSI</u> ProjectId : n6180 (CEI-001G) Description : Reconductor the ATSI owned portion of the Erie West-Ashtabula 3-point tap 345 kV Line. Type : Facility Total Cost : \$37,680,000 Time Estimate : 30.0 Months Ratings : 1639.0/1999.0/1856.0 <table><tr><th>Queue</th><th>MW</th><th>Cost %</th><th>Cost \$</th></tr><tr><td>AF1-304</td><td>19.83</td><td>50.42%</td><td>\$18,998,078</td></tr><tr><td>AF2-010</td><td>19.5</td><td>49.58%</td><td>\$18,681,922</td></tr></table> <u>PENELEC</u> ProjectId : n6180.1 (PN-AG1-F-0005) Description : Reconductor approximately 7 miles of line. Type : FAC Total Cost : \$42,800,000 Time Estimate : 40.0 Months Ratings : 1842.0/2136.0/2457.0 <table><tr><th>Queue</th><th>MW</th><th>Cost %</th><th>Cost \$</th></tr><tr><td>AF1-304</td><td>19.83</td><td>50.42%</td><td>\$21,579,558</td></tr><tr><td>AF2-010</td><td>19.5</td><td>49.58%</td><td>\$21,220,442</td></tr></table> Note 1: It should be noted that some of the contingencies taken in the analysis may not be valid due to system condition changes that were not captured in the model. This will be re-evaluated for validity in the Facilities Study phase.	Queue	MW	Cost %	Cost \$	AF1-304	19.83	50.42%	\$18,998,078	AF2-010	19.5	49.58%	\$18,681,922	Queue	MW	Cost %	Cost \$	AF1-304	19.83	50.42%	\$21,579,558	AF2-010	19.5	49.58%	\$21,220,442	\$37,680,000 + \$42,800,000	\$18,681,922+ \$21,220,442	n6180 n6180.1								
Queue	MW	Cost %	Cost \$																																			
AF1-304	19.83	50.42%	\$18,998,078																																			
AF2-010	19.5	49.58%	\$18,681,922																																			
Queue	MW	Cost %	Cost \$																																			
AF1-304	19.83	50.42%	\$21,579,558																																			
AF2-010	19.5	49.58%	\$21,220,442																																			

ID	Idx	Facility	Upgrade Description	Cost	Cost Allocated to AF2-010	Upgrade Number
111556533	15	AE2-344 TAP 115.0 kV - 26VENGO JT 115.0 kV Ckt 1	PENELEC Not a Violation. PENELEC applicable emergency rating on this facility is B: 282 MVA.	\$0	\$0	-
100432606,100 432605	10	01KISSNG 138.0 kV - 01KARNSC 138.0 kV Ckt 1	APS ProjectId : n7164.1 (WP-AG1-F-0003A) Description : Reconductor 7.5 miles of 556.5 ACSR from Karns City to Kissinger Jct (47 spans). Type : FAC Total Cost : \$19,537,875 Time Estimate : 42.0 Months Ratings : 293.0/306.0/306.0 Note: Per PJM cost allocation rules, Queue Project AF2-010 presently does not receive cost allocation for this upgrade. Note 1: as changes to the interconnection process occur, such as prior queued projects withdrawing from the queue, reducing in size, etc., Queue Project AF2-010 could receive cost allocation. Note 2: Although Queue Project AF2-010 may not have cost responsibility for this upgrade, Queue Project AF2-010 may need this upgrade in-service to be deliverable to the PJM system. If Queue Project AF2-010 comes into service prior to completion of the upgrade, Queue Project AF2-010 will need an interim study. ProjectId : n7164.2 (WP-AG1-F-0003B) Description : Replace relaying (RT, WT, MT, ZR, OR) at Karns City substation. Type : FAC Total Cost : \$455,884 Time Estimate : 12.0 Months Ratings : 294.0/350.0/350.0 Note: Per PJM cost allocation rules, Queue Project AF2-010 presently does not receive cost allocation for this upgrade. Note 1: as changes to the interconnection process occur, such as prior queued projects withdrawing from the queue, reducing in size, etc., Queue Project AF2-010 could receive cost allocation. Note 2: Although Queue Project AF2-010 may not have cost responsibility for this upgrade, Queue Project AF2-010 may need this upgrade in-service to be deliverable to the PJM system. If Queue Project AF2-010 comes into service prior to completion of the upgrade, Queue Project AF2-010 will need an interim study.	\$19,537,875 + \$455,884	\$0 + \$0	n7164.1 n7164.2

ID	Idx	Facility	Upgrade Description	Cost	Cost Allocated to AF2-010	Upgrade Number
100432606,100 432605	10	01KISSNG 138.0 kV - 01KARNSC 138.0 kV Ckt 1	APS Continued: ProjectId : n7164.3 (WP-AG1-F-0003C) Description : Replace 954 ACSR & 1024.5 ACAR bus conductor at Karns City substation. Type : FAC Total Cost : \$130,252 Time Estimate : 12.0 Months Ratings : 299.0/360.0/360.0 Note: Per PJM cost allocation rules, Queue Project AF2-010 presently does not receive cost allocation for this upgrade. Note 1: as changes to the interconnection process occur, such as prior queued projects withdrawing from the queue, reducing in size, etc., Queue Project AF2-010 could receive cost allocation. Note 2: Although Queue Project AF2-010 may not have cost responsibility for this upgrade, Queue Project AF2-010 may need this upgrade in-service to be deliverable to the PJM system. If Queue Project AF2-010 comes into service prior to completion of the upgrade, Queue Project AF2-010 will need an interim study. ProjectId : n7164.4 (WP-AG1-F-0003D) Description : Replace 1200 A circuit breaker at Karns City substation. Type : FAC Total Cost : \$911,768 Time Estimate : 12.0 Months Ratings : 308.0/376.0/376.0 Note: Per PJM cost allocation rules, Queue Project AF2-010 presently does not receive cost allocation for this upgrade. Note 1: as changes to the interconnection process occur, such as prior queued projects withdrawing from the queue, reducing in size, etc., Queue Project AF2-010 could receive cost allocation. Note 2: Although Queue Project AF2-010 may not have cost responsibility for this upgrade, Queue Project AF2-010 may need this upgrade in-service to be deliverable to the PJM system. If Queue Project AF2-010 comes into service prior to completion of the upgrade, Queue Project AF2-010 will need an interim study.	\$130,252 + \$911,768	\$0 + \$0	n7164.3 n7164.4

ID	Idx	Facility	Upgrade Description	Cost	Cost Allocated to AF2-010	Upgrade Number
98609743	13	02PERRY 345.0 kV - 02L.CENTER 345.0 kV Ckt 1	ATSI Description : Reconductor the Leroy Center-Perry S6 345 kV Line (~10 miles from Leroy Center to Perry). The existing conductor is (2) 954 ACSR conductor and the new conductor is (2) 954 kcmil ACSS conductor. Upgrade terminals as required. Type : FAC Total Cost : \$29,400,000 Time Estimate : 30 Months Ratings : 2527.0/2907.0/2907.0 Queue Project AF2-010 presently does not receive cost allocation for this upgrade as the driver for this upgrade is a derate of the line by ATSI. Note 1: as changes to the interconnection process occur, such as prior queued projects withdrawing from the queue, reducing in size, etc., Queue Project AF2-164 could receive cost allocation. Note 2: Although Queue Project AF2-010 may not have cost responsibility for this upgrade, Queue Project AF2-010 may need this upgrade in-service to be deliverable to the PJM system. If Queue Project AF2-010 comes into service prior to completion of the upgrade, Queue Project AF2-010 will need an interim study.	\$29,400,000	\$0	-
98609745	14	02PERRY 345.0 kV - 02EASTLK 345.0 kV Ckt 1	ATSI Description : Reconductor the Eastlake-Perry S8 345 kV Line (~22 miles from Eastlake to Perry). The existing conductor is (2) 954 ACSR conductor and the new conductor is (2) 954 kcmil ACSS conductor. Upgrade terminals as required. Type : FAC Total Cost : \$64,800,000 Time Estimate : 48 Months Ratings : 2527.0/2907.0/2907.0 Queue Project AF2-010 presently does not receive cost allocation for this upgrade as the driver for this upgrade is a derate of the line by ATSI. Note 1: as changes to the interconnection process occur, such as prior queued projects withdrawing from the queue, reducing in size, etc., Queue Project v could receive cost allocation. Note 2: Although Queue Project AF2-010 may not have cost responsibility for this upgrade, Queue Project AF2-010 may need this upgrade in-service to be deliverable to the PJM system. If Queue Project AF2-010 comes into service prior to completion of the upgrade, Queue Project AF2-010 will need an interim study.	\$64,800,000	\$0	-

ID	Idx	Facility	Upgrade Description	Cost	Cost Allocated to AF2-010	Upgrade Number																																																																												
98995959,98995960	4	26HOMER CT 230.0 kV - 26SHELOCTA 230.0 kV Ckt 1	<u>PENELEC</u> ProjectId : n6841 (PN-AF2-F-0005) Description : Construct 500 kV yard consisting of three-500 kV breakers configured in a breaker-and-a-half layout (initially a ring bus). Tap the Keystone - Conemaugh 500 kV line and loop into the new Homer City 500 kV yard Install one 500/345 kV transformer Install a new 345 kV breaker-and-a-half string using three new breakers Type : CON Total Cost : \$85,600,000 Time Estimate : 48.0 Months	\$85,600,000	\$1,351,579	n6841																																																																												
98995735,98996049	5	26HOMER CY 345.0 kV - STAR602 1.0 kV Ckt S																																																																																
98995950,98995951	7	26SHELOCTA 230.0 kV - 26KEYSTONE 230.0 kV Ckt 1																																																																																
98995979	6	26HOMER CY 345.0 kV - STAR601 1.0 kV Ckt N	<table><tr><th>Queue</th><th>MW</th><th>Cost %</th><th>Cost \$</th></tr><tr><td>AF1-000</td><td>210.81</td><td>34.67%</td><td>\$29,679,829</td></tr><tr><td>AF1-043</td><td>15.73</td><td>2.59%</td><td>\$2,214,618</td></tr><tr><td>AF1-086</td><td>17.91</td><td>2.95%</td><td>\$2,521,539</td></tr><tr><td>AF1-098</td><td>10.26</td><td>1.69%</td><td>\$1,444,500</td></tr><tr><td>AF1-099</td><td>18.4</td><td>3.03%</td><td>\$2,590,526</td></tr><tr><td>AF1-100</td><td>38.48</td><td>6.33%</td><td>\$5,417,579</td></tr><tr><td>AF1-142</td><td>29.44</td><td>4.84%</td><td>\$4,144,842</td></tr><tr><td>AF1-143</td><td>24.18</td><td>3.98%</td><td>\$3,404,289</td></tr><tr><td>AF1-220</td><td>26.53</td><td>4.36%</td><td>\$3,735,145</td></tr><tr><td>AF1-232</td><td>46.61</td><td>7.67%</td><td>\$6,562,197</td></tr><tr><td>AF1-273</td><td>24.83</td><td>4.08%</td><td>\$3,495,803</td></tr><tr><td>AF1-276</td><td>22.06</td><td>3.63%</td><td>\$3,105,816</td></tr><tr><td>AF1-277</td><td>22.06</td><td>3.63%</td><td>\$3,105,816</td></tr><tr><td>AF1-278</td><td>17.64</td><td>2.90%</td><td>\$2,483,526</td></tr><tr><td>AF1-304</td><td>13.7</td><td>2.25%</td><td>\$1,928,816</td></tr><tr><td>AF1-306</td><td>39.86</td><td>6.56%</td><td>\$5,611,868</td></tr><tr><td>AF1-307</td><td>19.9</td><td>3.27%</td><td>\$2,801,711</td></tr><tr><td>AF2-010</td><td>9.6</td><td>1.58%</td><td>\$1,351,579</td></tr></table>				Queue	MW	Cost %	Cost \$	AF1-000	210.81	34.67%	\$29,679,829	AF1-043	15.73	2.59%	\$2,214,618	AF1-086	17.91	2.95%	\$2,521,539	AF1-098	10.26	1.69%	\$1,444,500	AF1-099	18.4	3.03%	\$2,590,526	AF1-100	38.48	6.33%	\$5,417,579	AF1-142	29.44	4.84%	\$4,144,842	AF1-143	24.18	3.98%	\$3,404,289	AF1-220	26.53	4.36%	\$3,735,145	AF1-232	46.61	7.67%	\$6,562,197	AF1-273	24.83	4.08%	\$3,495,803	AF1-276	22.06	3.63%	\$3,105,816	AF1-277	22.06	3.63%	\$3,105,816	AF1-278	17.64	2.90%	\$2,483,526	AF1-304	13.7	2.25%	\$1,928,816	AF1-306	39.86	6.56%	\$5,611,868	AF1-307	19.9	3.27%	\$2,801,711	AF2-010	9.6	1.58%	\$1,351,579
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			This upgrade will need to be tested during the Facilities Study phase to ensure it does not cause any additional violations.																																																																															

ID	Idx	Facility	Upgrade Description	Cost	Cost Allocated to AF2-010	Upgrade Number																
100432694,100432695	2	01KITTAN 138.0 kV - 01AL&D6T 138.0 kV Ckt 1	<u>APS</u> ProjectId : n7165.1 (WP-AG1-F-0005A) Description : Replace 4/0 CU line exit at All Dam 6 Tap. Type : FAC Total Cost : \$130,252 Time Estimate : 12.0 Months Ratings : 148.0/151.0/151.0 This project is the driver: <table><tr><td>Queue</td><td>MW</td><td>Cost %</td><td>Cost \$</td></tr><tr><td>AF2-010</td><td>2.91</td><td>100.00%</td><td>\$130,252</td></tr></table> ProjectId : n7165.2 (WP-AG1-F-0005B) Description : Reconductor 6 miles of 4/0 CU from All Dam 6 Tap to Kittaning (49 spans). Type : FAC Total Cost : \$15,630,300 Time Estimate : 42.0 Months Ratings : 293.0/343.0/343.0 This project is the driver: <table><tr><td>Queue</td><td>MW</td><td>Cost %</td><td>Cost \$</td></tr><tr><td>AF2-010</td><td>2.47</td><td>100.00%</td><td>\$15,630,300</td></tr></table>	Queue	MW	Cost %	Cost \$	AF2-010	2.91	100.00%	\$130,252	Queue	MW	Cost %	Cost \$	AF2-010	2.47	100.00%	\$15,630,300	\$130,252 + \$15,630,300	\$130,252 + \$15,630,300	n7165.1 n7165.2
Queue	MW	Cost %	Cost \$																			
AF2-010	2.91	100.00%	\$130,252																			
Queue	MW	Cost %	Cost \$																			
AF2-010	2.47	100.00%	\$15,630,300																			

ID	Idx	Facility	Upgrade Description	Cost	Cost Allocated to AF2-010	Upgrade Number
100432689,100432688	8	01AL&D6T 138.0 kV - 01AL 4J 138.0 kV Ckt 1	<u>APS</u> ProjectId : n7166.1 (WP-AG1-F-0002A) Description : Replace 4/0 CU line exit at All Dam 6 Tap. Type : FAC Total Cost : \$130,252 Time Estimate : 12.0 Months Ratings : 148.0/151.0/151.0 Note: Per PJM cost allocation rules, Queue Project AF2-010 presently does not receive cost allocation for this upgrade. Note 1: as changes to the interconnection process occur, such as prior queued projects withdrawing from the queue, reducing in size, etc., Queue Project AF2-010 could receive cost allocation. Note 2: Although Queue Project AF2-010 may not have cost responsibility for this upgrade, Queue Project AF2-010 may need this upgrade in-service to be deliverable to the PJM system. If Queue Project AF2-010 comes into service prior to completion of the upgrade, Queue Project AF2-010 will need an interim study. ProjectId : n7166.2 (WP-AG1-F-0002B) Description : Reconductor 7.6 miles of 4/0 CU from All Dam 6 Tap to structure 84 (66 spans). Type : FAC Total Cost : \$19,928,632 Time Estimate : 42.0 Months Ratings : 297.0/358.0/358.0 Note: Per PJM cost allocation rules, Queue Project AF2-010 presently does not receive cost allocation for this upgrade. Note 1: as changes to the interconnection process occur, such as prior queued projects withdrawing from the queue, reducing in size, etc., Queue Project AF2-010 could receive cost allocation. Note 2: Although Queue Project AF2-010 may not have cost responsibility for this upgrade, Queue Project AF2-010 may need this upgrade in-service to be deliverable to the PJM system. If Queue Project AF2-010 comes into service prior to completion of the upgrade, Queue Project AF2-010 will need an interim study.	\$130,252 + \$19,928,632	\$0 + \$0	n7166.1 n7166.2

ID	Idx	Facility	Upgrade Description	Cost	Cost Allocated to AF2-010	Upgrade Number
100432568,100 432569,100432 286	9	01KARNSC 138.0 kV - 01BUTLER 138.0 kV Ckt 1	<u>APS</u> ProjectId : n7170.1 (WP-AG1-F-0008A) Description : Replace 336 ACSR, 954 ACSR, & 1024.5 ACAR bus conductor at Karns City substation. Type : FAC Total Cost : \$130,252 Time Estimate : 12.0 Months Ratings : 141.0/182.0/182.0 Note: Per PJM cost allocation rules, Queue Project AF2-010 presently does not receive cost allocation for this upgrade. Note 1: as changes to the interconnection process occur, such as prior queued projects withdrawing from the queue, reducing in size, etc., Queue Project AF2-010 could receive cost allocation. Note 2: Although Queue Project AF2-010 may not have cost responsibility for this upgrade, Queue Project AF2-010 may need this upgrade in-service to be deliverable to the PJM system. If Queue Project AF2-010 comes into service prior to completion of the upgrade, Queue Project AF2-010 will need an interim study. ProjectId : n7170.2 (WP-AG1-F-0008B) Description : Replace 350 CU, 1.00 IPS CU, & 954 ACSR bus conductor at Butler substation. Type : FAC Total Cost : \$130,252 Time Estimate : 12.0 Months Ratings : 160.0/192.0/192.0 Note: Per PJM cost allocation rules, Queue Project AF2-010 presently does not receive cost allocation for this upgrade. Note 1: as changes to the interconnection process occur, such as prior queued projects withdrawing from the queue, reducing in size, etc., Queue Project AF2-010 could receive cost allocation. Note 2: Although Queue Project AF2-010 may not have cost responsibility for this upgrade, Queue Project AF2-010 may need this upgrade in-service to be deliverable to the PJM system. If Queue Project AF2-010 comes into service prior to completion of the upgrade, Queue Project AF2-010 will need an interim study.	\$130,252 + \$130,252	\$0 + \$0	n7170.1 n7170.2

ID	Idx	Facility	Upgrade Description	Cost	Cost Allocated to AF2-010	Upgrade Number
100432568,100432569,100432286	9	01KARNSC 138.0 kV - 01BUTLER 138.0 kV Ckt 1	APS Continued: ProjectId : n7170.3 (WP-AG1-F-0008C) Description : Replace 336 ACSR line risers at Butler substation. Type : FAC Total Cost : \$130,252 Time Estimate : 12.0 Months Ratings : 160.0/192.0/192.0 Note: Per PJM cost allocation rules, Queue Project AF2-010 presently does not receive cost allocation for this upgrade. Note 1: as changes to the interconnection process occur, such as prior queued projects withdrawing from the queue, reducing in size, etc., Queue Project AF2-010 could receive cost allocation. Note 2: Although Queue Project AF2-010 may not have cost responsibility for this upgrade, Queue Project AF2-010 may need this upgrade in-service to be deliverable to the PJM system. If Queue Project AF2-010 comes into service prior to completion of the upgrade, Queue Project AF2-010 will need an interim study. ProjectId : n7170.4 (WP-AG1-F-0008D) Description : Replace 336 ACSR line risers at Karns City substation. Type : FAC Total Cost : \$130,252 Time Estimate : 12.0 Months Ratings : 160.0/192.0/192.0 Note: Per PJM cost allocation rules, Queue Project AF2-010 presently does not receive cost allocation for this upgrade. Note 1: as changes to the interconnection process occur, such as prior queued projects withdrawing from the queue, reducing in size, etc., Queue Project AF2-010 could receive cost allocation. Note 2: Although Queue Project AF2-010 may not have cost responsibility for this upgrade, Queue Project AF2-010 may need this upgrade in-service to be deliverable to the PJM system. If Queue Project AF2-010 comes into service prior to completion of the upgrade, Queue Project AF2-010 will need an interim study.	\$130,252 + \$130,252	\$0 + \$0	n7170.3 n7170.4

ID	Idx	Facility	Upgrade Description	Cost	Cost Allocated to AF2-010	Upgrade Number																																																																
100432568,100432569,100432286	9	01KARNSC 138.0 kV - 01BUTLER 138.0 kV Ckt 1	<u>APS Continued:</u> ProjectId : n7170.5 (WP-AG1-F-0008E) Description : Reconductor 15.6 miles of 336 ACSR on the Butler - Karns City 138 kV line (102 spans). Type : FAC Total Cost : \$40,638,780 Time Estimate : 60.0 Months Ratings : 164.0/206.0/206.0 <table><tr><th>Queue</th><th>MW</th><th>Cost %</th><th>Cost \$</th></tr><tr><td>AE2-000</td><td>9.79</td><td>7.31%</td><td>\$2,970,831</td></tr><tr><td>AE2-344</td><td>6.26</td><td>4.67%</td><td>\$1,899,632</td></tr><tr><td>AF1-043</td><td>5.74</td><td>4.29%</td><td>\$1,741,835</td></tr><tr><td>AF1-086</td><td>6.91</td><td>5.16%</td><td>\$2,096,879</td></tr><tr><td>AF1-098</td><td>5.79</td><td>4.32%</td><td>\$1,757,008</td></tr><tr><td>AF1-099</td><td>7.05</td><td>5.26%</td><td>\$2,139,362</td></tr><tr><td>AF1-100</td><td>15.18</td><td>11.34%</td><td>\$4,606,457</td></tr><tr><td>AF1-142</td><td>11.29</td><td>8.43%</td><td>\$3,426,014</td></tr><tr><td>AF1-170</td><td>6.42</td><td>4.79%</td><td>\$1,948,185</td></tr><tr><td>AF1-220</td><td>10.12</td><td>7.56%</td><td>\$3,070,971</td></tr><tr><td>AF1-276</td><td>6.19</td><td>4.62%</td><td>\$1,878,390</td></tr><tr><td>AF1-277</td><td>6.19</td><td>4.62%</td><td>\$1,878,390</td></tr><tr><td>AF1-304</td><td>12.18</td><td>9.09%</td><td>\$3,696,090</td></tr><tr><td>AF1-306</td><td>17.82</td><td>13.31%</td><td>\$5,407,580</td></tr><tr><td>AF2-010</td><td>6.99</td><td>5.22%</td><td>\$2,121,155</td></tr></table> ProjectId : n7170.6 (WP-AG1-F-0008F) Description : Replace 600 A line side and bus side disconnects at Butler substation. Type : FAC Total Cost : \$390,758 Time Estimate : 12.0 Months Ratings : 196.0/222.0/222.0 Note: Per PJM cost allocation rules, Queue Project AF2-010 presently does not receive cost allocation for this upgrade. Note 1: as changes to the interconnection process occur, such as prior queued projects withdrawing from the queue, reducing in size, etc., Queue Project AF2-010 could receive cost allocation. Note 2: Although Queue Project AF2-010 may not have cost responsibility for this upgrade, Queue Project AF2-010 may need this upgrade in-service to be deliverable to the PJM system. If Queue Project AF2-010 comes into service prior to completion of the upgrade, Queue Project AF2-010 will need an interim study.	Queue	MW	Cost %	Cost \$	AE2-000	9.79	7.31%	\$2,970,831	AE2-344	6.26	4.67%	\$1,899,632	AF1-043	5.74	4.29%	\$1,741,835	AF1-086	6.91	5.16%	\$2,096,879	AF1-098	5.79	4.32%	\$1,757,008	AF1-099	7.05	5.26%	\$2,139,362	AF1-100	15.18	11.34%	\$4,606,457	AF1-142	11.29	8.43%	\$3,426,014	AF1-170	6.42	4.79%	\$1,948,185	AF1-220	10.12	7.56%	\$3,070,971	AF1-276	6.19	4.62%	\$1,878,390	AF1-277	6.19	4.62%	\$1,878,390	AF1-304	12.18	9.09%	\$3,696,090	AF1-306	17.82	13.31%	\$5,407,580	AF2-010	6.99	5.22%	\$2,121,155	\$40,638,780 + \$390,758	\$638,780 + \$0	n7170.5 n7170.6
Queue	MW	Cost %	Cost \$																																																																			
AE2-000	9.79	7.31%	\$2,970,831																																																																			
AE2-344	6.26	4.67%	\$1,899,632																																																																			
AF1-043	5.74	4.29%	\$1,741,835																																																																			
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AF1-142	11.29	8.43%	\$3,426,014																																																																			
AF1-170	6.42	4.79%	\$1,948,185																																																																			
AF1-220	10.12	7.56%	\$3,070,971																																																																			
AF1-276	6.19	4.62%	\$1,878,390																																																																			
AF1-277	6.19	4.62%	\$1,878,390																																																																			
AF1-304	12.18	9.09%	\$3,696,090																																																																			
AF1-306	17.82	13.31%	\$5,407,580																																																																			
AF2-010	6.99	5.22%	\$2,121,155																																																																			

ID	Idx	Facility	Upgrade Description	Cost	Cost Allocated to AF2-010	Upgrade Number
100432568,100 432569,100432 286	9	01KARNSC 138.0 kV - 01BUTLER 138.0 kV Ckt 1	APS Continued: ProjectId : n7170.7 (WP-AG1-F-0008G) Description : Replace 350 CU wire to WT at Butler substation. Type : FAC Total Cost : \$130,252 Time Estimate : 12.0 Months Ratings : 196.0/222.0/222.0 Note: Per PJM cost allocation rules, Queue Project AF2-010 presently does not receive cost allocation for this upgrade. Note 1: as changes to the interconnection process occur, such as prior queued projects withdrawing from the queue, reducing in size, etc., Queue Project AF2-010 could receive cost allocation. Note 2: Although Queue Project AF2-010 may not have cost responsibility for this upgrade, Queue Project AF2-010 may need this upgrade in-service to be deliverable to the PJM system. If Queue Project AF2-010 comes into service prior to completion of the upgrade, Queue Project AF2-010 will need an interim study. ProjectId : n7170.8 (WP-AG1-F-0008H) Description : Replace relaying (WT, ZR) at Butler substation. Type : FAC Total Cost : \$455,884 Time Estimate : 12.0 Months Ratings : 299.0/306.0/306.0 Note: Per PJM cost allocation rules, Queue Project AF2-010 presently does not receive cost allocation for this upgrade. Note 1: as changes to the interconnection process occur, such as prior queued projects withdrawing from the queue, reducing in size, etc., Queue Project AF2-010 could receive cost allocation. Note 2: Although Queue Project AF2-010 may not have cost responsibility for this upgrade, Queue Project AF2-010 may need this upgrade in-service to be deliverable to the PJM system. If Queue Project AF2-010 comes into service prior to completion of the upgrade, Queue Project AF2-010 will need an interim study.	\$130,252 + \$455,884	\$0 + \$0	n7170.7 n7170.8

ID	Idx	Facility	Upgrade Description	Cost	Cost Allocated to AF2-010	Upgrade Number								
100432568,100432569,100432286	9	01KARNSC 138.0 kV - 01BUTLER 138.0 kV Ckt 1	APS Continued: ProjectId : n7170.9 (WP-AG1-F-0008I) Description : Replace relaying (RT, OR, WT, MT) at Karns City substation. Type : FAC Total Cost : \$455,884 Time Estimate : 12.0 Months Ratings : 299.0/360.0/360.0 Note: Per PJM cost allocation rules, Queue Project AF2-010 presently does not receive cost allocation for this upgrade. Note 1: as changes to the interconnection process occur, such as prior queued projects withdrawing from the queue, reducing in size, etc., Queue Project AF2-010 could receive cost allocation. Note 2: Although Queue Project AF2-010 may not have cost responsibility for this upgrade, Queue Project AF2-010 may need this upgrade in-service to be deliverable to the PJM system. If Queue Project AF2-010 comes into service prior to completion of the upgrade, Queue Project AF2-010 will need an interim study. ProjectId : n7170.10 (WP-AG1-F-0008J) Description : Replace 1200 A circuit breaker at Karns City substation. Type : FAC Total Cost : \$781,515 Time Estimate : 12.0 Months Ratings : 329.0/413.0/413.0 <table><tr><th>Queue</th><th>MW</th><th>Cost %</th><th>Cost \$</th></tr><tr><td>AF2-010</td><td>6.99</td><td>100.00%</td><td>\$781,515</td></tr></table>	Queue	MW	Cost %	Cost \$	AF2-010	6.99	100.00%	\$781,515	\$455,884 + \$781,515	\$0 + \$781,515	n7170.9 n7170.10
Queue	MW	Cost %	Cost \$											
AF2-010	6.99	100.00%	\$781,515											

ID	Idx	Facility	Upgrade Description	Cost	Cost Allocated to AF2-010	Upgrade Number																																				
100432612,100432613,100432304	11	01COLMBGPN 138.0 kV - 01KISKIV 138.0 kV Ckt 1	<u>APS</u> ProjectId : n7171.1 (WP-AG1-F-0018B) Description : Replace relays (RT) at Kiski Valley substation. Type : FAC Total Cost : \$455,884 Time Estimate : 12.0 Months Ratings : 225.0/295.0/295.0 Note: Per PJM cost allocation rules, Queue Project AF2-010 presently does not receive cost allocation for this upgrade. Note 1: as changes to the interconnection process occur, such as prior queued projects withdrawing from the queue, reducing in size, etc., Queue Project AF2-010 could receive cost allocation. Note 2: Although Queue Project AF2-010 may not have cost responsibility for this upgrade, Queue Project AF2-010 may need this upgrade in-service to be deliverable to the PJM system. If Queue Project AF2-010 comes into service prior to completion of the upgrade, Queue Project AF2-010 will need an interim study. ProjectId : n6214 (WP-AG1-F-0018A) Description : Reconductor 3.7 miles of 4/0 CU. Type : FAC Total Cost : \$9,638,685 Time Estimate : 36.0 Months Ratings : 191.0/191.0/191.0 Notes : Need to Retool to Determine the Driver <table><tr><th>Queue</th><th>MW</th><th>Cost %</th><th>Cost \$</th></tr><tr><td>AE2-000</td><td>13.95</td><td>22.09%</td><td>\$2,129,548</td></tr><tr><td>AE2-344</td><td>5.26</td><td>8.33%</td><td>\$802,969</td></tr><tr><td>AF1-170</td><td>5.57</td><td>8.82%</td><td>\$850,293</td></tr><tr><td>AF1-276</td><td>5.18</td><td>8.20%</td><td>\$790,757</td></tr><tr><td>AF1-277</td><td>5.18</td><td>8.20%</td><td>\$790,757</td></tr><tr><td>AF1-304</td><td>8.47</td><td>13.41%</td><td>\$1,292,994</td></tr><tr><td>AF1-306</td><td>14.19</td><td>22.47%</td><td>\$2,166,185</td></tr><tr><td>AF2-010</td><td>5.34</td><td>8.46%</td><td>\$815,182</td></tr></table>	Queue	MW	Cost %	Cost \$	AE2-000	13.95	22.09%	\$2,129,548	AE2-344	5.26	8.33%	\$802,969	AF1-170	5.57	8.82%	\$850,293	AF1-276	5.18	8.20%	\$790,757	AF1-277	5.18	8.20%	\$790,757	AF1-304	8.47	13.41%	\$1,292,994	AF1-306	14.19	22.47%	\$2,166,185	AF2-010	5.34	8.46%	\$815,182	\$455,884 + \$9,638,685	\$0 + \$815,182	n7171.1 n6214
Queue	MW	Cost %	Cost \$																																							
AE2-000	13.95	22.09%	\$2,129,548																																							
AE2-344	5.26	8.33%	\$802,969																																							
AF1-170	5.57	8.82%	\$850,293																																							
AF1-276	5.18	8.20%	\$790,757																																							
AF1-277	5.18	8.20%	\$790,757																																							
AF1-304	8.47	13.41%	\$1,292,994																																							
AF1-306	14.19	22.47%	\$2,166,185																																							
AF2-010	5.34	8.46%	\$815,182																																							
			TOTAL COST	\$397,696,385	\$62,191,000																																					

Note : For customers with System Reinforcements listed: If your present cost allocation to a System Reinforcement indicates \$0, then please be aware that as changes to the interconnection process occur, such as prior queued projects withdrawing from the queue, reducing in size, etc, the cost responsibilities can change and a cost allocation may be assigned to your project. In addition, although your present cost allocation to a System Reinforcement is presently \$0, your project may need this system reinforcement completed to be deliverable to the PJM system. If your project comes into service prior to completion of the system reinforcement, an interim deliverability study for your project will be required.

11.7 Flow Gate Details

The following indices contain additional information about each facility presented in the body of the report. For each index, a description of the flowgate and its contingency was included for convenience. The intent of the indices is to provide more details on which projects/generators have contributions to the flowgate in question. All New Service Queue Requests, through the end of the Queue under study, that are contributors to a flowgate will be listed in the indices. Please note that there may be contributors that are subsequently queued after the queue under study that are not listed in the indices. Although this information is not used "as is" for cost allocation purposes, it can be used to gage the impact of other projects/generators. It should be noted the project/generator MW contributions presented in the body of the report are Full MW Impact contributions which are also noted in the indices column named "Full MW Impact", whereas the loading percentages reported in the body of the report, take into consideration the PJM Generator Deliverability Test rules such as commercial probability of each project as well as the ramping impact of "Adder" contributions. The MW Impact found and used in the analysis is shown in the indices column named "Gendeliv MW Impact".

11.7.1 Index 1

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
98608978	200599	26ERIE W	PENELEC	238547	02AT	ATSI	1	PN-P2-2-PN-230-007T	bus	1900.0	107.34	109.07	AC	33.29

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
200828	26HNSMLK 1	3.2057	50/50	3.2057
200829	26HNSMLK 2	3.2057	50/50	3.2057
200830	26HNSMLK 3	3.2057	50/50	3.2057
200831	26HNSMLK 4	3.2057	50/50	3.2057
200832	26HNSMLK 5	3.2057	50/50	3.2057
200849	26LAKVU GN	0.4178	50/50	0.4178
200894	26K02	8.2015	Adder	9.65
200898	26AA1-106	2.8346	Adder	3.33
201477	26Y2-055	8.7739	Adder	10.32
203910	26Z1-091	2.5790	Adder	3.03
203999	P-047 E	11.6884	Adder	13.75
235003	AC1-025 E	0.2555	Adder	0.3
236828	01GRAYMONT	0.6839	Adder	0.8
290086	Q-036 E	5.4332	Adder	6.39
915951	Y3-092 FTIR	572.1100	50/50	572.1100
916202	Z1-069 E	9.6857	Adder	11.39
918682	AA1-082 E	7.4642	Adder	8.78
919201	AA1-144 OP	21.3215	Adder	25.08
919491	AA2-000	75.6521	Adder	89.0
920341	AA2-132 (Withdrawn : 12/07/2020)	2.7543	Adder	3.24
923443	AB1-160 E	2.7674	Adder	3.26
930511	AB1-092	2.7777	Adder	3.27
935191	AD1-154 (Withdrawn : 02/04/2021)	3.2710	Adder	3.85
936421	AD2-055	5.7188	Adder	6.73
936991	AD2-133 C	2.4449	Adder	2.88
936992	AD2-133 E	11.1829	Adder	13.16
938951	AE1-123	3.5031	Adder	4.12
939171	AE1-147 C	2.0530	Adder	2.42
939172	AE1-147 E	1.3686	Adder	1.61
939291	AE1-160 C	3.8416	Adder	4.52
939292	AE1-160 E	2.2082	Adder	2.6
940201	AE2-001 C	2.0522	Adder	2.41
940202	AE2-001 E	1.3682	Adder	1.61
940681	AE2-055 C (Suspended)	2.0418	Adder	2.4
940682	AE2-055 E (Suspended)	1.3612	Adder	1.6
940861	AE2-074 C	2.8419	Adder	3.34
940862	AE2-074 E	3.7409	Adder	4.4
941191	AE2-113 C	12.5501	Adder	14.76
941192	AE2-113 E	13.5124	Adder	15.9

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
941261	AE2-120 C	2.0519	Adder	2.41
941262	AE2-120 E	1.3680	Adder	1.61
941271	AE2-121 C	1.0921	Adder	1.28
941272	AE2-121 E	0.7292	Adder	0.86
941321	AE2-126 C (Suspended)	2.1985	Adder	2.59
941322	AE2-126 E (Suspended)	1.4657	Adder	1.72
941331	AE2-129 C	1.7951	Adder	2.11
941332	AE2-129 E	1.1967	Adder	1.41
941351	AE2-131 C (Suspended)	1.7951	Adder	2.11
941352	AE2-131 E (Suspended)	1.1967	Adder	1.41
941421	AE2-139 C	8.1854	Adder	9.63
941422	AE2-139 E	5.4569	Adder	6.42
942351	AE2-248 C	1.6370	Adder	1.93
942352	AE2-248 E	1.0913	Adder	1.28
942491	AE2-262 C	8.6980	Adder	10.23
942492	AE2-262 E	5.8451	Adder	6.88
942501	AE2-263 C	8.1762	Adder	9.62
942502	AE2-263 E	5.4589	Adder	6.42
942811	AE2-299 C	15.1763	50/50	15.1763
942812	AE2-299 E	60.7053	50/50	60.7053
942961	AE2-316 C	7.2495	Adder	8.53
942962	AE2-316 E	10.3378	Adder	12.16
943151	AE2-344 C	29.9529	50/50	29.9529
943152	AE2-344 E	19.9686	50/50	19.9686
943351	AF1-006 C	6.4677	50/50	6.4677
943352	AF1-006 E	3.6381	50/50	3.6381
943751	AF1-043	17.1565	Adder	20.18
944001	AF1-068 C O1 (Withdrawn : 12/15/2020)	2.1812	Adder	2.57
944002	AF1-068 E O1 (Withdrawn : 12/15/2020)	1.2269	Adder	1.44
944181	AF1-086 C O1	2.7653	Adder	3.25
944182	AF1-086 E O1	12.0307	Adder	14.15
944261	AF1-094 C	5.2337	50/50	5.2337
944262	AF1-094 E	3.4891	50/50	3.4891
944281	AF1-096 C	3.6299	Adder	4.27
944282	AF1-096 E	2.4199	Adder	2.85
944301	AF1-098 C	22.0282	50/50	22.0282
944302	AF1-098 E	14.6854	50/50	14.6854
944311	AF1-099 C	13.2036	Adder	15.53
944312	AF1-099 E	8.8024	Adder	10.36
944321	AF1-100 C	26.3912	Adder	31.05
944322	AF1-100 E	17.5942	Adder	20.7
944381	AF1-103 O1	6.0551	Adder	7.12
944391	AF1-104 O1	10.7850	50/50	10.7850
944411	AF1-106 O1	2.9077	Adder	3.42
944471	AF1-112 C	2.1838	Adder	2.57
944472	AF1-112 E	1.2284	Adder	1.45
944671	AF1-132 C O1 (Withdrawn : 12/15/2020)	2.0442	Adder	2.4

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
944672	AF1-132 E O1 (Withdrawn : 12/15/2020)	1.3628	Adder	1.6
944691	AF1-134 C O1	1.7798	Adder	2.09
944692	AF1-134 E O1	1.1865	Adder	1.4
944771	AF1-142 C	21.1258	Adder	24.85
944772	AF1-142 E	14.0839	Adder	16.57
944881	AF1-153 C O1	2.2358	Adder	2.63
944882	AF1-153 E O1	1.4906	Adder	1.75
944901	AF1-155 C	2.2340	Adder	2.63
944902	AF1-155 E	1.4893	Adder	1.75
945021	AF1-167 C	1.2936	Adder	1.52
945022	AF1-167 E	0.8640	Adder	1.02
945051	AF1-170 C	22.8952	50/50	22.8952
945052	AF1-170 E	15.2635	50/50	15.2635
945121	AF1-177	1.5138	Adder	1.78
945161	AF1-181	0.1407	Adder	0.17
945171	AF1-182	0.7038	Adder	0.83
945181	AF1-183	0.1714	Adder	0.2
945331	AF1-198	0.3722	Adder	0.44
945451	AF1-210 C	1.4114	Adder	1.66
945452	AF1-210 E	0.9409	Adder	1.11
945491	AF1-214 C (Withdrawn : 12/03/2020)	2.0519	Adder	2.41
945492	AF1-214 E (Withdrawn : 12/03/2020)	1.3680	Adder	1.61
945521	AF1-217 C O2	2.4734	Adder	2.91
945522	AF1-217 E O2	1.6489	Adder	1.94
945751	AF1-240 C O1	1.9626	Adder	2.31
945752	AF1-240 E O1	1.3084	Adder	1.54
946111	AF1-276 C	14.7842	Adder	17.39
946112	AF1-276 E	7.2818	Adder	8.57
946121	AF1-277 C	14.7842	Adder	17.39
946122	AF1-277 E	7.2818	Adder	8.57
946131	AF1-278 C	11.7832	Adder	13.86
946132	AF1-278 E	5.8607	Adder	6.89
946211	AF1-286 C O1	1.1776	Adder	1.39
946212	AF1-286 E O1	0.7996	Adder	0.94
946221	AF1-287 C	2.4979	Adder	2.94
946222	AF1-287 E	1.6653	Adder	1.96
946381	AF1-302 C	3.1657	Adder	3.72
946382	AF1-302 E	4.2210	Adder	4.97
946401	AF1-304 C	17.2839	Adder	20.33
946402	AF1-304 E	11.5226	Adder	13.56
946421	AF1-306 C	9.5409	Adder	11.22
946422	AF1-306 E	38.1637	Adder	44.9
957161	AF2-010 C	19.8867	50/50	19.8867
957162	AF2-010 E	13.4019	50/50	13.4019
957451	AF2-039 C	1.3486	Adder	1.59
957452	AF2-039 E	0.8991	Adder	1.06
957571	AF2-051 C	13.7059	50/50	13.7059
957572	AF2-051 E	7.0606	50/50	7.0606
957941	AF2-088 C	0.6493	Adder	0.76
957942	AF2-088 E	0.4329	Adder	0.51

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
958271	AF2-121 C	1.7951	Adder	2.11
958272	AF2-121 E	1.1967	Adder	1.41
958361	AF2-130 C	2.7620	Adder	3.25
958362	AF2-130 E	1.8413	Adder	2.17
958521	AF2-145 C2	3.1230	Adder	3.67
958522	AF2-145 E2	2.0820	Adder	2.45
958731	AF2-164 C O1	17.1158	50/50	17.1158
958732	AF2-164 E O1	11.4106	50/50	11.4106
958741	AF2-165 C	3.2561	Adder	3.83
958742	AF2-165 E	2.1708	Adder	2.55
958751	AF2-166 C	2.7620	Adder	3.25
958752	AF2-166 E	1.8413	Adder	2.17
959061	AF2-197 C O1	5.2916	Adder	6.23
959062	AF2-197 E O1	7.9374	Adder	9.34
959441	AF2-235 C	1.8149	Adder	2.14
959442	AF2-235 E	1.2100	Adder	1.42
959471	AF2-238 C	1.7204	Adder	2.02
959472	AF2-238 E	1.1470	Adder	1.35
959481	AF2-239 C	1.4441	Adder	1.7
959482	AF2-239 E	1.1579	Adder	1.36
959491	AF2-240 C	0.5496	Adder	0.65
959492	AF2-240 E	0.4681	Adder	0.55
959501	AF2-241 C	1.6428	Adder	1.93
959502	AF2-241 E	1.2648	Adder	1.49
959521	AF2-243 C	2.4421	Adder	2.87
959522	AF2-243 E	1.6281	Adder	1.92
959741	AF2-265 C	1.2330	Adder	1.45
959742	AF2-265 E	0.9176	Adder	1.08
959802	AF2-271 E	0.3857	Adder	0.45
959822	AF2-273 E	0.5667	Adder	0.67
960022	AF2-293 E	0.1995	Adder	0.23
960031	AF2-294 C	1.6609	Adder	1.95
960032	AF2-294 E	1.1072	Adder	1.3
960041	AF2-295 C	2.2340	Adder	2.63
960042	AF2-295 E	1.4893	Adder	1.75
960051	AF2-296 C	1.7798	Adder	2.09
960052	AF2-296 E	1.1865	Adder	1.4
960271	AF2-318 C	1.5612	Adder	1.84
960272	AF2-318 E	1.0408	Adder	1.22
961141	AF2-405	1.4538	Adder	1.71
961151	AF2-406	10.9038	Adder	12.83
961201	AF2-411 O1 (Withdrawn : 12/08/2020)	45.1171	Adder	53.08
961211	AF2-412	19.5740	Adder	23.03
NEWTON	NEWTON	3.9617	Confirmed LTF	3.9617
BLUEG	BLUEG	6.3104	Confirmed LTF	6.3104
G-007A	G-007A	11.3546	Confirmed LTF	11.3546
VFT	VFT	31.1922	Confirmed LTF	31.1922
DUCKCREEK /* 35% REVERSE 3023630 4273103 TURNED OFF IN CASE FOR DEACTIVATION	DUCKCREEK /* 35% REVERSE 3023630 4273103 TURNED OFF IN CASE FOR DEACTIVATION	0.0000	LTF	0.0000
PRAIRIE	PRAIRIE	9.1955	Confirmed LTF	9.1955

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
LGE-TSR-0092018	LGE-TSR-0092018	0.0000	LTF	0.0000
COFFEEN	COFFEEN	0.7401	Confirmed LTF	0.7401
CHEOAH	CHEOAH	1.3864	Confirmed LTF	1.3864
EDWARDS	EDWARDS	1.3580	Confirmed LTF	1.3580
TILTON	TILTON	2.4097	Confirmed LTF	2.4097
MADISON	MADISON	0.1290	Confirmed LTF	0.1290
GIBSON	GIBSON	2.0175	Confirmed LTF	2.0175
CALDERWOOD	CALDERWOOD	1.3876	Confirmed LTF	1.3876
FARMERCITY	FARMERCITY	0.2035	Confirmed LTF	0.2035
TRIMBLE	TRIMBLE	2.0268	Confirmed LTF	2.0268
LGE-GI-0012019	LGE-GI-0012019	4.0062	LTF	4.0062
CATAWBA	CATAWBA	0.7539	Confirmed LTF	0.7539

11.7.2 Index 2

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
100432695	235204	01KITTAN	AP	235139	01AL&D6T	AP	1	PN-P1-2-PN-345-107T	single	151.0	99.72	101.55	AC	2.91

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
200608	26PINEY #1	0.4372	80/20	0.4372
200642	26SENECA#1	2.0081	80/20	2.0081
200643	26SENECA#2	1.8966	80/20	1.8966
200644	26SENECA#3	0.1845	80/20	0.1845
200662	26SCRUB GR	1.0501	80/20	1.0501
200828	26HNSMLK 1	0.6595	80/20	0.6595
200829	26HNSMLK 2	0.6595	80/20	0.6595
200830	26HNSMLK 3	0.6595	80/20	0.6595
200831	26HNSMLK 4	0.6595	80/20	0.6595
200832	26HNSMLK 5	0.6595	80/20	0.6595
200849	26LAKVU GN	0.0493	80/20	0.0493
201201	26WRREN CT	0.5083	80/20	0.5083
201477	26Y2-055	1.5387	80/20	1.5387
235030	01MHNG-T155	0.1680	80/20	0.1680
915951	Y3-092 FTIR	61.5700	80/20	61.5700
935191	AD1-154 (Withdrawn : 02/04/2021)	1.2310	80/20	1.2310
936881	AD2-112 C	-3.7652	Adder	-4.43
938951	AE1-123	1.4517	80/20	1.4517
939291	AE1-160 C	0.8948	80/20	0.8948
942811	AE2-299 C	1.8339	80/20	1.8339
942961	AE2-316 C	3.4571	80/20	3.4571
943151	AE2-344 C	3.7963	80/20	3.7963
943351	AF1-006 C	0.7584	80/20	0.7584
944261	AF1-094 C	0.7552	80/20	0.7552
944281	AF1-096 C	0.8455	80/20	0.8455
944301	AF1-098 C	2.7542	80/20	2.7542
944381	AF1-103 O1	1.0614	80/20	1.0614
944391	AF1-104 O1	1.2068	80/20	1.2068
944881	AF1-153 C O1	0.6878	80/20	0.6878
944901	AF1-155 C	0.6928	80/20	0.6928
945021	AF1-167 C	0.8141	80/20	0.8141
945051	AF1-170 C	2.7156	80/20	2.7156
945121	AF1-177	0.2654	80/20	0.2654
945451	AF1-210 C	1.0776	80/20	1.0776
945751	AF1-240 C O1	0.7386	80/20	0.7386
946381	AF1-302 C	1.5097	80/20	1.5097
946401	AF1-304 C	4.3230	80/20	4.3230
946421	AF1-306 C	2.5168	80/20	2.5168
957161	AF2-010 C	2.9113	80/20	2.9113
957571	AF2-051 C	2.1014	80/20	2.1014

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
958361	AF2-130 C	0.9306	80/20	0.9306
958731	AF2-164 C O1	3.5213	80/20	3.5213
958741	AF2-165 C	0.8891	80/20	0.8891
958751	AF2-166 C	0.9306	80/20	0.9306
959441	AF2-235 C	0.4228	80/20	0.4228
959521	AF2-243 C	0.6668	80/20	0.6668
960041	AF2-295 C	0.6928	80/20	0.6928
960891	AF2-380 C	0.5270	80/20	0.5270
NEWTON	NEWTON	0.4867	Confirmed LTF	0.4867
FARMERCITY	FARMERCITY	0.0250	Confirmed LTF	0.0250
G-007A	G-007A	1.5895	Confirmed LTF	1.5895
VFT	VFT	4.4053	Confirmed LTF	4.4053
GIBSON	GIBSON	0.2484	Confirmed LTF	0.2484
PRAIRIE	PRAIRIE	1.1494	Confirmed LTF	1.1494
COFFEEN	COFFEEN	0.0902	Confirmed LTF	0.0902
CHEOAH	CHEOAH	0.2042	Confirmed LTF	0.2042
EDWARDS	EDWARDS	0.1599	Confirmed LTF	0.1599
TILTON	TILTON	0.2885	Confirmed LTF	0.2885
MADISON	MADISON	0.0302	Confirmed LTF	0.0302
CALDERWOOD	CALDERWOOD	0.2033	Confirmed LTF	0.2033
BLUEG	BLUEG	0.7916	Confirmed LTF	0.7916
TRIMBLE	TRIMBLE	0.2543	Confirmed LTF	0.2543
CATAWBA	CATAWBA	0.1288	Confirmed LTF	0.1288

11.7.3 Index 3

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
100886088	957160	AF2-010 TAP	PENELEC	200571	26UNION CY	PENELEC	1	PN-P1-2-PN-345-001	single	120.0	82.03	106.82	AC	32.41

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
200608	26PINEY #1	0.4414	80/20	0.4414
200662	26SCRUB GR	1.8261	80/20	1.8261
935191	AD1-154 (Withdrawn : 02/04/2021)	1.1522	80/20	1.1522
938951	AE1-123	1.4657	80/20	1.4657
939291	AE1-160 C	4.6012	80/20	4.6012
944281	AF1-096 C	4.3476	80/20	4.3476
945751	AF1-240 C O1	0.6913	80/20	0.6913
946401	AF1-304 C	17.1816	80/20	17.1816
957161	AF2-010 C	32.4116	80/20	32.4116
958361	AF2-130 C	1.5571	80/20	1.5571
958741	AF2-165 C	1.9696	80/20	1.9696
958751	AF2-166 C	1.5571	80/20	1.5571
959441	AF2-235 C	2.1738	80/20	2.1738
959521	AF2-243 C	1.4772	80/20	1.4772
NEWTON	NEWTON	0.2138	Confirmed LTF	0.2138
FARMERCITY	FARMERCITY	0.0110	Confirmed LTF	0.0110
G-007A	G-007A	0.4843	Confirmed LTF	0.4843
VFT	VFT	1.2964	Confirmed LTF	1.2964
GIBSON	GIBSON	0.1087	Confirmed LTF	0.1087
PRAIRIE	PRAIRIE	0.4908	Confirmed LTF	0.4908
COFFEEN	COFFEEN	0.0401	Confirmed LTF	0.0401
CHEOAH	CHEOAH	0.0671	Confirmed LTF	0.0671
EDWARDS	EDWARDS	0.0749	Confirmed LTF	0.0749
TILTON	TILTON	0.1317	Confirmed LTF	0.1317
MADISON	MADISON	0.0040	Confirmed LTF	0.0040
CALDERWOOD	CALDERWOOD	0.0671	Confirmed LTF	0.0671
BLUEG	BLUEG	0.3368	Confirmed LTF	0.3368
TRIMBLE	TRIMBLE	0.1085	Confirmed LTF	0.1085
CATAWBA	CATAWBA	0.0322	Confirmed LTF	0.0322

11.7.4 Index 4

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
98995960	200767	26HOMER CT	PENELEC	200795	26SHELOCTA	PENELEC	1	PN_P4-500-002F	breaker	917.0	158.04	158.92	AC	9.6

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
200503	26C.SLOPE (Deactivation : 07/06/2019)	3.1457	50/50	3.1457
200794	26CONEMAGH	0.5533	50/50	0.5533
200809	26SITHE	2.6776	50/50	2.6776
200823	26MHP_X3-003	1.9091	Adder	2.25
200833	26SEWRDB34	32.6294	50/50	32.6294
200834	26SW_E13_K22	0.0866	50/50	0.0866
200835	26DSGENWIN	0.4222	50/50	0.4222
200837	26HOMER C1	46.1150	50/50	46.1150
200838	26HOMER C2	38.2576	50/50	38.2576
200839	26HOMER C3	40.5008	50/50	40.5008
200846	26FORWARD	0.3094	50/50	0.3094
200864	K-013 E	7.3660	50/50	7.3660
200883	Q-053 E	13.4016	50/50	13.4016
200886	26ARWF_N39	0.8270	50/50	0.8270
200888	26HIGHLAND	0.6802	50/50	0.6802
200889	26STNY CRK	0.5182	50/50	0.5182
200890	26BF_G21_K23	0.2070	50/50	0.2070
200891	26CSLMN_L13	0.3248	50/50	0.3248
200892	26LOOKOUT	0.3085	50/50	0.3085
200894	26K02	6.9744	Adder	8.21
200898	26AA1-106	2.6810	Adder	3.15
200915	26CHSTN_FL	0.3566	50/50	0.3566
200925	26R32	0.7849	50/50	0.7849
201477	26Y2-055	4.0283	Adder	4.74
202225	26SCI_S29B	0.1347	50/50	0.1347
203034	26NA_O38_P22	0.7236	50/50	0.7236
203910	26Z1-091	2.4503	Adder	2.88
203999	P-047 E	12.8906	Adder	15.17
235003	AC1-025 E	0.1713	Adder	0.2
236828	01GRAYMONT	0.4633	Adder	0.55
290086	Q-036 E	7.6435	Adder	8.99
292350	K-023	3.5997	50/50	3.5997
292542	L-013 1	7.7309	50/50	7.7309
293301	N-039 E	19.6851	50/50	19.6851
293393	V3-030E	5.2482	Adder	6.17
293432	R-040 E	0.3696	Adder	0.43
293603	O-018 E	16.1918	50/50	16.1918
293902	O-048 E	6.9578	50/50	6.9578
294515	O38_P22	17.2245	50/50	17.2245
294573	P-028 E	11.4546	Adder	13.48
294903	P-060 E	12.3350	50/50	12.3350

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
296332	R-032 E	18.6828	50/50	18.6828
913142	Y1-033 E OP1	5.3760	Adder	6.32
916202	Z1-069 E	10.2308	Adder	12.04
917672	Z2-108 E	4.3486	50/50	4.3486
918682	AA1-082 E	6.9442	Adder	8.17
919201	AA1-144 OP	19.7281	Adder	23.21
919491	AA2-000	58.8869	Adder	69.28
920341	AA2-132 (Withdrawn : 12/07/2020)	2.7160	Adder	3.2
923443	AB1-160 E	2.9231	Adder	3.44
930511	AB1-092	2.1622	Adder	2.54
935191	AD1-154 (Withdrawn : 02/04/2021)	2.4834	Adder	2.92
936421	AD2-055	4.4515	Adder	5.24
936991	AD2-133 C	3.4396	Adder	4.05
936992	AD2-133 E	15.7323	Adder	18.51
938351	AE1-053	2.0535	Adder	2.42
938881	AE1-116	1.2886	50/50	1.2886
938951	AE1-123	2.1594	Adder	2.54
938991	AE1-128 C	21.8088	50/50	21.8088
938992	AE1-128 E	14.5392	50/50	14.5392
939171	AE1-147 C	1.3928	Adder	1.64
939172	AE1-147 E	0.9285	Adder	1.09
939291	AE1-160 C	1.4533	Adder	1.71
939292	AE1-160 E	0.8354	Adder	0.98
940201	AE2-001 C	1.3911	Adder	1.64
940202	AE2-001 E	0.9274	Adder	1.09
940681	AE2-055 C (Suspended)	1.3643	Adder	1.61
940682	AE2-055 E (Suspended)	0.9095	Adder	1.07
940861	AE2-074 C	2.8116	Adder	3.31
940862	AE2-074 E	3.7010	Adder	4.35
941191	AE2-113 C	10.7394	Adder	12.63
941192	AE2-113 E	11.5628	Adder	13.6
941231	AE2-117 C	1.9740	Adder	2.32
941232	AE2-117 E	1.3160	Adder	1.55
941241	AE2-118 C	1.9740	Adder	2.32
941242	AE2-118 E	1.3160	Adder	1.55
941261	AE2-120 C	1.3902	Adder	1.64
941262	AE2-120 E	0.9268	Adder	1.09
941271	AE2-121 C	0.7421	Adder	0.87
941272	AE2-121 E	0.4955	Adder	0.58
941321	AE2-126 C (Suspended)	1.7600	Adder	2.07
941322	AE2-126 E (Suspended)	1.1733	Adder	1.38
941331	AE2-129 C	2.0399	Adder	2.4
941332	AE2-129 E	1.3599	Adder	1.6
941351	AE2-131 C (Suspended)	2.0399	Adder	2.4
941352	AE2-131 E (Suspended)	1.3599	Adder	1.6
941421	AE2-139 C	7.3771	Adder	8.68
941422	AE2-139 E	4.9181	Adder	5.79
942121	AE2-224 C	21.0858	50/50	21.0858
942122	AE2-224 E	14.0572	50/50	14.0572
942351	AE2-248 C	1.1004	Adder	1.29
942352	AE2-248 E	0.7336	Adder	0.86

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
942361	AE2-249 C (Suspended)	2.4535	50/50	2.4535
942362	AE2-249 E (Suspended)	1.6357	50/50	1.6357
942491	AE2-262 C	6.1736	Adder	7.26
942492	AE2-262 E	4.1486	Adder	4.88
942501	AE2-263 C	5.8031	Adder	6.83
942502	AE2-263 E	3.8745	Adder	4.56
942511	AE2-264 C	9.4970	Adder	11.17
942512	AE2-264 E	6.3313	Adder	7.45
942811	AE2-299 C	3.4340	Adder	4.04
942812	AE2-299 E	13.7360	Adder	16.16
942961	AE2-316 C	5.5726	Adder	6.56
942962	AE2-316 E	7.9466	Adder	9.35
943351	AF1-006 C	1.2313	Adder	1.45
943352	AF1-006 E	0.6926	Adder	0.81
943711	AF1-039 C O1	1.4222	Adder	1.67
943712	AF1-039 E O1	0.9481	Adder	1.12
943751	AF1-043	13.3545	Adder	15.71
944001	AF1-068 C O1 (Withdrawn : 12/15/2020)	1.4636	Adder	1.72
944002	AF1-068 E O1 (Withdrawn : 12/15/2020)	0.8233	Adder	0.97
944181	AF1-086 C O1	2.8428	Adder	3.34
944182	AF1-086 E O1	12.3679	Adder	14.55
944261	AF1-094 C	1.2664	Adder	1.49
944262	AF1-094 E	0.8443	Adder	0.99
944281	AF1-096 C	1.3732	Adder	1.62
944282	AF1-096 E	0.9155	Adder	1.08
944301	AF1-098 C	5.2261	Adder	6.15
944302	AF1-098 E	3.4840	Adder	4.1
944311	AF1-099 C	9.3714	Adder	11.03
944312	AF1-099 E	6.2476	Adder	7.35
944321	AF1-100 C	19.6019	Adder	23.06
944322	AF1-100 E	13.0679	Adder	15.37
944381	AF1-103 O1	2.7768	Adder	3.27
944411	AF1-106 O1	2.9575	Adder	3.48
944471	AF1-112 C	1.4703	Adder	1.73
944472	AF1-112 E	0.8271	Adder	0.97
944671	AF1-132 C O1 (Withdrawn : 12/15/2020)	1.3702	Adder	1.61
944672	AF1-132 E O1 (Withdrawn : 12/15/2020)	0.9134	Adder	1.07
944691	AF1-134 C O1	1.6898	Adder	1.99
944692	AF1-134 E O1	1.1266	Adder	1.33
944751	AF1-140 C	2.9093	50/50	2.9093
944752	AF1-140 E	1.9395	50/50	1.9395
944771	AF1-142 C	14.9943	Adder	17.64
944772	AF1-142 E	9.9962	Adder	11.76
944781	AF1-143 C	14.4954	50/50	14.4954
944782	AF1-143 E	9.6636	50/50	9.6636
944881	AF1-153 C O1	1.6058	Adder	1.89
944882	AF1-153 E O1	1.0705	Adder	1.26

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
944901	AF1-155 C	1.6049	Adder	1.89
944902	AF1-155 E	1.0699	Adder	1.26
945021	AF1-167 C	0.9225	Adder	1.09
945022	AF1-167 E	0.6162	Adder	0.72
945121	AF1-177	0.6942	Adder	0.82
945161	AF1-181	0.1049	Adder	0.12
945171	AF1-182	0.5246	Adder	0.62
945181	AF1-183	0.1373	Adder	0.16
945331	AF1-198	0.3959	Adder	0.47
945451	AF1-210 C	1.0101	Adder	1.19
945452	AF1-210 E	0.6734	Adder	0.79
945491	AF1-214 C (Withdrawn : 12/03/2020)	1.3902	Adder	1.64
945492	AF1-214 E (Withdrawn : 12/03/2020)	0.9268	Adder	1.09
945671	AF1-232 C O2 (Withdrawn : 01/19/2021)	30.2765	50/50	30.2765
945672	AF1-232 E O2 (Withdrawn : 01/19/2021)	16.3027	50/50	16.3027
945751	AF1-240 C O1	1.4900	Adder	1.75
945752	AF1-240 E O1	0.9933	Adder	1.17
946111	AF1-276 C	12.5512	Adder	14.77
946112	AF1-276 E	6.1819	Adder	7.27
946121	AF1-277 C	12.5512	Adder	14.77
946122	AF1-277 E	6.1819	Adder	7.27
946131	AF1-278 C	10.0035	Adder	11.77
946132	AF1-278 E	4.9755	Adder	5.85
946211	AF1-286 C O1	1.1978	Adder	1.41
946212	AF1-286 E O1	0.8133	Adder	0.96
946381	AF1-302 C	2.4335	Adder	2.86
946382	AF1-302 E	3.2446	Adder	3.82
946401	AF1-304 C	6.9788	Adder	8.21
946402	AF1-304 E	4.6526	Adder	5.47
946421	AF1-306 C	6.7689	Adder	7.96
946422	AF1-306 E	27.0757	Adder	31.85
946431	AF1-307 C (Withdrawn : 10/28/2020)	11.9345	50/50	11.9345
946432	AF1-307 E (Withdrawn : 10/28/2020)	7.9563	50/50	7.9563
946571	AF1-321 C O1	3.8536	50/50	3.8536
946572	AF1-321 E O1	2.5690	50/50	2.5690
957001	AF2-001 C O1	3.8536	50/50	3.8536
957002	AF2-001 E O1	2.5690	50/50	2.5690
957011	AF2-002 C O1	1.9268	50/50	1.9268
957012	AF2-002 E O1	1.2845	50/50	1.2845
957161	AF2-010 C	4.8742	Adder	5.73
957162	AF2-010 E	3.2848	Adder	3.86
957451	AF2-039 C	1.1668	Adder	1.37
957452	AF2-039 E	0.7779	Adder	0.92
957512	AF2-045 E	5.4434	50/50	5.4434
957561	AF2-050 C	10.5429	50/50	10.5429
957562	AF2-050 E	7.0286	50/50	7.0286

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
957571	AF2-051 C	3.6681	Adder	4.32
957572	AF2-051 E	1.8896	Adder	2.22
957931	AF2-087 C (Suspended)	0.8013	50/50	0.8013
957932	AF2-087 E (Suspended)	1.1034	50/50	1.1034
957941	AF2-088 C	0.5618	Adder	0.66
957942	AF2-088 E	0.3745	Adder	0.44
957981	AF2-092 C	2.1722	50/50	2.1722
957982	AF2-092 E	1.4481	50/50	1.4481
958101	AF2-104 C (Withdrawn : 12/08/2020)	0.6873	50/50	0.6873
958102	AF2-104 E (Withdrawn : 12/08/2020)	0.4582	50/50	0.4582
958271	AF2-121 C	2.0399	Adder	2.4
958272	AF2-121 E	1.3599	Adder	1.6
958361	AF2-130 C	1.4335	Adder	1.69
958362	AF2-130 E	0.9557	Adder	1.12
958471	AF2-141	1.6428	Adder	1.93
958731	AF2-164 C O1	7.5011	Adder	8.82
958732	AF2-164 E O1	5.0007	Adder	5.88
958741	AF2-165 C	1.4341	Adder	1.69
958742	AF2-165 E	0.9561	Adder	1.12
958751	AF2-166 C	1.4335	Adder	1.69
958752	AF2-166 E	0.9557	Adder	1.12
959061	AF2-197 C O1	5.0548	Adder	5.95
959062	AF2-197 E O1	7.5822	Adder	8.92
959441	AF2-235 C	0.6866	Adder	0.81
959442	AF2-235 E	0.4577	Adder	0.54
959471	AF2-238 C	1.8198	Adder	2.14
959472	AF2-238 E	1.2132	Adder	1.43
959481	AF2-239 C	1.3727	Adder	1.61
959482	AF2-239 E	1.1006	Adder	1.29
959491	AF2-240 C	0.5590	Adder	0.66
959492	AF2-240 E	0.4762	Adder	0.56
959501	AF2-241 C	1.6710	Adder	1.97
959502	AF2-241 E	1.2865	Adder	1.51
959521	AF2-243 C	1.0756	Adder	1.27
959522	AF2-243 E	0.7171	Adder	0.84
959741	AF2-265 C	1.3042	Adder	1.53
959742	AF2-265 E	0.9706	Adder	1.14
959792	AF2-270 E	1.1271	50/50	1.1271
959802	AF2-271 E	0.6066	Adder	0.71
959822	AF2-273 E	0.4050	Adder	0.48
960022	AF2-293 E	0.1354	Adder	0.16
960031	AF2-294 C	1.6378	Adder	1.93
960032	AF2-294 E	1.0919	Adder	1.28
960041	AF2-295 C	1.6049	Adder	1.89
960042	AF2-295 E	1.0699	Adder	1.26
960051	AF2-296 C	1.6898	Adder	1.99
960052	AF2-296 E	1.1266	Adder	1.33
960271	AF2-318 C	1.4840	Adder	1.75
960272	AF2-318 E	0.9893	Adder	1.16
960451	AF2-336 C	3.6203	50/50	3.6203
960452	AF2-336 E	2.4135	50/50	2.4135

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
960461	AF2-337 C	3.6203	50/50	3.6203
960462	AF2-337 E	2.4135	50/50	2.4135
960471	AF2-338 C	3.6203	50/50	3.6203
960472	AF2-338 E	2.4135	50/50	2.4135
960481	AF2-339 C	3.6203	50/50	3.6203
960482	AF2-339 E	2.4135	50/50	2.4135
960901	AF2-381 C	25.6208	50/50	25.6208
960902	AF2-381 E	13.4830	50/50	13.4830
961141	AF2-405	1.4787	Adder	1.74
961151	AF2-406	11.0906	Adder	13.05
961201	AF2-411 O1 (Withdrawn : 12/08/2020)	56.9186	Adder	66.96
961211	AF2-412	22.6448	Adder	26.64
961451	AF2-436	0.1664	Adder	0.2
NEWTON	NEWTON	0.8005	Confirmed LTF	0.8005
FARMERCITY	FARMERCITY	0.0419	Confirmed LTF	0.0419
GIBSON	GIBSON	0.4084	Confirmed LTF	0.4084
PRAIRIE	PRAIRIE	1.9708	Confirmed LTF	1.9708
O-066	O-066	0.0269	Confirmed LTF	0.0269
COFFEEN	COFFEEN	0.1474	Confirmed LTF	0.1474
CHEOAH	CHEOAH	0.4444	Confirmed LTF	0.4444
EDWARDS	EDWARDS	0.2471	Confirmed LTF	0.2471
TILTON	TILTON	0.4542	Confirmed LTF	0.4542
G-007	G-007	0.1290	Confirmed LTF	0.1290
MADISON	MADISON	0.0282	Confirmed LTF	0.0282
CALDERWOOD	CALDERWOOD	0.4403	Confirmed LTF	0.4403
BLUEG	BLUEG	1.3280	Confirmed LTF	1.3280
TRIMBLE	TRIMBLE	0.4257	Confirmed LTF	0.4257
CATAWBA	CATAWBA	0.3388	Confirmed LTF	0.3388

11.7.5 Index 5

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
98996049	200769	26HOMER CY	PENELEC	999391	STAR602	PENELEC	S	PN-P2-3-PN-230-9I	breaker	824.0	123.99	125.0	AC	11.21

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
200823	26MHP_X3-003	2.2698	Adder	2.67
200830	26HNSMLK 3	2.1914	50/50	2.1914
200831	26HNSMLK 4	2.1914	50/50	2.1914
200832	26HNSMLK 5	2.1914	50/50	2.1914
200838	26HOMER C2	53.7485	50/50	53.7485
200839	26HOMER C3	56.8999	50/50	56.8999
200894	26K02	6.8335	Adder	8.04
200898	26AA1-106	3.0060	Adder	3.54
201477	26Y2-055	4.5928	Adder	5.4
203349	26Z1-069 C	0.5917	50/50	0.5917
203907	26Y2-042	1.4981	Adder	1.76
203909	26Z1-038	2.2380	Adder	2.63
203910	26Z1-091	3.0229	Adder	3.56
203999	P-047 E	16.1777	Adder	19.03
294573	P-028 E	13.6190	Adder	16.02
915951	Y3-092 FTIR	131.9000	Merchant Transmission	131.9000
916202	Z1-069 E	15.0125	50/50	15.0125
917672	Z2-108 E	-1.9231	Adder	-2.26
917673	Z2-108 BAT	2.2624	Merchant Transmission	2.2624
918682	AA1-082 E	8.1567	Adder	9.6
919201	AA1-144 OP	22.9425	Adder	26.99
919491	AA2-000	45.9312	Adder	54.04
920341	AA2-132 (Withdrawn : 12/07/2020)	3.3915	Adder	3.99
923442	AB1-160 C	0.1691	50/50	0.1691
923443	AB1-160 E	4.2893	50/50	4.2893
930511	AB1-092	1.6865	Adder	1.98
936421	AD2-055	3.4721	Adder	4.08
938352	AE1-053 BAT	1.2569	Merchant Transmission	1.2569
938882	AE1-116 BAT	1.3112	Merchant Transmission	1.3112
938951	AE1-123	1.8724	Adder	2.2
939291	AE1-160 C	1.5625	Adder	1.84
939292	AE1-160 E	0.8981	Adder	1.06
940861	AE2-074 C	4.0572	50/50	4.0572
940862	AE2-074 E	5.3407	50/50	5.3407
941191	AE2-113 C	11.5067	Adder	13.54
941192	AE2-113 E	12.3890	Adder	14.58
941321	AE2-126 C (Suspended)	1.1007	Adder	1.29
941322	AE2-126 E (Suspended)	0.7338	Adder	0.86
941421	AE2-139 C	8.1567	Adder	9.6
941422	AE2-139 E	5.4378	Adder	6.4
942811	AE2-299 C	4.2913	Adder	5.05

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
942812	AE2-299 E	17.1654	Adder	20.19
942961	AE2-316 C	5.4637	Adder	6.43
942962	AE2-316 E	7.7912	Adder	9.17
943151	AE2-344 C	7.7584	Adder	9.13
943152	AE2-344 E	5.1723	Adder	6.09
943351	AF1-006 C	1.5444	Adder	1.82
943352	AF1-006 E	0.8687	Adder	1.02
943751	AF1-043	10.4164	Adder	12.25
944261	AF1-094 C	1.4855	Adder	1.75
944262	AF1-094 E	0.9904	Adder	1.17
944281	AF1-096 C	1.4763	Adder	1.74
944282	AF1-096 E	0.9842	Adder	1.16
944301	AF1-098 C	6.3481	Adder	7.47
944302	AF1-098 E	4.2320	Adder	4.98
944321	AF1-100 C	11.4814	Adder	13.51
944322	AF1-100 E	7.6542	Adder	9.0
944381	AF1-103 O1	3.1663	Adder	3.73
944391	AF1-104 O1	2.2902	Adder	2.69
944411	AF1-106 O1	3.6786	Adder	4.33
944881	AF1-153 C O1	1.3033	Adder	1.53
944882	AF1-153 E O1	0.8688	Adder	1.02
944901	AF1-155 C	1.3070	Adder	1.54
944902	AF1-155 E	0.8714	Adder	1.03
945021	AF1-167 C	0.9777	Adder	1.15
945022	AF1-167 E	0.6530	Adder	0.77
945051	AF1-170 C	5.3351	Adder	6.28
945052	AF1-170 E	3.5567	Adder	4.18
945121	AF1-177	0.7916	Adder	0.93
945161	AF1-181	0.0598	Adder	0.07
945171	AF1-182	0.2984	Adder	0.35
945331	AF1-198	0.4822	Adder	0.57
945451	AF1-210 C	1.0981	Adder	1.29
945452	AF1-210 E	0.7321	Adder	0.86
945673	AF1-232 BAT (Withdrawn : 01/19/2021)	2.9128	Merchant Transmission	2.9128
946111	AF1-276 C	18.0786	50/50	18.0786
946112	AF1-276 E	8.9044	50/50	8.9044
946121	AF1-277 C	18.0786	50/50	18.0786
946122	AF1-277 E	8.9044	50/50	8.9044
946131	AF1-278 C	14.4089	50/50	14.4089
946132	AF1-278 E	7.1667	50/50	7.1667
946211	AF1-286 C O1	1.4898	Adder	1.75
946212	AF1-286 E O1	1.0116	Adder	1.19
946381	AF1-302 C	2.3859	Adder	2.81
946382	AF1-302 E	3.1812	Adder	3.74
946401	AF1-304 C	7.3721	Adder	8.67
946402	AF1-304 E	4.9147	Adder	5.78
946421	AF1-306 C	5.4133	Adder	6.37
946422	AF1-306 E	21.6532	Adder	25.47
957161	AF2-010 C	5.6930	Adder	6.7
957162	AF2-010 E	3.8366	Adder	4.51
957513	AF2-045 BAT	2.4806	Merchant Transmission	2.4806
957571	AF2-051 C	4.5865	Adder	5.4

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
957572	AF2-051 E	2.3627	Adder	2.78
957931	AF2-087 C (Suspended)	-0.2664	Adder	-0.31
958361	AF2-130 C	1.3563	Adder	1.6
958362	AF2-130 E	0.9042	Adder	1.06
958472	AF2-141 BAT	2.5138	Merchant Transmission	2.5138
958731	AF2-164 C O1	9.9454	Adder	11.7
958732	AF2-164 E O1	6.6303	Adder	7.8
958741	AF2-165 C	1.4976	Adder	1.76
958742	AF2-165 E	0.9984	Adder	1.17
958751	AF2-166 C	1.3563	Adder	1.6
958752	AF2-166 E	0.9042	Adder	1.06
959061	AF2-197 C O1	6.0517	Adder	7.12
959062	AF2-197 E O1	9.0776	Adder	10.68
959441	AF2-235 C	0.7382	Adder	0.87
959442	AF2-235 E	0.4921	Adder	0.58
959471	AF2-238 C	2.2303	Adder	2.62
959472	AF2-238 E	1.4869	Adder	1.75
959481	AF2-239 C	1.6428	Adder	1.93
959482	AF2-239 E	1.3172	Adder	1.55
959491	AF2-240 C	0.6953	Adder	0.82
959492	AF2-240 E	0.5923	Adder	0.7
959501	AF2-241 C	2.0784	Adder	2.45
959502	AF2-241 E	1.6002	Adder	1.88
959521	AF2-243 C	1.1232	Adder	1.32
959522	AF2-243 E	0.7488	Adder	0.88
959741	AF2-265 C	1.5984	Adder	1.88
959742	AF2-265 E	1.1895	Adder	1.4
959793	AF2-270 BAT	0.2183	Merchant Transmission	0.2183
959822	AF2-273 E	0.4360	Adder	0.51
960031	AF2-294 C	2.0451	Adder	2.41
960032	AF2-294 E	1.3634	Adder	1.6
960041	AF2-295 C	1.3070	Adder	1.54
960042	AF2-295 E	0.8714	Adder	1.03
960271	AF2-318 C	1.7760	Adder	2.09
960272	AF2-318 E	1.1840	Adder	1.39
960891	AF2-380 C	0.6942	Adder	0.82
960892	AF2-380 E	0.4628	Adder	0.54
961141	AF2-405	1.8393	Adder	2.16
961151	AF2-406	13.7949	Adder	16.23
961201	AF2-411 O1 (Withdrawn : 12/08/2020)	87.3930	50/50	87.3930
961211	AF2-412	33.9534	50/50	33.9534
961451	AF2-436	0.1985	Adder	0.23
961461	AF2-437	0.1228	Adder	0.14
WEC	WEC	0.0425	Confirmed LTF	0.0425
LGEE	LGEE	0.0188	Confirmed LTF	0.0188
O-066	O-066	2.8291	Confirmed LTF	2.8291
CHEOAH	CHEOAH	0.1146	Confirmed LTF	0.1146
G-007	G-007	0.5793	Confirmed LTF	0.5793
MEC	MEC	0.1112	Confirmed LTF	0.1112
CALDERWOOD	CALDERWOOD	0.1108	Confirmed LTF	0.1108
CATAWBA	CATAWBA	0.1460	Confirmed LTF	0.1460
CBM-W1	CBM-W1	3.4277	Confirmed LTF	3.4277

11.7.6 Index 6

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
98995979	200769	26HOMER CY	PENELEC	999392	STAR601	PENELEC	N	PN-P2-3-PN-230-9E	breaker	807.0	143.09	144.12	AC	11.21

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
200823	26MHP_X3-003	2.2680	Adder	2.67
200830	26HNSMLK 3	2.1960	50/50	2.1960
200831	26HNSMLK 4	2.1960	50/50	2.1960
200832	26HNSMLK 5	2.1960	50/50	2.1960
200838	26HOMER C2	53.9610	50/50	53.9610
200839	26HOMER C3	57.1249	50/50	57.1249
200894	26K02	6.8311	Adder	8.04
200898	26AA1-106	3.0083	Adder	3.54
201477	26Y2-055	4.5975	Adder	5.41
203349	26Z1-069 C	0.5931	50/50	0.5931
203907	26Y2-042	1.4933	Adder	1.76
203909	26Z1-038	2.2362	Adder	2.63
203910	26Z1-091	3.0252	Adder	3.56
203999	P-047 E	16.2099	Adder	19.07
294573	P-028 E	13.6078	Adder	16.01
915951	Y3-092 FTIR	131.7900	Merchant Transmission	131.7900
916202	Z1-069 E	15.0476	50/50	15.0476
917672	Z2-108 E	-1.9468	Adder	-2.29
917673	Z2-108 BAT	2.2903	Merchant Transmission	2.2903
918682	AA1-082 E	8.1607	Adder	9.6
919201	AA1-144 OP	22.9522	Adder	27.0
919491	AA2-000	45.8525	Adder	53.94
920341	AA2-132 (Withdrawn : 12/07/2020)	3.3960	Adder	4.0
923442	AB1-160 C	0.1695	50/50	0.1695
923443	AB1-160 E	4.2993	50/50	4.2993
930511	AB1-092	1.6836	Adder	1.98
936421	AD2-055	3.4662	Adder	4.08
938352	AE1-053 BAT	1.2724	Merchant Transmission	1.2724
938882	AE1-116 BAT	1.3262	Merchant Transmission	1.3262
938951	AE1-123	1.8700	Adder	2.2
939291	AE1-160 C	1.5620	Adder	1.84
939292	AE1-160 E	0.8979	Adder	1.06
940861	AE2-074 C	4.0670	50/50	4.0670
940862	AE2-074 E	5.3536	50/50	5.3536
941191	AE2-113 C	11.5258	Adder	13.56
941192	AE2-113 E	12.4096	Adder	14.6
941321	AE2-126 C (Suspended)	1.0981	Adder	1.29
941322	AE2-126 E (Suspended)	0.7321	Adder	0.86
941421	AE2-139 C	8.1588	Adder	9.6
941422	AE2-139 E	5.4392	Adder	6.4
942811	AE2-299 C	4.2922	Adder	5.05

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
942812	AE2-299 E	17.1686	Adder	20.2
942961	AE2-316 C	5.4644	Adder	6.43
942962	AE2-316 E	7.7922	Adder	9.17
943151	AE2-344 C	7.7513	Adder	9.12
943152	AE2-344 E	5.1675	Adder	6.08
943351	AF1-006 C	1.5438	Adder	1.82
943352	AF1-006 E	0.8684	Adder	1.02
943751	AF1-043	10.3985	Adder	12.23
944261	AF1-094 C	1.4851	Adder	1.75
944262	AF1-094 E	0.9901	Adder	1.16
944281	AF1-096 C	1.4759	Adder	1.74
944282	AF1-096 E	0.9840	Adder	1.16
944301	AF1-098 C	6.3489	Adder	7.47
944302	AF1-098 E	4.2326	Adder	4.98
944321	AF1-100 C	11.4367	Adder	13.45
944322	AF1-100 E	7.6245	Adder	8.97
944381	AF1-103 O1	3.1695	Adder	3.73
944391	AF1-104 O1	2.2885	Adder	2.69
944411	AF1-106 O1	3.6847	Adder	4.33
944881	AF1-153 C O1	1.3018	Adder	1.53
944882	AF1-153 E O1	0.8679	Adder	1.02
944901	AF1-155 C	1.3056	Adder	1.54
944902	AF1-155 E	0.8704	Adder	1.02
945021	AF1-167 C	0.9773	Adder	1.15
945022	AF1-167 E	0.6527	Adder	0.77
945051	AF1-170 C	5.3315	Adder	6.27
945052	AF1-170 E	3.5544	Adder	4.18
945121	AF1-177	0.7924	Adder	0.93
945161	AF1-181	0.0595	Adder	0.07
945171	AF1-182	0.2971	Adder	0.35
945331	AF1-198	0.4831	Adder	0.57
945451	AF1-210 C	1.0974	Adder	1.29
945452	AF1-210 E	0.7316	Adder	0.86
945673	AF1-232 BAT (Withdrawn : 01/19/2021)	2.9460	Merchant Transmission	2.9460
946111	AF1-276 C	18.1222	50/50	18.1222
946112	AF1-276 E	8.9258	50/50	8.9258
946121	AF1-277 C	18.1222	50/50	18.1222
946122	AF1-277 E	8.9258	50/50	8.9258
946131	AF1-278 C	14.4436	50/50	14.4436
946132	AF1-278 E	7.1839	50/50	7.1839
946211	AF1-286 C O1	1.4923	Adder	1.76
946212	AF1-286 E O1	1.0133	Adder	1.19
946381	AF1-302 C	2.3862	Adder	2.81
946382	AF1-302 E	3.1816	Adder	3.74
946401	AF1-304 C	7.3700	Adder	8.67
946402	AF1-304 E	4.9133	Adder	5.78
946421	AF1-306 C	5.4078	Adder	6.36
946422	AF1-306 E	21.6314	Adder	25.45
957161	AF2-010 C	5.6914	Adder	6.7
957162	AF2-010 E	3.8355	Adder	4.51
957513	AF2-045 BAT	2.5097	Merchant Transmission	2.5097
957571	AF2-051 C	4.5881	Adder	5.4

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
957572	AF2-051 E	2.3636	Adder	2.78
957931	AF2-087 C (Suspended)	-0.2701	Adder	-0.32
958361	AF2-130 C	1.3553	Adder	1.59
958362	AF2-130 E	0.9035	Adder	1.06
958472	AF2-141 BAT	2.5448	Merchant Transmission	2.5448
958731	AF2-164 C O1	9.9662	Adder	11.72
958732	AF2-164 E O1	6.6441	Adder	7.82
958741	AF2-165 C	1.4973	Adder	1.76
958742	AF2-165 E	0.9982	Adder	1.17
958751	AF2-166 C	1.3553	Adder	1.59
958752	AF2-166 E	0.9035	Adder	1.06
959061	AF2-197 C O1	6.0565	Adder	7.13
959062	AF2-197 E O1	9.0848	Adder	10.69
959441	AF2-235 C	0.7380	Adder	0.87
959442	AF2-235 E	0.4920	Adder	0.58
959471	AF2-238 C	2.2342	Adder	2.63
959472	AF2-238 E	1.4895	Adder	1.75
959481	AF2-239 C	1.6440	Adder	1.93
959482	AF2-239 E	1.3181	Adder	1.55
959491	AF2-240 C	0.6964	Adder	0.82
959492	AF2-240 E	0.5932	Adder	0.7
959501	AF2-241 C	2.0819	Adder	2.45
959502	AF2-241 E	1.6029	Adder	1.89
959521	AF2-243 C	1.1229	Adder	1.32
959522	AF2-243 E	0.7486	Adder	0.88
959741	AF2-265 C	1.6012	Adder	1.88
959742	AF2-265 E	1.1916	Adder	1.4
959793	AF2-270 BAT	0.2208	Merchant Transmission	0.2208
959822	AF2-273 E	0.4357	Adder	0.51
960031	AF2-294 C	2.0479	Adder	2.41
960032	AF2-294 E	1.3652	Adder	1.61
960041	AF2-295 C	1.3056	Adder	1.54
960042	AF2-295 E	0.8704	Adder	1.02
960271	AF2-318 C	1.7772	Adder	2.09
960272	AF2-318 E	1.1848	Adder	1.39
960891	AF2-380 C	0.6916	Adder	0.81
960892	AF2-380 E	0.4611	Adder	0.54
961141	AF2-405	1.8424	Adder	2.17
961151	AF2-406	13.8178	Adder	16.26
961201	AF2-411 O1 (Withdrawn : 12/08/2020)	87.6270	50/50	87.6270
961211	AF2-412	34.0327	50/50	34.0327
961451	AF2-436	0.1986	Adder	0.23
961461	AF2-437	0.1224	Adder	0.14
WEC	WEC	0.0158	Confirmed LTF	0.0158
NEWTON	NEWTON	0.0258	Confirmed LTF	0.0258
FARMERCITY	FARMERCITY	0.0022	Confirmed LTF	0.0022
GIBSON	GIBSON	0.0126	Confirmed LTF	0.0126
PRAIRIE	PRAIRIE	0.1653	Confirmed LTF	0.1653
O-066	O-066	3.4205	Confirmed LTF	3.4205
COFFEEN	COFFEEN	0.0032	Confirmed LTF	0.0032
CHEOAH	CHEOAH	0.1582	Confirmed LTF	0.1582
G-007	G-007	0.6708	Confirmed LTF	0.6708

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
CALDERWOOD	CALDERWOOD	0.1536	Confirmed LTF	0.1536
BLUEG	BLUEG	0.0764	Confirmed LTF	0.0764
TRIMBLE	TRIMBLE	0.0234	Confirmed LTF	0.0234
CATAWBA	CATAWBA	0.1767	Confirmed LTF	0.1767
CBM-W1	CBM-W1	2.3894	Confirmed LTF	2.3894

11.7.7 Index 7

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
98995951	200795	26SHELOCTA	PENELEC	200810	26KEYSTONE	PENELEC	1	PN_P4-500-002A	breaker	917.0	153.48	154.49	AC	10.68

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
200636	26IUP CO-G	0.8069	50/50	0.8069
200794	26CONEMAGH	0.6539	50/50	0.6539
200809	26SITHE	2.5865	50/50	2.5865
200833	26SEWRDB34	33.8900	50/50	33.8900
200837	26HOMER C1	44.5450	50/50	44.5450
200838	26HOMER C2	37.3487	50/50	37.3487
200839	26HOMER C3	39.5385	50/50	39.5385
200846	26FORWARD	0.3358	50/50	0.3358
200864	K-013 E	7.9940	50/50	7.9940
200883	Q-053 E	12.7716	Adder	15.03
200888	26HIGHLAND	0.7488	50/50	0.7488
200894	26K02	7.3214	Adder	8.61
200898	26AA1-106	2.7470	Adder	3.23
200925	26R32	0.8640	50/50	0.8640
201477	26Y2-055	4.2309	Adder	4.98
202158	26CON.GEN1	0.1356	50/50	0.1356
203910	26Z1-091	2.4549	Adder	2.89
203999	P-047 E	12.9200	Adder	15.2
235003	AC1-025 E	0.1892	Adder	0.22
236828	01GRAYMONT	0.5125	Adder	0.6
290086	Q-036 E	8.7951	Adder	10.35
292350	K-023	3.3303	Adder	3.92
292542	L-013 1	7.1522	Adder	8.41
293301	N-039 E	17.3090	Adder	20.36
293393	V3-030E	7.5618	Adder	8.9
293432	R-040 E	0.4023	Adder	0.47
293603	O-018 E	17.8235	50/50	17.8235
293902	O-048 E	6.4370	Adder	7.57
294515	O38_P22	15.1454	Adder	17.82
294903	P-060 E	11.4340	Adder	13.45
296332	R-032 E	20.5656	50/50	20.5656
913142	Y1-033 E OP1	5.8745	Adder	6.91
916202	Z1-069 E	10.3210	Adder	12.14
917672	Z2-108 E	4.0231	Adder	4.73
918682	AA1-082 E	7.0287	Adder	8.27
919201	AA1-144 OP	20.0122	Adder	23.54
919491	AA2-000	64.0699	Adder	75.38
920341	AA2-132 (Withdrawn : 12/07/2020)	2.7199	Adder	3.2
923443	AB1-160 E	2.9488	Adder	3.47
930511	AB1-092	2.3525	Adder	2.77

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
935191	AD1-154 (Withdrawn : 02/04/2021)	4.1109	Adder	4.84
936421	AD2-055	4.8433	Adder	5.7
936991	AD2-133 C	3.9578	Adder	4.66
936992	AD2-133 E	18.1026	Adder	21.3
938351	AE1-053	2.2351	Adder	2.63
938881	AE1-116	1.1897	Adder	1.4
938951	AE1-123	2.8365	Adder	3.34
938991	AE1-128 C	20.3129	Adder	23.9
938992	AE1-128 E	13.5419	Adder	15.93
939171	AE1-147 C	1.5412	Adder	1.81
939172	AE1-147 E	1.0275	Adder	1.21
939291	AE1-160 C	1.7137	Adder	2.02
939292	AE1-160 E	0.9850	Adder	1.16
940201	AE2-001 C	1.5391	Adder	1.81
940202	AE2-001 E	1.0261	Adder	1.21
940681	AE2-055 C (Suspended)	1.5069	Adder	1.77
940682	AE2-055 E (Suspended)	1.0046	Adder	1.18
940861	AE2-074 C	2.8558	Adder	3.36
940862	AE2-074 E	3.7592	Adder	4.42
941191	AE2-113 C	11.2837	Adder	13.27
941192	AE2-113 E	12.1489	Adder	14.29
941231	AE2-117 C	2.0722	Adder	2.44
941232	AE2-117 E	1.3815	Adder	1.63
941241	AE2-118 C	2.0722	Adder	2.44
941242	AE2-118 E	1.3815	Adder	1.63
941261	AE2-120 C	1.5380	Adder	1.81
941262	AE2-120 E	1.0253	Adder	1.21
941271	AE2-121 C	0.8214	Adder	0.97
941272	AE2-121 E	0.5484	Adder	0.65
941321	AE2-126 C (Suspended)	2.0226	Adder	2.38
941322	AE2-126 E (Suspended)	1.3484	Adder	1.59
941331	AE2-129 C	2.3914	Adder	2.81
941332	AE2-129 E	1.5943	Adder	1.88
941351	AE2-131 C (Suspended)	2.3914	Adder	2.81
941352	AE2-131 E (Suspended)	1.5943	Adder	1.88
941421	AE2-139 C	7.5652	Adder	8.9
941422	AE2-139 E	5.0435	Adder	5.93
942121	AE2-224 C	21.6888	50/50	21.6888
942122	AE2-224 E	14.4592	50/50	14.4592
942351	AE2-248 C	1.2163	Adder	1.43
942352	AE2-248 E	0.8109	Adder	0.95
942361	AE2-249 C (Suspended)	2.2852	Adder	2.69
942362	AE2-249 E (Suspended)	1.5235	Adder	1.79
942491	AE2-262 C	6.8655	Adder	8.08
942492	AE2-262 E	4.6136	Adder	5.43
942501	AE2-263 C	6.4535	Adder	7.59
942502	AE2-263 E	4.3088	Adder	5.07
942511	AE2-264 C	9.9405	Adder	11.69
942512	AE2-264 E	6.6270	Adder	7.8
942811	AE2-299 C	3.6054	Adder	4.24
942812	AE2-299 E	14.4214	Adder	16.97
942961	AE2-316 C	6.1609	Adder	7.25

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
942962	AE2-316 E	8.7855	Adder	10.34
943711	AF1-039 C O1	1.5541	Adder	1.83
943712	AF1-039 E O1	1.0361	Adder	1.22
943751	AF1-043	14.5299	Adder	17.09
944001	AF1-068 C O1 (Withdrawn : 12/15/2020)	1.6174	Adder	1.9
944002	AF1-068 E O1 (Withdrawn : 12/15/2020)	0.9098	Adder	1.07
944181	AF1-086 C O1	4.9708	Adder	5.85
944182	AF1-086 E O1	21.6254	Adder	25.44
944261	AF1-094 C	1.4041	Adder	1.65
944262	AF1-094 E	0.9361	Adder	1.1
944281	AF1-096 C	1.6192	Adder	1.9
944282	AF1-096 E	1.0795	Adder	1.27
944301	AF1-098 C	5.5251	Adder	6.5
944302	AF1-098 E	3.6834	Adder	4.33
944311	AF1-099 C	10.4218	Adder	12.26
944312	AF1-099 E	6.9478	Adder	8.17
944321	AF1-100 C	22.1327	Adder	26.04
944322	AF1-100 E	14.7552	Adder	17.36
944381	AF1-103 O1	2.9164	Adder	3.43
944411	AF1-106 O1	2.9687	Adder	3.49
944471	AF1-112 C	1.6255	Adder	1.91
944472	AF1-112 E	0.9143	Adder	1.08
944671	AF1-132 C O1 (Withdrawn : 12/15/2020)	1.5141	Adder	1.78
944672	AF1-132 E O1 (Withdrawn : 12/15/2020)	1.0094	Adder	1.19
944691	AF1-134 C O1	2.5539	Adder	3.0
944692	AF1-134 E O1	1.7026	Adder	2.0
944751	AF1-140 C	2.7397	Adder	3.22
944752	AF1-140 E	1.8265	Adder	2.15
944771	AF1-142 C	16.6748	Adder	19.62
944772	AF1-142 E	11.1165	Adder	13.08
944781	AF1-143 C	13.4105	Adder	15.78
944782	AF1-143 E	8.9403	Adder	10.52
944881	AF1-153 C O1	1.7941	Adder	2.11
944882	AF1-153 E O1	1.1961	Adder	1.41
944901	AF1-155 C	1.7937	Adder	2.11
944902	AF1-155 E	1.1958	Adder	1.41
945021	AF1-167 C	1.1048	Adder	1.3
945022	AF1-167 E	0.7379	Adder	0.87
945121	AF1-177	0.7291	Adder	0.86
945161	AF1-181	0.1189	Adder	0.14
945171	AF1-182	0.5946	Adder	0.7
945181	AF1-183	0.1608	Adder	0.19
945331	AF1-198	0.3997	Adder	0.47
945451	AF1-210 C	1.2110	Adder	1.42
945452	AF1-210 E	0.8073	Adder	0.95
945491	AF1-214 C (Withdrawn : 12/03/2020)	1.5380	Adder	1.81

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
945492	AF1-214 E (Withdrawn : 12/03/2020)	1.0253	Adder	1.21
945671	AF1-232 C O2 (Withdrawn : 01/19/2021)	28.0263	Adder	32.97
945672	AF1-232 E O2 (Withdrawn : 01/19/2021)	15.0911	Adder	17.75
945751	AF1-240 C O1	2.4666	Adder	2.9
945752	AF1-240 E O1	1.6444	Adder	1.93
946071	AF1-272 C O1	25.5677	50/50	25.5677
946072	AF1-272 E O1	17.0452	50/50	17.0452
946111	AF1-276 C	12.8012	Adder	15.06
946112	AF1-276 E	6.3051	Adder	7.42
946121	AF1-277 C	12.8012	Adder	15.06
946122	AF1-277 E	6.3051	Adder	7.42
946131	AF1-278 C	10.2028	Adder	12.0
946132	AF1-278 E	5.0746	Adder	5.97
946211	AF1-286 C O1	1.2023	Adder	1.41
946212	AF1-286 E O1	0.8164	Adder	0.96
946381	AF1-302 C	2.6904	Adder	3.17
946382	AF1-302 E	3.5871	Adder	4.22
946401	AF1-304 C	8.3237	Adder	9.79
946402	AF1-304 E	5.5491	Adder	6.53
946421	AF1-306 C	7.4800	Adder	8.8
946422	AF1-306 E	29.9202	Adder	35.2
946431	AF1-307 C (Withdrawn : 10/28/2020)	14.0448	50/50	14.0448
946432	AF1-307 E (Withdrawn : 10/28/2020)	9.3632	50/50	9.3632
946571	AF1-321 C O1	4.1726	50/50	4.1726
946572	AF1-321 E O1	2.7818	50/50	2.7818
957001	AF2-001 C O1	4.1726	50/50	4.1726
957002	AF2-001 E O1	2.7818	50/50	2.7818
957011	AF2-002 C O1	2.0863	50/50	2.0863
957012	AF2-002 E O1	1.3909	50/50	1.3909
957161	AF2-010 C	5.4216	Adder	6.38
957162	AF2-010 E	3.6537	Adder	4.3
957451	AF2-039 C	1.4098	Adder	1.66
957452	AF2-039 E	0.9398	Adder	1.11
957512	AF2-045 E	5.2025	Adder	6.12
957561	AF2-050 C	10.8444	50/50	10.8444
957562	AF2-050 E	7.2296	50/50	7.2296
957571	AF2-051 C	3.9416	Adder	4.64
957572	AF2-051 E	2.0305	Adder	2.39
957931	AF2-087 C (Suspended)	0.7424	Adder	0.87
957932	AF2-087 E (Suspended)	1.0223	Adder	1.2
957941	AF2-088 C	0.6788	Adder	0.8
957942	AF2-088 E	0.4525	Adder	0.53
957981	AF2-092 C	2.0281	Adder	2.39
957982	AF2-092 E	1.3521	Adder	1.59
958101	AF2-104 C (Withdrawn : 12/08/2020)	0.6345	Adder	0.75

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
958102	AF2-104 E (Withdrawn : 12/08/2020)	0.4230	Adder	0.5
958271	AF2-121 C	2.3914	Adder	2.81
958272	AF2-121 E	1.5943	Adder	1.88
958361	AF2-130 C	1.8119	Adder	2.13
958362	AF2-130 E	1.2080	Adder	1.42
958471	AF2-141	1.7881	Adder	2.1
958731	AF2-164 C O1	7.7585	Adder	9.13
958732	AF2-164 E O1	5.1724	Adder	6.09
958741	AF2-165 C	1.7255	Adder	2.03
958742	AF2-165 E	1.1504	Adder	1.35
958751	AF2-166 C	1.8119	Adder	2.13
958752	AF2-166 E	1.2080	Adder	1.42
959061	AF2-197 C O1	5.0993	Adder	6.0
959062	AF2-197 E O1	7.6490	Adder	9.0
959441	AF2-235 C	0.8096	Adder	0.95
959442	AF2-235 E	0.5397	Adder	0.63
959471	AF2-238 C	1.8325	Adder	2.16
959472	AF2-238 E	1.2217	Adder	1.44
959481	AF2-239 C	1.3844	Adder	1.63
959482	AF2-239 E	1.1100	Adder	1.31
959491	AF2-240 C	0.5611	Adder	0.66
959492	AF2-240 E	0.4780	Adder	0.56
959501	AF2-241 C	1.6773	Adder	1.97
959502	AF2-241 E	1.2914	Adder	1.52
959521	AF2-243 C	1.2942	Adder	1.52
959522	AF2-243 E	0.8628	Adder	1.02
959741	AF2-265 C	1.3133	Adder	1.55
959742	AF2-265 E	0.9774	Adder	1.15
959792	AF2-270 E	1.0498	Adder	1.24
959802	AF2-271 E	0.6922	Adder	0.81
959822	AF2-273 E	0.4853	Adder	0.57
960022	AF2-293 E	0.1498	Adder	0.18
960031	AF2-294 C	1.6402	Adder	1.93
960032	AF2-294 E	1.0934	Adder	1.29
960041	AF2-295 C	1.7937	Adder	2.11
960042	AF2-295 E	1.1958	Adder	1.41
960051	AF2-296 C	2.5539	Adder	3.0
960052	AF2-296 E	1.7026	Adder	2.0
960271	AF2-318 C	1.4966	Adder	1.76
960272	AF2-318 E	0.9978	Adder	1.17
960451	AF2-336 C	3.3802	Adder	3.98
960452	AF2-336 E	2.2535	Adder	2.65
960461	AF2-337 C	3.3802	Adder	3.98
960462	AF2-337 E	2.2535	Adder	2.65
960471	AF2-338 C	3.3802	Adder	3.98
960472	AF2-338 E	2.2535	Adder	2.65
960481	AF2-339 C	3.3802	Adder	3.98
960482	AF2-339 E	2.2535	Adder	2.65
960901	AF2-381 C	27.8886	50/50	27.8886
960902	AF2-381 E	14.6764	50/50	14.6764
961141	AF2-405	1.4844	Adder	1.75
961151	AF2-406	11.1327	Adder	13.1

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
961201	AF2-411 O1 (Withdrawn : 12/08/2020)	56.5972	Adder	66.58
961211	AF2-412	22.6426	Adder	26.64
961451	AF2-436	0.1675	Adder	0.2
NEWTON	NEWTON	0.6769	Confirmed LTF	0.6769
FARMERCITY	FARMERCITY	0.0356	Confirmed LTF	0.0356
GIBSON	GIBSON	0.3451	Confirmed LTF	0.3451
PRAIRIE	PRAIRIE	1.6841	Confirmed LTF	1.6841
O-066	O-066	1.7069	Confirmed LTF	1.7069
COFFEEN	COFFEEN	0.1245	Confirmed LTF	0.1245
CHEOAH	CHEOAH	0.3999	Confirmed LTF	0.3999
EDWARDS	EDWARDS	0.2054	Confirmed LTF	0.2054
TILTON	TILTON	0.3793	Confirmed LTF	0.3793
G-007	G-007	0.3910	Confirmed LTF	0.3910
MADISON	MADISON	0.0161	Confirmed LTF	0.0161
CALDERWOOD	CALDERWOOD	0.3951	Confirmed LTF	0.3951
BLUEG	BLUEG	1.1284	Confirmed LTF	1.1284
TRIMBLE	TRIMBLE	0.3612	Confirmed LTF	0.3612
CATAWBA	CATAWBA	0.3143	Confirmed LTF	0.3143

11.7.8 Index 8

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
100432689	235139	01AL&D6T	AP	235138	01AL 4J	AP	1	ATSI-P1-2-CEI-345-700T	single	151.0	103.68	105.51	AC	2.91

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
200608	26PINEY #1	0.4370	80/20	0.4370
200642	26SENECA#1	2.0066	80/20	2.0066
200643	26SENECA#2	1.8951	80/20	1.8951
200644	26SENECA#3	0.1843	80/20	0.1843
200662	26SCRUB GR	1.0495	80/20	1.0495
200828	26HNSMLK 1	0.6592	80/20	0.6592
200829	26HNSMLK 2	0.6592	80/20	0.6592
200830	26HNSMLK 3	0.6592	80/20	0.6592
200831	26HNSMLK 4	0.6592	80/20	0.6592
200832	26HNSMLK 5	0.6592	80/20	0.6592
200849	26LAKVU GN	0.0493	80/20	0.0493
201201	26WRREN CT	0.5079	80/20	0.5079
201477	26Y2-055	1.5376	80/20	1.5376
235030	01MHNG-T155	0.1680	80/20	0.1680
235134	01AL&D6	0.5103	80/20	0.5103
915951	Y3-092 FTIR	61.5400	80/20	61.5400
935191	AD1-154 (Withdrawn : 02/04/2021)	1.2304	80/20	1.2304
936881	AD2-112 C	-3.7669	Adder	-4.43
938951	AE1-123	1.4510	80/20	1.4510
939291	AE1-160 C	0.8943	80/20	0.8943
942811	AE2-299 C	1.8326	80/20	1.8326
942961	AE2-316 C	3.4559	80/20	3.4559
943151	AE2-344 C	3.7935	80/20	3.7935
943351	AF1-006 C	0.7579	80/20	0.7579
944261	AF1-094 C	0.7547	80/20	0.7547
944281	AF1-096 C	0.8450	80/20	0.8450
944301	AF1-098 C	2.7523	80/20	2.7523
944381	AF1-103 O1	1.0606	80/20	1.0606
944391	AF1-104 O1	1.2060	80/20	1.2060
944881	AF1-153 C O1	0.6875	80/20	0.6875
944901	AF1-155 C	0.6924	80/20	0.6924
945021	AF1-167 C	0.8138	80/20	0.8138
945051	AF1-170 C	2.7142	80/20	2.7142
945121	AF1-177	0.2651	80/20	0.2651
945451	AF1-210 C	1.0773	80/20	1.0773
945751	AF1-240 C O1	0.7382	80/20	0.7382
946381	AF1-302 C	1.5091	80/20	1.5091
946401	AF1-304 C	4.3206	80/20	4.3206
946421	AF1-306 C	2.5148	80/20	2.5148

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
957161	AF2-010 C	2.9095	80/20	2.9095
957571	AF2-051 C	2.1001	80/20	2.1001
958361	AF2-130 C	0.9302	80/20	0.9302
958731	AF2-164 C O1	3.5194	80/20	3.5194
958741	AF2-165 C	0.8887	80/20	0.8887
958751	AF2-166 C	0.9302	80/20	0.9302
959441	AF2-235 C	0.4225	80/20	0.4225
959521	AF2-243 C	0.6665	80/20	0.6665
960041	AF2-295 C	0.6924	80/20	0.6924
960891	AF2-380 C	0.5266	80/20	0.5266
NEWTON	NEWTON	0.4900	Confirmed LTF	0.4900
FARMERCITY	FARMERCITY	0.0252	Confirmed LTF	0.0252
G-007A	G-007A	1.5823	Confirmed LTF	1.5823
VFT	VFT	4.3860	Confirmed LTF	4.3860
GIBSON	GIBSON	0.2501	Confirmed LTF	0.2501
PRAIRIE	PRAIRIE	1.1572	Confirmed LTF	1.1572
COFFEEN	COFFEEN	0.0908	Confirmed LTF	0.0908
CHEOAH	CHEOAH	0.2057	Confirmed LTF	0.2057
EDWARDS	EDWARDS	0.1610	Confirmed LTF	0.1610
TILTON	TILTON	0.2904	Confirmed LTF	0.2904
MADISON	MADISON	0.0302	Confirmed LTF	0.0302
CALDERWOOD	CALDERWOOD	0.2048	Confirmed LTF	0.2048
BLUEG	BLUEG	0.7968	Confirmed LTF	0.7968
TRIMBLE	TRIMBLE	0.2560	Confirmed LTF	0.2560
CATAWBA	CATAWBA	0.1299	Confirmed LTF	0.1299

11.7.9 Index 9

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
100432286	235197	01KARNSC	AP	235152	01BUTLER	AP	1	ATSI-P2-3-CEI-345-004D	breaker	179.0	199.56	202.88	AC	6.99

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
200608	26PINEY #1	0.8526	50/50	0.8526
200662	26SCRUB GR	1.7816	50/50	1.7816
201477	26Y2-055	1.6819	Adder	1.98
235030	01MHNG-T155	0.1725	50/50	0.1725
293393	V3-030E	1.2541	Adder	1.48
915951	Y3-092 FTIR	92.1100	Merchant Transmission	92.1100
916202	Z1-069 E	2.5032	Adder	2.94
919491	AA2-000	21.4760	Adder	25.27
923443	AB1-160 E	0.7152	Adder	0.84
930511	AB1-092	0.7885	Adder	0.93
935191	AD1-154 (Withdrawn : 02/04/2021)	2.3410	50/50	2.3410
936421	AD2-055	1.6235	Adder	1.91
938951	AE1-123	2.8310	50/50	2.8310
939291	AE1-160 C	1.4743	50/50	1.4743
939292	AE1-160 E	0.8475	50/50	0.8475
940861	AE2-074 C	0.7260	Adder	0.85
940862	AE2-074 E	0.9557	Adder	1.12
941191	AE2-113 C	3.3852	Adder	3.98
941192	AE2-113 E	3.6448	Adder	4.29
941321	AE2-126 C (Suspended)	0.6515	Adder	0.77
941322	AE2-126 E (Suspended)	0.4343	Adder	0.51
941331	AE2-129 C	0.5508	Adder	0.65
941332	AE2-129 E	0.3672	Adder	0.43
941351	AE2-131 C (Suspended)	0.5508	Adder	0.65
941352	AE2-131 E (Suspended)	0.3672	Adder	0.43
942491	AE2-262 C	2.3673	Adder	2.79
942492	AE2-262 E	1.5908	Adder	1.87
942501	AE2-263 C	2.2252	Adder	2.62
942502	AE2-263 E	1.4857	Adder	1.75
942811	AE2-299 C	1.8996	Adder	2.23
942812	AE2-299 E	7.5986	Adder	8.94
942961	AE2-316 C	4.3232	50/50	4.3232
942962	AE2-316 E	6.1648	50/50	6.1648
943151	AE2-344 C	4.3878	Adder	5.16
943152	AE2-344 E	2.9252	Adder	3.44
943351	AF1-006 C	0.8631	Adder	1.02
943352	AF1-006 E	0.4855	Adder	0.57
943751	AF1-043	4.8704	Adder	5.73
944181	AF1-086 C O1	1.0969	Adder	1.29

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
944182	AF1-086 E O1	4.7723	Adder	5.61
944261	AF1-094 C	0.9127	Adder	1.07
944262	AF1-094 E	0.6085	Adder	0.72
944281	AF1-096 C	1.3931	50/50	1.3931
944282	AF1-096 E	0.9287	50/50	0.9287
944301	AF1-098 C	2.9490	Adder	3.47
944302	AF1-098 E	1.9660	Adder	2.31
944311	AF1-099 C	3.5935	Adder	4.23
944312	AF1-099 E	2.3957	Adder	2.82
944321	AF1-100 C	7.7354	Adder	9.1
944322	AF1-100 E	5.1569	Adder	6.07
944381	AF1-103 O1	1.1601	Adder	1.36
944391	AF1-104 O1	1.4831	Adder	1.74
944691	AF1-134 C O1	0.6278	Adder	0.74
944692	AF1-134 E O1	0.4185	Adder	0.49
944771	AF1-142 C	5.7496	Adder	6.76
944772	AF1-142 E	3.8331	Adder	4.51
944881	AF1-153 C O1	0.8874	Adder	1.04
944882	AF1-153 E O1	0.5916	Adder	0.7
944901	AF1-155 C	0.8962	Adder	1.05
944902	AF1-155 E	0.5974	Adder	0.7
945021	AF1-167 C	1.5108	50/50	1.5108
945022	AF1-167 E	1.0091	50/50	1.0091
945051	AF1-170 C	3.2735	Adder	3.85
945052	AF1-170 E	2.1823	Adder	2.57
945121	AF1-177	0.2900	Adder	0.34
945161	AF1-181	0.0413	Adder	0.05
945171	AF1-182	0.2064	Adder	0.24
945181	AF1-183	0.0512	Adder	0.06
945451	AF1-210 C	1.7109	50/50	1.7109
945452	AF1-210 E	1.1406	50/50	1.1406
945751	AF1-240 C O1	1.4046	50/50	1.4046
945752	AF1-240 E O1	0.9364	50/50	0.9364
946111	AF1-276 C	3.5189	Adder	4.14
946112	AF1-276 E	1.7332	Adder	2.04
946121	AF1-277 C	3.5189	Adder	4.14
946122	AF1-277 E	1.7332	Adder	2.04
946131	AF1-278 C	2.8046	Adder	3.3
946132	AF1-278 E	1.3950	Adder	1.64
946381	AF1-302 C	1.8878	50/50	1.8878
946382	AF1-302 E	2.5171	50/50	2.5171
946401	AF1-304 C	7.3032	50/50	7.3032
946402	AF1-304 E	4.8688	50/50	4.8688
946421	AF1-306 C	3.0276	Adder	3.56
946422	AF1-306 E	12.1105	Adder	14.25
957161	AF2-010 C	3.5483	Adder	4.17
957162	AF2-010 E	2.3913	Adder	2.81
957451	AF2-039 C	0.4120	Adder	0.48
957452	AF2-039 E	0.2747	Adder	0.32
957571	AF2-051 C	2.5767	Adder	3.03
957572	AF2-051 E	1.3274	Adder	1.56
957941	AF2-088 C	0.1984	Adder	0.23

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
957942	AF2-088 E	0.1322	Adder	0.16
958271	AF2-121 C	0.5508	Adder	0.65
958272	AF2-121 E	0.3672	Adder	0.43
958361	AF2-130 C	1.7261	50/50	1.7261
958362	AF2-130 E	1.1507	50/50	1.1507
958731	AF2-164 C O1	3.9711	Adder	4.67
958732	AF2-164 E O1	2.6474	Adder	3.11
958741	AF2-165 C	1.5401	50/50	1.5401
958742	AF2-165 E	1.0267	50/50	1.0267
958751	AF2-166 C	1.7261	50/50	1.7261
958752	AF2-166 E	1.1507	50/50	1.1507
959441	AF2-235 C	0.6965	50/50	0.6965
959442	AF2-235 E	0.4644	50/50	0.4644
959521	AF2-243 C	1.1551	50/50	1.1551
959522	AF2-243 E	0.7700	50/50	0.7700
959822	AF2-273 E	0.6772	50/50	0.6772
960041	AF2-295 C	0.8962	Adder	1.05
960042	AF2-295 E	0.5974	Adder	0.7
960051	AF2-296 C	0.6278	Adder	0.74
960052	AF2-296 E	0.4185	Adder	0.49
960891	AF2-380 C	3.6787	50/50	3.6787
960892	AF2-380 E	2.4525	50/50	2.4525
NEWTON	NEWTON	0.7178	Confirmed LTF	0.7178
FARMERCITY	FARMERCITY	0.0368	Confirmed LTF	0.0368
G-007A	G-007A	1.9995	Confirmed LTF	1.9995
VFT	VFT	5.5341	Confirmed LTF	5.5341
GIBSON	GIBSON	0.3669	Confirmed LTF	0.3669
PRAIRIE	PRAIRIE	1.6841	Confirmed LTF	1.6841
COFFEEN	COFFEEN	0.1333	Confirmed LTF	0.1333
CHEOAH	CHEOAH	0.2828	Confirmed LTF	0.2828
EDWARDS	EDWARDS	0.2390	Confirmed LTF	0.2390
TILTON	TILTON	0.4290	Confirmed LTF	0.4290
MADISON	MADISON	0.0524	Confirmed LTF	0.0524
CALDERWOOD	CALDERWOOD	0.2823	Confirmed LTF	0.2823
BLUEG	BLUEG	1.1649	Confirmed LTF	1.1649
TRIMBLE	TRIMBLE	0.3740	Confirmed LTF	0.3740
CATAWBA	CATAWBA	0.1694	Confirmed LTF	0.1694

11.7.10 Index 10

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
100432606	235203	01KISSNG	AP	235197	01KARNSC	AP	1	PN-P1-2-PN-345-107T	single	268.0	121.78	123.45	AC	4.63

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
200608	26PINEY #1	0.8948	80/20	0.8948
200642	26SENECA#1	2.9404	80/20	2.9404
200643	26SENECA#2	2.7771	80/20	2.7771
200644	26SENECA#3	0.2701	80/20	0.2701
200662	26SCRUB GR	1.8513	80/20	1.8513
200828	26HNSMLK 1	0.8843	80/20	0.8843
200829	26HNSMLK 2	0.8843	80/20	0.8843
200830	26HNSMLK 3	0.8843	80/20	0.8843
200831	26HNSMLK 4	0.8843	80/20	0.8843
200832	26HNSMLK 5	0.8843	80/20	0.8843
200849	26LAKVU GN	0.0714	80/20	0.0714
201201	26WRREN CT	0.7423	80/20	0.7423
201477	26Y2-055	2.2469	80/20	2.2469
915951	Y3-092 FTIR	87.8500	80/20	87.8500
919491	AA2-000	23.1486	Adder	27.23
930511	AB1-092	0.8499	Adder	1.0
935191	AD1-154 (Withdrawn : 02/04/2021)	2.4594	80/20	2.4594
936421	AD2-055	1.7499	Adder	2.06
936991	AD2-133 C	0.8054	Adder	0.95
938951	AE1-123	2.9711	80/20	2.9711
939171	AE1-147 C	0.5421	Adder	0.64
939291	AE1-160 C	1.5698	80/20	1.5698
940201	AE2-001 C	0.5408	Adder	0.64
940681	AE2-055 C (Suspended)	0.5199	Adder	0.61
940861	AE2-074 C	0.7917	Adder	0.93
941191	AE2-113 C	3.6859	Adder	4.34
941261	AE2-120 C	0.5400	Adder	0.64
941271	AE2-121 C	0.2895	Adder	0.34
941321	AE2-126 C (Suspended)	0.7033	Adder	0.83
941331	AE2-129 C	0.5900	Adder	0.69
941351	AE2-131 C (Suspended)	0.5900	Adder	0.69
942351	AE2-248 C	0.4229	Adder	0.5
942491	AE2-262 C	2.5479	Adder	3.0
942501	AE2-263 C	2.3950	Adder	2.82
942811	AE2-299 C	2.6518	80/20	2.6518
942961	AE2-316 C	4.5742	80/20	4.5742
943151	AE2-344 C	5.5410	80/20	5.5410
943351	AF1-006 C	1.1004	80/20	1.1004
943751	AF1-043	5.2497	Adder	6.18

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
944001	AF1-068 C O1 (Withdrawn : 12/15/2020)	0.5611	Adder	0.66
944181	AF1-086 C O1	1.1619	Adder	1.37
944261	AF1-094 C	1.1948	80/20	1.1948
944281	AF1-096 C	1.4833	80/20	1.4833
944301	AF1-098 C	4.0147	80/20	4.0147
944311	AF1-099 C	3.8677	Adder	4.55
944321	AF1-100 C	8.3219	Adder	9.79
944381	AF1-103 O1	1.5500	80/20	1.5500
944391	AF1-104 O1	1.7332	80/20	1.7332
944471	AF1-112 C	0.5663	Adder	0.67
944671	AF1-132 C O1 (Withdrawn : 12/15/2020)	0.5245	Adder	0.62
944691	AF1-134 C O1	0.6688	Adder	0.79
944771	AF1-142 C	6.1883	Adder	7.28
944881	AF1-153 C O1	1.1154	80/20	1.1154
944901	AF1-155 C	1.1261	80/20	1.1261
945021	AF1-167 C	1.4391	80/20	1.4391
945051	AF1-170 C	3.9292	80/20	3.9292
945121	AF1-177	0.3875	80/20	0.3875
945161	AF1-181	0.0444	Adder	0.05
945171	AF1-182	0.2221	Adder	0.26
945181	AF1-183	0.0551	Adder	0.06
945451	AF1-210 C	1.2842	80/20	1.2842
945491	AF1-214 C (Withdrawn : 12/03/2020)	0.5401	Adder	0.64
945751	AF1-240 C O1	1.4756	80/20	1.4756
946111	AF1-276 C	3.8754	Adder	4.56
946121	AF1-277 C	3.8754	Adder	4.56
946131	AF1-278 C	3.0888	Adder	3.63
946381	AF1-302 C	1.9975	80/20	1.9975
946401	AF1-304 C	7.7220	80/20	7.7220
946421	AF1-306 C	3.8410	80/20	3.8410
957161	AF2-010 C	4.6327	80/20	4.6327
957451	AF2-039 C	0.4423	Adder	0.52
957571	AF2-051 C	3.1109	80/20	3.1109
957941	AF2-088 C	0.2130	Adder	0.25
958271	AF2-121 C	0.5900	Adder	0.69
958361	AF2-130 C	1.8104	80/20	1.8104
958731	AF2-164 C O1	4.7213	80/20	4.7213
958741	AF2-165 C	1.6102	80/20	1.6102
958751	AF2-166 C	1.8104	80/20	1.8104
959441	AF2-235 C	0.7417	80/20	0.7417
959521	AF2-243 C	1.2076	80/20	1.2076
960041	AF2-295 C	1.1261	80/20	1.1261
960051	AF2-296 C	0.6688	Adder	0.79
961201	AF2-411 O1 (Withdrawn : 12/08/2020)	12.9948	Adder	15.29
961211	AF2-412	5.5648	Adder	6.55
NEWTON	NEWTON	0.7801	Confirmed LTF	0.7801
FARMERCITY	FARMERCITY	0.0400	Confirmed LTF	0.0400

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
G-007A	G-007A	2.1961	Confirmed LTF	2.1961
VFT	VFT	6.0759	Confirmed LTF	6.0759
GIBSON	GIBSON	0.3991	Confirmed LTF	0.3991
PRAIRIE	PRAIRIE	1.8313	Confirmed LTF	1.8313
COFFEEN	COFFEEN	0.1450	Confirmed LTF	0.1450
CHEOAH	CHEOAH	0.3063	Confirmed LTF	0.3063
EDWARDS	EDWARDS	0.2601	Confirmed LTF	0.2601
TILTON	TILTON	0.4668	Confirmed LTF	0.4668
MADISON	MADISON	0.0564	Confirmed LTF	0.0564
CALDERWOOD	CALDERWOOD	0.3057	Confirmed LTF	0.3057
BLUEG	BLUEG	1.2655	Confirmed LTF	1.2655
TRIMBLE	TRIMBLE	0.4068	Confirmed LTF	0.4068
CATAWBA	CATAWBA	0.1827	Confirmed LTF	0.1827

11.7.11 Index 11

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
100432304	235240	01COLMBGPN	AP	235202	01KISKIV	AP	1	ATSI-P2-3-CEI-345-004D	breaker	151.0	162.34	165.34	AC	5.34

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
201477	26Y2-055	1.3831	Adder	1.63
235030	01MHNG-T155	0.1947	50/50	0.1947
235134	01AL&D6	0.1817	50/50	0.1817
915951	Y3-092 FTIR	82.4500	Merchant Transmission	82.4500
935191	AD1-154 (Withdrawn : 02/04/2021)	1.1934	Adder	1.4
938951	AE1-123	1.4679	Adder	1.73
939291	AE1-160 C	0.8838	Adder	1.04
939292	AE1-160 E	0.5080	Adder	0.6
941191	AE2-113 C	2.7696	Adder	3.26
941192	AE2-113 E	2.9819	Adder	3.51
942811	AE2-299 C	1.5697	Adder	1.85
942812	AE2-299 E	6.2788	Adder	7.39
942961	AE2-316 C	3.5012	Adder	4.12
942962	AE2-316 E	4.9928	Adder	5.87
943151	AE2-344 C	3.7128	Adder	4.37
943152	AE2-344 E	2.4752	Adder	2.91
943351	AF1-006 C	0.7333	Adder	0.86
943352	AF1-006 E	0.4125	Adder	0.49
944261	AF1-094 C	0.7007	Adder	0.82
944262	AF1-094 E	0.4672	Adder	0.55
944281	AF1-096 C	0.8351	Adder	0.98
944282	AF1-096 E	0.5567	Adder	0.65
944301	AF1-098 C	2.4345	Adder	2.86
944302	AF1-098 E	1.6230	Adder	1.91
944381	AF1-103 O1	0.9540	Adder	1.12
944391	AF1-104 O1	1.3056	Adder	1.54
944881	AF1-153 C O1	0.6707	Adder	0.79
944882	AF1-153 E O1	0.4471	Adder	0.53
944901	AF1-155 C	0.6761	Adder	0.8
944902	AF1-155 E	0.4507	Adder	0.53
945021	AF1-167 C	0.9702	50/50	0.9702
945022	AF1-167 E	0.6480	50/50	0.6480
945051	AF1-170 C	2.8395	Adder	3.34
945052	AF1-170 E	1.8930	Adder	2.23
945121	AF1-177	0.2385	Adder	0.28
945451	AF1-210 C	1.2703	50/50	1.2703
945452	AF1-210 E	0.8468	50/50	0.8468
945751	AF1-240 C O1	0.7160	Adder	0.84
945752	AF1-240 E O1	0.4774	Adder	0.56

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
946111	AF1-276 C	2.9449	Adder	3.46
946112	AF1-276 E	1.4505	Adder	1.71
946121	AF1-277 C	2.9449	Adder	3.46
946122	AF1-277 E	1.4505	Adder	1.71
946131	AF1-278 C	2.3471	Adder	2.76
946132	AF1-278 E	1.1674	Adder	1.37
946381	AF1-302 C	1.5289	Adder	1.8
946382	AF1-302 E	2.0386	Adder	2.4
946401	AF1-304 C	4.3172	Adder	5.08
946402	AF1-304 E	2.8781	Adder	3.39
946421	AF1-306 C	2.4100	Adder	2.84
946422	AF1-306 E	9.6398	Adder	11.34
957161	AF2-010 C	2.7104	Adder	3.19
957162	AF2-010 E	1.8266	Adder	2.15
957571	AF2-051 C	2.1868	Adder	2.57
957572	AF2-051 E	1.1265	Adder	1.33
958361	AF2-130 C	0.9419	Adder	1.11
958362	AF2-130 E	0.6279	Adder	0.74
958731	AF2-164 C O1	3.7361	Adder	4.4
958732	AF2-164 E O1	2.4907	Adder	2.93
958741	AF2-165 C	0.9045	Adder	1.06
958742	AF2-165 E	0.6030	Adder	0.71
958751	AF2-166 C	0.9419	Adder	1.11
958752	AF2-166 E	0.6279	Adder	0.74
959441	AF2-235 C	0.4175	Adder	0.49
959442	AF2-235 E	0.2784	Adder	0.33
959521	AF2-243 C	0.6784	Adder	0.8
959522	AF2-243 E	0.4523	Adder	0.53
959822	AF2-273 E	0.4771	50/50	0.4771
960041	AF2-295 C	0.6761	Adder	0.8
960042	AF2-295 E	0.4507	Adder	0.53
960891	AF2-380 C	0.5148	Adder	0.61
960892	AF2-380 E	0.3432	Adder	0.4
NEWTON	NEWTON	0.5372	Confirmed LTF	0.5372
FARMERCITY	FARMERCITY	0.0277	Confirmed LTF	0.0277
G-007A	G-007A	1.6111	Confirmed LTF	1.6111
VFT	VFT	4.4763	Confirmed LTF	4.4763
GIBSON	GIBSON	0.2746	Confirmed LTF	0.2746
PRAIRIE	PRAIRIE	1.2708	Confirmed LTF	1.2708
COFFEEN	COFFEEN	0.0998	Confirmed LTF	0.0998
CHEOAH	CHEOAH	0.2232	Confirmed LTF	0.2232
EDWARDS	EDWARDS	0.1774	Confirmed LTF	0.1774
TILTON	TILTON	0.3194	Confirmed LTF	0.3194
MADISON	MADISON	0.0343	Confirmed LTF	0.0343
CALDERWOOD	CALDERWOOD	0.2227	Confirmed LTF	0.2227
BLUEG	BLUEG	0.8749	Confirmed LTF	0.8749
TRIMBLE	TRIMBLE	0.2810	Confirmed LTF	0.2810
CATAWBA	CATAWBA	0.1400	Confirmed LTF	0.1400

11.7.12 Index 12

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
100432296	235282	01GAR RN	AP	235240	01COLMBGPN	AP	1	ATSI-P2-3-CEI-345-004D	breaker	151.0	169.88	172.88	AC	5.34

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
201477	26Y2-055	1.3831	Adder	1.63
235030	01MHNG-T155	0.1947	50/50	0.1947
235134	01AL&D6	0.1817	50/50	0.1817
915951	Y3-092 FTIR	82.4500	Merchant Transmission	82.4500
935191	AD1-154 (Withdrawn : 02/04/2021)	1.1934	Adder	1.4
938951	AE1-123	1.4679	Adder	1.73
939291	AE1-160 C	0.8838	Adder	1.04
939292	AE1-160 E	0.5080	Adder	0.6
941191	AE2-113 C	2.7696	Adder	3.26
941192	AE2-113 E	2.9819	Adder	3.51
942811	AE2-299 C	1.5697	Adder	1.85
942812	AE2-299 E	6.2788	Adder	7.39
942961	AE2-316 C	3.5012	Adder	4.12
942962	AE2-316 E	4.9928	Adder	5.87
943151	AE2-344 C	3.7128	Adder	4.37
943152	AE2-344 E	2.4752	Adder	2.91
943351	AF1-006 C	0.7333	Adder	0.86
943352	AF1-006 E	0.4125	Adder	0.49
944261	AF1-094 C	0.7007	Adder	0.82
944262	AF1-094 E	0.4672	Adder	0.55
944281	AF1-096 C	0.8351	Adder	0.98
944282	AF1-096 E	0.5567	Adder	0.65
944301	AF1-098 C	2.4345	Adder	2.86
944302	AF1-098 E	1.6230	Adder	1.91
944381	AF1-103 O1	0.9540	Adder	1.12
944391	AF1-104 O1	1.3056	Adder	1.54
944881	AF1-153 C O1	0.6707	Adder	0.79
944882	AF1-153 E O1	0.4471	Adder	0.53
944901	AF1-155 C	0.6761	Adder	0.8
944902	AF1-155 E	0.4507	Adder	0.53
945021	AF1-167 C	0.9702	50/50	0.9702
945022	AF1-167 E	0.6480	50/50	0.6480
945051	AF1-170 C	2.8395	Adder	3.34
945052	AF1-170 E	1.8930	Adder	2.23
945121	AF1-177	0.2385	Adder	0.28
945451	AF1-210 C	1.2703	50/50	1.2703
945452	AF1-210 E	0.8468	50/50	0.8468
945751	AF1-240 C O1	0.7160	Adder	0.84
945752	AF1-240 E O1	0.4774	Adder	0.56

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
946111	AF1-276 C	2.9449	Adder	3.46
946112	AF1-276 E	1.4505	Adder	1.71
946121	AF1-277 C	2.9449	Adder	3.46
946122	AF1-277 E	1.4505	Adder	1.71
946131	AF1-278 C	2.3471	Adder	2.76
946132	AF1-278 E	1.1674	Adder	1.37
946381	AF1-302 C	1.5289	Adder	1.8
946382	AF1-302 E	2.0386	Adder	2.4
946401	AF1-304 C	4.3172	Adder	5.08
946402	AF1-304 E	2.8781	Adder	3.39
946421	AF1-306 C	2.4100	Adder	2.84
946422	AF1-306 E	9.6398	Adder	11.34
957161	AF2-010 C	2.7104	Adder	3.19
957162	AF2-010 E	1.8266	Adder	2.15
957571	AF2-051 C	2.1868	Adder	2.57
957572	AF2-051 E	1.1265	Adder	1.33
958361	AF2-130 C	0.9419	Adder	1.11
958362	AF2-130 E	0.6279	Adder	0.74
958731	AF2-164 C O1	3.7361	Adder	4.4
958732	AF2-164 E O1	2.4907	Adder	2.93
958741	AF2-165 C	0.9045	Adder	1.06
958742	AF2-165 E	0.6030	Adder	0.71
958751	AF2-166 C	0.9419	Adder	1.11
958752	AF2-166 E	0.6279	Adder	0.74
959441	AF2-235 C	0.4175	Adder	0.49
959442	AF2-235 E	0.2784	Adder	0.33
959521	AF2-243 C	0.6784	Adder	0.8
959522	AF2-243 E	0.4523	Adder	0.53
959822	AF2-273 E	0.4771	50/50	0.4771
960041	AF2-295 C	0.6761	Adder	0.8
960042	AF2-295 E	0.4507	Adder	0.53
960891	AF2-380 C	0.5148	Adder	0.61
960892	AF2-380 E	0.3432	Adder	0.4
NEWTON	NEWTON	0.5372	Confirmed LTF	0.5372
FARMERCITY	FARMERCITY	0.0277	Confirmed LTF	0.0277
G-007A	G-007A	1.6111	Confirmed LTF	1.6111
VFT	VFT	4.4763	Confirmed LTF	4.4763
GIBSON	GIBSON	0.2746	Confirmed LTF	0.2746
PRAIRIE	PRAIRIE	1.2708	Confirmed LTF	1.2708
COFFEEN	COFFEEN	0.0998	Confirmed LTF	0.0998
CHEOAH	CHEOAH	0.2232	Confirmed LTF	0.2232
EDWARDS	EDWARDS	0.1774	Confirmed LTF	0.1774
TILTON	TILTON	0.3194	Confirmed LTF	0.3194
MADISON	MADISON	0.0343	Confirmed LTF	0.0343
CALDERWOOD	CALDERWOOD	0.2227	Confirmed LTF	0.2227
BLUEG	BLUEG	0.8749	Confirmed LTF	0.8749
TRIMBLE	TRIMBLE	0.2810	Confirmed LTF	0.2810
CATAWBA	CATAWBA	0.1400	Confirmed LTF	0.1400

11.7.13 Index 13

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
98609743	239036	02PERRY	ATSI	239334	02L.CENTER	ATSI	1	ATSI-P7-1-CEI-345-012	tower	1667.0	129.98	131.33	AC	22.1

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
200828	26HNSMLK 1	2.1169	50/50	2.1169
200829	26HNSMLK 2	2.1169	50/50	2.1169
200830	26HNSMLK 3	2.1169	50/50	2.1169
200831	26HNSMLK 4	2.1169	50/50	2.1169
200832	26HNSMLK 5	2.1169	50/50	2.1169
200849	26LAKVU GN	0.2788	50/50	0.2788
200894	26K02	4.6867	Adder	5.51
200898	26AA1-106	1.6761	Adder	1.97
201477	26Y2-055	5.6490	Adder	6.65
203999	P-047 E	7.1641	Adder	8.43
239035	02PERRG1 (Deactivation : 31/05/2021)	99.6812	50/50	99.6812
915951	Y3-092 FTIR	384.7400	50/50	384.7400
916202	Z1-069 E	5.9815	Adder	7.04
919491	AA2-000	41.2047	Adder	48.48
920341	AA2-132 (Withdrawn : 12/07/2020)	1.6995	Adder	2.0
923443	AB1-160 E	1.7090	Adder	2.01
930511	AB1-092	1.5129	Adder	1.78
935191	AD1-154 (Withdrawn : 02/04/2021)	2.0432	Adder	2.4
936421	AD2-055	3.1148	Adder	3.66
938951	AE1-123	2.2297	Adder	2.62
939291	AE1-160 C	2.9722	50/50	2.9722
939292	AE1-160 E	1.7084	50/50	1.7084
940861	AE2-074 C	1.7588	Adder	2.07
940862	AE2-074 E	2.3152	Adder	2.72
941191	AE2-113 C	7.5755	Adder	8.91
941192	AE2-113 E	8.1564	Adder	9.6
941321	AE2-126 C (Suspended)	1.2452	Adder	1.46
941322	AE2-126 E (Suspended)	0.8301	Adder	0.98
942491	AE2-262 C	4.3240	Adder	5.09
942492	AE2-262 E	2.9057	Adder	3.42
942501	AE2-263 C	4.0645	Adder	4.78
942502	AE2-263 E	2.7137	Adder	3.19
942811	AE2-299 C	10.1027	50/50	10.1027
942812	AE2-299 E	40.4109	50/50	40.4109
942961	AE2-316 C	4.2752	Adder	5.03
942962	AE2-316 E	6.0965	Adder	7.17
943151	AE2-344 C	19.9774	50/50	19.9774
943152	AE2-344 E	13.3183	50/50	13.3183

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
943351	AF1-006 C	4.3237	50/50	4.3237
943352	AF1-006 E	2.4321	50/50	2.4321
943751	AF1-043	9.3445	Adder	10.99
944261	AF1-094 C	3.4752	50/50	3.4752
944262	AF1-094 E	2.3168	50/50	2.3168
944281	AF1-096 C	2.8084	50/50	2.8084
944282	AF1-096 E	1.8722	50/50	1.8722
944301	AF1-098 C	14.6069	50/50	14.6069
944302	AF1-098 E	9.7379	50/50	9.7379
944311	AF1-099 C	6.5638	Adder	7.72
944312	AF1-099 E	4.3758	Adder	5.15
944321	AF1-100 C	13.9001	Adder	16.35
944322	AF1-100 E	9.2667	Adder	10.9
944381	AF1-103 O1	3.8990	Adder	4.59
944391	AF1-104 O1	7.2374	50/50	7.2374
944411	AF1-106 O1	1.7874	Adder	2.1
944691	AF1-134 C O1	1.0020	Adder	1.18
944692	AF1-134 E O1	0.6680	Adder	0.79
944771	AF1-142 C	10.5020	Adder	12.36
944772	AF1-142 E	7.0013	Adder	8.24
944881	AF1-153 C O1	1.2756	Adder	1.5
944882	AF1-153 E O1	0.8504	Adder	1.0
944901	AF1-155 C	1.2758	Adder	1.5
944902	AF1-155 E	0.8505	Adder	1.0
945021	AF1-167 C	0.8066	Adder	0.95
945022	AF1-167 E	0.5387	Adder	0.63
945051	AF1-170 C	15.3287	50/50	15.3287
945052	AF1-170 E	10.2192	50/50	10.2192
945121	AF1-177	0.9747	Adder	1.15
945161	AF1-181	0.0740	Adder	0.09
945171	AF1-182	0.3702	Adder	0.44
945181	AF1-183	0.0916	Adder	0.11
945331	AF1-198	0.2257	Adder	0.27
945451	AF1-210 C	0.8772	Adder	1.03
945452	AF1-210 E	0.5848	Adder	0.69
945521	AF1-217 C O2	1.6386	Adder	1.93
945522	AF1-217 E O2	1.0924	Adder	1.29
945751	AF1-240 C O1	1.2259	Adder	1.44
945752	AF1-240 E O1	0.8173	Adder	0.96
946111	AF1-276 C	9.2772	Adder	10.91
946112	AF1-276 E	4.5693	Adder	5.38
946121	AF1-277 C	9.2772	Adder	10.91
946122	AF1-277 E	4.5693	Adder	5.38
946131	AF1-278 C	7.3940	Adder	8.7
946132	AF1-278 E	3.6776	Adder	4.33
946211	AF1-286 C O1	0.7239	Adder	0.85
946212	AF1-286 E O1	0.4915	Adder	0.58
946221	AF1-287 C	1.6552	Adder	1.95
946222	AF1-287 E	1.1034	Adder	1.3
946381	AF1-302 C	1.8669	Adder	2.2
946382	AF1-302 E	2.4892	Adder	2.93
946401	AF1-304 C	11.3363	Adder	13.34

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
946402	AF1-304 E	7.5575	Adder	8.89
946421	AF1-306 C	5.4309	Adder	6.39
946422	AF1-306 E	21.7236	Adder	25.56
957161	AF2-010 C	13.1997	50/50	13.1997
957162	AF2-010 E	8.8954	50/50	8.8954
957451	AF2-039 C	0.7333	Adder	0.86
957452	AF2-039 E	0.4888	Adder	0.58
957571	AF2-051 C	9.1172	50/50	9.1172
957572	AF2-051 E	4.6968	50/50	4.6968
957941	AF2-088 C	0.3530	Adder	0.42
957942	AF2-088 E	0.2354	Adder	0.28
958361	AF2-130 C	1.7840	Adder	2.1
958362	AF2-130 E	1.1893	Adder	1.4
958521	AF2-145 C2	2.0687	Adder	2.43
958522	AF2-145 E2	1.3791	Adder	1.62
958731	AF2-164 C O1	11.3026	50/50	11.3026
958732	AF2-164 E O1	7.5350	50/50	7.5350
958741	AF2-165 C	2.1289	Adder	2.5
958742	AF2-165 E	1.4193	Adder	1.67
958751	AF2-166 C	1.7840	Adder	2.1
958752	AF2-166 E	1.1893	Adder	1.4
959441	AF2-235 C	1.4042	50/50	1.4042
959442	AF2-235 E	0.9361	50/50	0.9361
959471	AF2-238 C	1.0458	Adder	1.23
959472	AF2-238 E	0.6972	Adder	0.82
959491	AF2-240 C	0.3378	Adder	0.4
959492	AF2-240 E	0.2878	Adder	0.34
959501	AF2-241 C	1.0099	Adder	1.19
959502	AF2-241 E	0.7775	Adder	0.91
959521	AF2-243 C	1.5967	Adder	1.88
959522	AF2-243 E	1.0645	Adder	1.25
959741	AF2-265 C	0.7495	Adder	0.88
959742	AF2-265 E	0.5578	Adder	0.66
959822	AF2-273 E	0.3527	Adder	0.41
960031	AF2-294 C	1.0248	Adder	1.21
960032	AF2-294 E	0.6832	Adder	0.8
960041	AF2-295 C	1.2758	Adder	1.5
960042	AF2-295 E	0.8505	Adder	1.0
960051	AF2-296 C	1.0020	Adder	1.18
960052	AF2-296 E	0.6680	Adder	0.79
961141	AF2-405	0.8937	Adder	1.05
961151	AF2-406	6.7027	Adder	7.89
961201	AF2-411 O1 (Withdrawn : 12/08/2020)	27.9250	Adder	32.85
961211	AF2-412	12.0544	Adder	14.18
NEWTON	NEWTON	3.4137	Confirmed LTF	3.4137
BLUEG	BLUEG	5.4528	Confirmed LTF	5.4528
G-007A	G-007A	6.9959	Confirmed LTF	6.9959
VFT	VFT	19.2533	Confirmed LTF	19.2533
PRAIRIE	PRAIRIE	7.9711	Confirmed LTF	7.9711
LGE-TSR-0092018	LGE-TSR-0092018	0.0000	LTF	0.0000
COFFEEN	COFFEEN	0.6372	Confirmed LTF	0.6372
CHEOAH	CHEOAH	1.2613	Confirmed LTF	1.2613

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
EDWARDS	EDWARDS	1.1602	Confirmed LTF	1.1602
TILTON	TILTON	2.0639	Confirmed LTF	2.0639
MADISON	MADISON	0.0968	Confirmed LTF	0.0968
GIBSON	GIBSON	1.7385	Confirmed LTF	1.7385
CALDERWOOD	CALDERWOOD	1.2604	Confirmed LTF	1.2604
FARMERCITY	FARMERCITY	0.1758	Confirmed LTF	0.1758
TRIMBLE	TRIMBLE	1.7507	Confirmed LTF	1.7507
LGE-GI-0012019	LGE-GI-0012019	3.4606	LTF	3.4606
CATAWBA	CATAWBA	0.7270	Confirmed LTF	0.7270

11.7.14 Index 14

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
98609745	239036	02PERRY	ATSI	238684	02EASTLK	ATSI	1	ATSI-P7-1-CEI-345-016	tower	1667.0	121.97	123.26	AC	21.37

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
200828	26HNSMLK 1	2.0470	50/50	2.0470
200829	26HNSMLK 2	2.0470	50/50	2.0470
200830	26HNSMLK 3	2.0470	50/50	2.0470
200831	26HNSMLK 4	2.0470	50/50	2.0470
200832	26HNSMLK 5	2.0470	50/50	2.0470
200849	26LAKVU GN	0.2696	50/50	0.2696
200894	26K02	4.5306	Adder	5.33
200898	26AA1-106	1.6205	Adder	1.91
201477	26Y2-055	5.4627	Adder	6.43
203999	P-047 E	6.9260	Adder	8.15
239035	02PERRG1 (Deactivation : 31/05/2021)	93.8239	50/50	93.8239
239276	02COLLW 11	-2.0552	Adder	-2.42
915951	Y3-092 FTIR	372.1000	50/50	372.1000
916202	Z1-069 E	5.7829	Adder	6.8
919491	AA2-000	39.8351	Adder	46.86
920341	AA2-132 (Withdrawn : 12/07/2020)	1.6430	Adder	1.93
923443	AB1-160 E	1.6523	Adder	1.94
930511	AB1-092	1.4626	Adder	1.72
935191	AD1-154 (Withdrawn : 02/04/2021)	1.9751	Adder	2.32
936421	AD2-055	3.0113	Adder	3.54
938951	AE1-123	2.1555	Adder	2.54
939291	AE1-160 C	2.8740	50/50	2.8740
939292	AE1-160 E	1.6520	50/50	1.6520
940861	AE2-074 C	1.7005	Adder	2.0
940862	AE2-074 E	2.2384	Adder	2.63
941191	AE2-113 C	7.3238	Adder	8.62
941192	AE2-113 E	7.8854	Adder	9.28
941321	AE2-126 C (Suspended)	1.2038	Adder	1.42
941322	AE2-126 E (Suspended)	0.8025	Adder	0.94
942491	AE2-262 C	4.1803	Adder	4.92
942492	AE2-262 E	2.8092	Adder	3.3
942501	AE2-263 C	3.9295	Adder	4.62
942502	AE2-263 E	2.6236	Adder	3.09
942811	AE2-299 C	9.7702	50/50	9.7702
942812	AE2-299 E	39.0810	50/50	39.0810
942961	AE2-316 C	4.1330	Adder	4.86
942962	AE2-316 E	5.8936	Adder	6.93
943151	AE2-344 C	19.3204	50/50	19.3204
943152	AE2-344 E	12.8802	50/50	12.8802

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
943351	AF1-006 C	4.1815	50/50	4.1815
943352	AF1-006 E	2.3521	50/50	2.3521
943751	AF1-043	9.0339	Adder	10.63
944261	AF1-094 C	3.3607	50/50	3.3607
944262	AF1-094 E	2.2405	50/50	2.2405
944281	AF1-096 C	2.7156	50/50	2.7156
944282	AF1-096 E	1.8104	50/50	1.8104
944301	AF1-098 C	14.1259	50/50	14.1259
944302	AF1-098 E	9.4173	50/50	9.4173
944311	AF1-099 C	6.3457	Adder	7.47
944312	AF1-099 E	4.2305	Adder	4.98
944321	AF1-100 C	13.4372	Adder	15.81
944322	AF1-100 E	8.9581	Adder	10.54
944381	AF1-103 O1	3.7703	Adder	4.44
944391	AF1-104 O1	6.9996	50/50	6.9996
944411	AF1-106 O1	1.7280	Adder	2.03
944691	AF1-134 C O1	0.9685	Adder	1.14
944692	AF1-134 E O1	0.6457	Adder	0.76
944771	AF1-142 C	10.1531	Adder	11.94
944772	AF1-142 E	6.7687	Adder	7.96
944881	AF1-153 C O1	1.2333	Adder	1.45
944882	AF1-153 E O1	0.8222	Adder	0.97
944901	AF1-155 C	1.2334	Adder	1.45
944902	AF1-155 E	0.8223	Adder	0.97
945021	AF1-167 C	0.7796	Adder	0.92
945022	AF1-167 E	0.5207	Adder	0.61
945051	AF1-170 C	14.8247	50/50	14.8247
945052	AF1-170 E	9.8831	50/50	9.8831
945121	AF1-177	0.9426	Adder	1.11
945161	AF1-181	0.0716	Adder	0.08
945171	AF1-182	0.3579	Adder	0.42
945181	AF1-183	0.0885	Adder	0.1
945331	AF1-198	0.2182	Adder	0.26
945451	AF1-210 C	0.8478	Adder	1.0
945452	AF1-210 E	0.5652	Adder	0.66
945521	AF1-217 C O2	1.5846	Adder	1.86
945522	AF1-217 E O2	1.0564	Adder	1.24
945751	AF1-240 C O1	1.1850	Adder	1.39
945752	AF1-240 E O1	0.7900	Adder	0.93
946111	AF1-276 C	8.9702	Adder	10.55
946112	AF1-276 E	4.4182	Adder	5.2
946121	AF1-277 C	8.9702	Adder	10.55
946122	AF1-277 E	4.4182	Adder	5.2
946131	AF1-278 C	7.1494	Adder	8.41
946132	AF1-278 E	3.5559	Adder	4.18
946211	AF1-286 C O1	0.6999	Adder	0.82
946212	AF1-286 E O1	0.4752	Adder	0.56
946221	AF1-287 C	1.6005	Adder	1.88
946222	AF1-287 E	1.0670	Adder	1.26
946381	AF1-302 C	1.8048	Adder	2.12
946382	AF1-302 E	2.4064	Adder	2.83
946401	AF1-304 C	10.9619	Adder	12.9

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
946402	AF1-304 E	7.3080	Adder	8.6
946421	AF1-306 C	5.2507	Adder	6.18
946422	AF1-306 E	21.0028	Adder	24.71
957161	AF2-010 C	12.7650	50/50	12.7650
957162	AF2-010 E	8.6025	50/50	8.6025
957451	AF2-039 C	0.7088	Adder	0.83
957452	AF2-039 E	0.4725	Adder	0.56
957571	AF2-051 C	8.8166	50/50	8.8166
957572	AF2-051 E	4.5419	50/50	4.5419
957941	AF2-088 C	0.3413	Adder	0.4
957942	AF2-088 E	0.2275	Adder	0.27
958361	AF2-130 C	1.7248	Adder	2.03
958362	AF2-130 E	1.1499	Adder	1.35
958521	AF2-145 C2	2.0004	Adder	2.35
958522	AF2-145 E2	1.3336	Adder	1.57
958731	AF2-164 C O1	10.9296	50/50	10.9296
958732	AF2-164 E O1	7.2864	50/50	7.2864
958741	AF2-165 C	2.0586	Adder	2.42
958742	AF2-165 E	1.3724	Adder	1.61
958751	AF2-166 C	1.7248	Adder	2.03
958752	AF2-166 E	1.1499	Adder	1.35
959441	AF2-235 C	1.3578	50/50	1.3578
959442	AF2-235 E	0.9052	50/50	0.9052
959471	AF2-238 C	1.0110	Adder	1.19
959472	AF2-238 E	0.6740	Adder	0.79
959491	AF2-240 C	0.3266	Adder	0.38
959492	AF2-240 E	0.2782	Adder	0.33
959501	AF2-241 C	0.9763	Adder	1.15
959502	AF2-241 E	0.7517	Adder	0.88
959521	AF2-243 C	1.5439	Adder	1.82
959522	AF2-243 E	1.0293	Adder	1.21
959741	AF2-265 C	0.7246	Adder	0.85
959742	AF2-265 E	0.5392	Adder	0.63
959822	AF2-273 E	0.3408	Adder	0.4
960031	AF2-294 C	0.9907	Adder	1.17
960032	AF2-294 E	0.6605	Adder	0.78
960041	AF2-295 C	1.2334	Adder	1.45
960042	AF2-295 E	0.8223	Adder	0.97
960051	AF2-296 C	0.9685	Adder	1.14
960052	AF2-296 E	0.6457	Adder	0.76
961141	AF2-405	0.8640	Adder	1.02
961151	AF2-406	6.4802	Adder	7.62
961201	AF2-411 O1 (Withdrawn : 12/08/2020)	26.9969	Adder	31.76
961211	AF2-412	11.6533	Adder	13.71
NEWTON	NEWTON	3.2676	Confirmed LTF	3.2676
BLUEG	BLUEG	5.2219	Confirmed LTF	5.2219
G-007A	G-007A	6.7610	Confirmed LTF	6.7610
VFT	VFT	18.6018	Confirmed LTF	18.6018
PRAIRIE	PRAIRIE	7.6302	Confirmed LTF	7.6302
COFFEEN	COFFEEN	0.6097	Confirmed LTF	0.6097
CHEOAH	CHEOAH	1.2082	Confirmed LTF	1.2082
EDWARDS	EDWARDS	1.1102	Confirmed LTF	1.1102

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
TILTON	TILTON	1.9751	Confirmed LTF	1.9751
MADISON	MADISON	0.0988	Confirmed LTF	0.0988
GIBSON	GIBSON	1.6642	Confirmed LTF	1.6642
CALDERWOOD	CALDERWOOD	1.2072	Confirmed LTF	1.2072
FARMERCITY	FARMERCITY	0.1682	Confirmed LTF	0.1682
TRIMBLE	TRIMBLE	1.6762	Confirmed LTF	1.6762
LGE-GI-0012019	LGE-GI-0012019	3.3132	LTF	3.3132
CATAWBA	CATAWBA	0.6965	Confirmed LTF	0.6965

11.7.15 Index 15

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
111556533	943150	AE2-344 TAP	PENELEC	200573	26VENGO JT	PENELEC	1	ATSI-P2-3-CEI-345-004D	breaker	179.0	122.92	124.48	AC	4.03

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
915952	Y3-092 NFTWR	115.0500	50/50	115.0500
923821	AB2-019 FTWR	3.2214	50/50	3.2214
942811	AE2-299 C	2.0501	Adder	2.41
942812	AE2-299 E	8.2003	Adder	9.65
943151	AE2-344 C	28.7275	50/50	28.7275
943152	AE2-344 E	19.1517	50/50	19.1517
944261	AF1-094 C	0.5659	Adder	0.67
944262	AF1-094 E	0.3773	Adder	0.44
944301	AF1-098 C	3.0657	Adder	3.61
944302	AF1-098 E	2.0438	Adder	2.4
944392	AF1-104 BAT	2.7082	50/50	2.7082
945511	AF1-216 C1O1	2.6052	Adder	3.06
945512	AF1-216 E1O1	1.7348	Adder	2.04
945521	AF1-217 C O2	1.9870	50/50	1.9870
945522	AF1-217 E O2	1.3246	50/50	1.3246
946221	AF1-287 C	2.0114	50/50	2.0114
946222	AF1-287 E	1.3410	50/50	1.3410
957161	AF2-010 C	2.0453	Adder	2.41
957162	AF2-010 E	1.3784	Adder	1.62
958511	AF2-145 C1	0.9280	Adder	1.09
958512	AF2-145 E1	0.6187	Adder	0.73
958521	AF2-145 C2	2.5029	50/50	2.5029
958522	AF2-145 E2	1.6686	50/50	1.6686
NEWTON	NEWTON	0.2579	Confirmed LTF	0.2579
FARMERCITY	FARMERCITY	0.0133	Confirmed LTF	0.0133
G-007A	G-007A	1.3714	Confirmed LTF	1.3714
VFT	VFT	3.8378	Confirmed LTF	3.8378
GIBSON	GIBSON	0.1316	Confirmed LTF	0.1316
PRAIRIE	PRAIRIE	0.6070	Confirmed LTF	0.6070
COFFEEN	COFFEEN	0.0479	Confirmed LTF	0.0479
CHEOAH	CHEOAH	0.1056	Confirmed LTF	0.1056
EDWARDS	EDWARDS	0.0851	Confirmed LTF	0.0851
TILTON	TILTON	0.1531	Confirmed LTF	0.1531
MADISON	MADISON	0.0161	Confirmed LTF	0.0161
CALDERWOOD	CALDERWOOD	0.1054	Confirmed LTF	0.1054
BLUEG	BLUEG	0.4184	Confirmed LTF	0.4184
TRIMBLE	TRIMBLE	0.1347	Confirmed LTF	0.1347
CATAWBA	CATAWBA	0.0651	Confirmed LTF	0.0651

11.8 Queue Dependencies

The Queue Projects below are listed in one or more indices for the overloads identified in your report. These projects contribute to the loading of the overloaded facilities identified in your report. The percent overload of a facility and cost allocation you may have towards a particular reinforcement could vary depending on the action of these earlier projects. The status of each project at the time of the analysis is presented in the table. This list may change as earlier projects withdraw or modify their requests.

Queue Number	Project Name	Status
AA1-082	E. Towanda 230kV	In Service
AA1-106	Grover II 34.5kV	Withdrawn
AA1-144	East Towanda-Grover 230kV	Engineering and Procurement
AA2-000	N/A	N/A
AA2-132	Thompson 34.5kV	Withdrawn
AB1-092	Moshannon-East Towanda 230kV	Active
AB1-160	Gold-Sabinsville 115kV	In Service
AB2-019	Erie West 345kV	Engineering and Procurement
AC1-025	Dale Summit	In Service
AD1-154	Timblin 34.5 kV	Withdrawn
AD2-055	Moshannon-East Towanda 230 kV	Active
AD2-112	Springdale CC II 138kV	Engineering and Procurement
AD2-133	Eagle Valley 115kV	Active
AE1-053	Meyersdale North	Active
AE1-116	Somerset Windpower 23 kV	Active
AE1-123	Emlenton 34.5 kV	Engineering and Procurement
AE1-128	Bedford North-Wills Mounain 115 kV	Active
AE1-147	Bellefonte 46 kV	Engineering and Procurement
AE1-160	Venango 34.5 kV	Engineering and Procurement
AE2-001	Nittany-Zion 46 kV	Active
AE2-055	Shingletown-Boalsburg 46 kV	Suspended
AE2-074	Potter 46 kV	Active
AE2-113	Farmers Valley-Ridgeway 115 kV	Active
AE2-117	ABW Tap-Alexandria 46 kV	Active
AE2-118	ABW Tap-Williamsburg 46 kV	Active
AE2-120	Graymont-Zion 46 kV	Active
AE2-121	Milesburg-Tanney Junction 46 kV	Active
AE2-126	Dubois-Curwensville 34.5 kV	Suspended
AE2-129	Philipsburg-Clarence 34.5 kV	Engineering and Procurement
AE2-131	Philipsburg-Karthaus 34.5	Suspended
AE2-139	East Towanda-Grover 230 kV	Active
AE2-224	Bearrock-Johnstown 230 kV	Active
AE2-248	Fillmore-Thompson Farm 46 kV	Active
AE2-249	Bedford North-Pennsylvania Hollow 23 kV	Suspended
AE2-262	Moshannon-Milesburg 230 kV	Active
AE2-263	Moshannon-Milesburg 230 kV	Active
AE2-264	Altoona-Raystown 230 kV	Active
AE2-299	Erie East 230 kV	Active
AE2-316	Brookville-Squab Hollow 138 kV	Active

Queue Number	Project Name	Status
AE2-344	Edinboro South-Venango Junction 115 kV	Active
AF1-006	Fairview East 34.5 kV	Active
AF1-039	Listonburg-Highpoint 24.9 kV	Active
AF1-043	Moshannon-East Towanda 230 kV	Active
AF1-068	Boalsburg-Centre Hall 46 kV	Withdrawn
AF1-086	Madera-Westover South 115 kV	Active
AF1-094	Union City-Cambridge Springs 34.5 kV	Active
AF1-096	Titusville-Oil Creek 34.5 kV	Active
AF1-098	Four Mile Jct-Corry East 115 kV	Active
AF1-099	Moshannon-Milesburg 230 kV	Active
AF1-100	Shawville-Moshannon 230 kV	Active
AF1-103	Warren 34.5 kV	Active
AF1-104	Erie West 34.5 kV	Active
AF1-106	East Sayre 34.5 kV	Active
AF1-112	Centre Hall-Boalsburg 46 kV	Active
AF1-132	Shingletown-Boalsburg 46 kV	Withdrawn
AF1-134	Philipsburg-Madera 34.5 kV	Active
AF1-140	Claysburg 23 kV	Engineering and Procurement
AF1-142	Moshannon-Milesburg 230 kV	Active
AF1-143	Lick Run 115 kV	Active
AF1-153	Motion-Ridgeway 46 kV	Active
AF1-155	Paper City-Wilcox 46 kV	Engineering and Procurement
AF1-167	West Freedom-C&K Coal 25 kV	Active
AF1-170	Springboro-Venango Junction 115 kV	Active
AF1-177	Warren 115 kV	Partially in Service - Under Construction
AF1-181	Shawville 3 230 kV	Partially in Service - Under Construction
AF1-182	Shawville 4 230 kV	Partially in Service - Under Construction
AF1-183	Shawville 1 230 kV	Partially in Service - Under Construction
AF1-198	Blossburg #1 CT 34.5 kV	Partially in Service - Under Construction
AF1-210	Burma 23 kV	Engineering and Procurement
AF1-214	Nittany-Zion 46 kV	Withdrawn
AF1-216	Lycoming-Lock Haven 69 kV	Active
AF1-217	Edinboro -Cambridge Springs 34.5 kV	Engineering and Procurement
AF1-232	Allegheny-Somerset 115 kV	Withdrawn
AF1-240	Timblin 34.5 kV	Active
AF1-272	Lucerne 115 kV	Active
AF1-276	Lewis Run-Pierce Brook 230 kV	Active
AF1-277	Lewis Run-Pierce Brook 2 230 kV	Active
AF1-278	Lewis Run-Pierce Brook 3 230 kV	Active
AF1-286	East Sayre 34.5 kV II	Active
AF1-287	Edinboro South 34.5 kV II	Active
AF1-302	Brookville-Squab Hollow 138 kV	Active
AF1-304	Titusville-Grandview 115 kV	Active
AF1-306	Squab Hollow 230 kV	Active
AF1-307	Seward-Jackson Rd 115 kV	Withdrawn
AF1-321	Hooversville 115 kV I	Active
AF2-001	Hooversville 115 kV II	Active
AF2-002	Hooversville 115 kV III	Active
AF2-010	Union City-Titusville 115 kV	Active
AF2-039	Shawville-Clearfield 34.5 kV	Active
AF2-045	Cambria Nug 115 kV	Engineering and Procurement
AF2-050	Bearrock-Johnstown 230 kV	Active

Queue Number	Project Name	Status
AF2-051	Geneva 115 kV	Active
AF2-087	East Altoona-Pinecroft 12.47 kV	Suspended
AF2-088	Shawville-Clearfield 34.5 kV II	Active
AF2-092	Snake Spring-Bedford Area 23 kV	Active
AF2-104	Somerset 23 kV	Withdrawn
AF2-121	Philipsburg-Shawville 34.5 kV	Active
AF2-130	Wolfs Corners 34.5 kV	Active
AF2-141	Lick Run 115 kV	Active
AF2-145	Lycoming-Lock Haven 69 kV	Active
AF2-164	Handsome Lake 345 kV	Active
AF2-165	Clark Summit-Emlenton 34.5 kV	Active
AF2-166	Clark Summit-Emlenton 34 kV	Active
AF2-197	East Towanda 115 kV	Active
AF2-235	Titusville-Oil Creek 34.5 kV	Active
AF2-238	Mansfield-South Troy 34.5 kV	Active
AF2-239	Wyalusing-Hollenback WRC 34.5 kV	Active
AF2-240	North Orwell 12.47 kV	Active
AF2-241	Athens-Milan 34.5 kV	Active
AF2-243	Clark Summit 34.5 kV	Active
AF2-265	South Troy-Athens 34.5 kV	Engineering and Procurement
AF2-270	Bedford South RCB-Bedford Area 23 kV	Engineering and Procurement
AF2-271	Pemberton-Sinking Valley 12.47 kV	Engineering and Procurement
AF2-273	Sligo 25 kV	Engineering and Procurement
AF2-293	Beech Creek 12.47 kV	Active
AF2-294	Thompson 34.5 kV	Active
AF2-295	Wilcox-Paper City 46 kV	Active
AF2-296	Madera 34.5 kV	Active
AF2-318	East Towanda-New Albany 34.5 kV	Active
AF2-336	Snake Spring 115 kV I	Active
AF2-337	Snake Spring 115 kV II	Active
AF2-338	Snake Spring 115 kV III	Active
AF2-339	Snake Spring 115 kV IV	Active
AF2-380	Karns City-Bear Creek 25 kV	Active
AF2-381	Bedford North-Central City West 115 kV	Active
AF2-405	East Sayre 34.5 kV III	Active
AF2-406	Sayre 115 kV	Active
AF2-411	Mainesburg 345 kV	Withdrawn
AF2-412	Mainesburg 115 kV	Active
AF2-436	Wyalusing 34.5 kV II	Engineering and Procurement
AF2-437	Oxbow 34.5 kV III	In Service
V3-030	St. Benedict-Patton 46kV	In Service
X3-003	Mehoopany II 115 kV	In Service
Y1-033	Penn Mar-Rock Wood 115kV	In Service
Y2-042	Oxbow 25kV	In Service
Y2-055	Elm Street 34.5kV	Withdrawn
Y3-092	Erie West 345kV	Engineering and Procurement
Z1-038	Florey Knob 34.5kV	Withdrawn
Z1-069	Gold-Sabinsville 115kV	In Service
Z1-091	Lenox 34kV	Withdrawn
Z2-108	Meyersdale North 115kV	In Service

11.9 Contingency Descriptions

Contingency Name	Contingency Definition
PN-P2-2-PN-230-001T	CONTINGENCY 'PN-P2-2-PN-230-001T' /* HOMER CITY NORTH 230KV BUS DISCONNECT BRANCH FROM BUS 200769 TO BUS 200767 TO BUS 202640 CKT N/* 26HOMER CY 345 26HOMER CT 230 26HOMERCITYN 23.00 DISCONNECT BRANCH FROM BUS 200767 TO BUS 200807 CKT 1 /* 26HOMER CT 230 26PIERC230 230 REMOVE MACHINE 1 FROM BUS 200809 /* 26SITHE 138 DISCONNECT BUS 200807 /* 26PIERC230 230 DISCONNECT BUS 200808 /* 26PIERC138 138 DISCONNECT BUS 200809 /* 26SITHE 138 END
PN-P2-2-PN-230-007T	CONTINGENCY 'PN-P2-2-PN-230-007T' /* LEWISTOWN #2 230KV BUS DISCONNECT BRANCH FROM BUS 200513 TO BUS 200531 CKT 2 /* 26LEWISTWN 230 26YEAGRTWN 230 DISCONNECT BRANCH FROM BUS 200513 TO BUS 235248 CKT 1 /* 26LEWISTWN 230 01SHINGL 230 DISCONNECT BRANCH FROM BUS 200513 TO BUS 200517 CKT 1 /* 26LEWISTWN 230 26RAYSTOWN 230 END
PN_P4-500-002A	CONTINGENCY 'PN_P4-500-002A' /* CONEMAUGH 500KV BKR 1 DISCONNECT BRANCH FROM BUS 200005 TO BUS 200912 CKT 3 /* CONEM-GH 500 26CONEMAGH 230 DISCONNECT BUS 200031 /* CONE G2 22 END
PN_P4-500-002F	CONTINGENCY 'PN_P4-500-002F' /* CONEMAUGH 500KV BKR 6 DISCONNECT BRANCH FROM BUS 200005 TO BUS 200912 CKT 3 /* CONEM-GH 500 26CONEMAGH 230 DISCONNECT BUS 200030 /* CONE G1 22 END
ATSI-P2-3-CEI-345-004D	CONTINGENCY 'ATSI-P2-3-CEI-345-004D' /* ERIE WEST 345KV BKR 8 DISCONNECT BRANCH FROM BUS 200599 TO BUS 200600 CKT 1 /* 26ERIE W 345 26ERIE SO 345 DISCONNECT BRANCH FROM BUS 200599 TO BUS 238547 CKT 1 /* 26ERIE W 345 02AT 345 DISCONNECT BRANCH FROM BUS 238547 TO BUS 239036 CKT 1 /* 02AT 345 02PERRY 345 DISCONNECT BRANCH FROM BUS 238547 TO BUS 239082 CKT 1 /* 02AT 345 02S8-ATT 345 DISCONNECT BUS 200600 /* 26ERIE SO 345 DISCONNECT BUS 238547 /* 02AT 345 END

Contingency Name	Contingency Definition
ATSI-P7-1-CEI-345-012	CONTINGENCY 'ATSI-P7-1-CEI-345-012' /* PERRY-EASTLAKE AND PERRY-NORTHFIELD 345KV LINE OUTAGES DISCONNECT BRANCH FROM BUS 238684 TO BUS 239036 CKT 1 /* 02EASTLK 345 02PERRY 345 DISCONNECT BRANCH FROM BUS 239358 TO BUS 239036 CKT 1 /* 02NFIELD 345 02PERRY 345 END
ATSI-P7-1-CEI-345-016	CONTINGENCY 'ATSI-P7-1-CEI-345-016' /* PERRY-NORTHFEILD AND PERRY-LC 345KV LINE OUTAGES DISCONNECT BRANCH FROM BUS 239036 TO BUS 239358 CKT 1 /* 02PERRY 345 02NFIELD 345 DISCONNECT BRANCH FROM BUS 239036 TO BUS 239334 CKT 1 /* 02PERRY 345 02L.CENTER 345 END
PN-P1-2-PN-115-021-B	CONTINGENCY 'PN-P1-2-PN-115-021-B' /* UNION CITY - AF2-101 TAP 115KV DISCONNECT BRANCH FROM BUS 200571 TO BUS 957160 CKT 1 /* 26UNION CY 115 AF2-101 TAP 115 END
PN-P2-3-PN-230-9I	CONTINGENCY 'PN-P2-3-PN-230-9I' /* 579 DISCONNECT BRANCH FROM BUS 200767 TO BUS 200807 CKT 1 /* 26HOMER CT 230 26PIERC230 230 DISCONNECT BRANCH FROM BUS 200807 TO BUS 200808 CKT 1 /* 26PIERC230 230 26PIERC138 138 DISCONNECT BRANCH FROM BUS 200808 TO BUS 200809 CKT 1 /* 26PIERC138 138 26SITHE 138 DISCONNECT BRANCH FROM BUS 200767 TO BUS 200769 TO BUS 202640 CKT N/* 26HOMER CT 230 26HOMER CY 345 26HOMERCITYN 23.00 REMOVE MACHINE 1 FROM BUS 200809 /* 26SITHE 138 END
PN-P1-2-PN-345-107T	CONTINGENCY 'PN-P1-2-PN-345-107T' /* ERIE WEST - ASHTABULA - PERRY 345KV DISCONNECT BRANCH FROM BUS 200599 TO BUS 238547 CKT 1 /* 26ERIE W 345 02AT 345 DISCONNECT BRANCH FROM BUS 238547 TO BUS 239082 CKT 1 /* 02AT 345 02S8-ATT 345 DISCONNECT BRANCH FROM BUS 238547 TO BUS 239036 CKT 1 /* 02AT 345 02PERRY 345 DISCONNECT BUS 238547 /* 02AT 345 END

Contingency Name	Contingency Definition
PN-P2-3-PN-230-9E	CONTINGENCY 'PN-P2-3-PN-230-9E' /* 575 DISCONNECT BRANCH FROM BUS 200767 TO BUS 200769 TO BUS 202641 CKT S/* 26HOMER CT 230 26HOMER CY 345 26HOMERCITYS 23.00 DISCONNECT BRANCH FROM BUS 200767 TO BUS 200837 CKT 1 /* 26HOMER CT 230 26HOMER C1 20 REMOVE MACHINE 1 FROM BUS 200837 /* 26HOMER C1 20 END
ATSI-P1-3-SYS-345-722	CONTINGENCY 'ATSI-P1-3-SYS-345-722' /* TRAN 02S8-ATT 345 TO 02ASH_3 138 CK 8 DISCONNECT BRANCH FROM BUS 239082 TO BUS 238544 CKT 8 /* 02S8-ATT 345 02ASH_3 138 END
PN-P1-3-PN-230-006	CONTINGENCY 'PN-P1-3-PN-230-006' /* ERIE SOUTH #8 345/230KV XFMR DISCONNECT BRANCH FROM BUS 200600 TO BUS 200819 CKT 8 /* 26ERIE SO 345 26ERIE SE 230 END
PN-P1-3-PN-230-003A	CONTINGENCY 'PN-P1-3-PN-230-003A' /* HOMER CITY NORTH 345/230KV XFMR DISCONNECT BRANCH FROM BUS 200769 TO BUS 200767 TO BUS 202640 CKT N/* 26HOMER CY 345 26HOMER CT 230 26HOMERCITYN 23.00 END
PN-P1-3-PN-230-004A	CONTINGENCY 'PN-P1-3-PN-230-004A' /* HOMER CITY SOUTH 345/230KV XFMR DISCONNECT BRANCH FROM BUS 200769 TO BUS 200767 TO BUS 202641 CKT S/* 26HOMER CY 345 26HOMER CT 230 26HOMERCITYS 23.00 END
Base Case	
PN-P1-2-PN-230-025	CONTINGENCY 'PN-P1-2-PN-230-025' /* CONEMAUGH - SEWARD 230KV DISCONNECT BRANCH FROM BUS 200912 TO BUS 200793 CKT 1 /* 26CONEMAGH 230 26SEWARD 2 230 END
PN-P1-2-PN-345-003	CONTINGENCY 'PN-P1-2-PN-345-003' /* HANDSOME LAKE - WAYNE 345KV DISCONNECT BRANCH FROM BUS 200826 TO BUS 200595 CKT 1 /* 26HANDSMLK 345 26WAYNE 345 END
PN-P1-2-PN-345-001	CONTINGENCY 'PN-P1-2-PN-345-001' /* ERIE WEST - WAYNE 345KV DISCONNECT BRANCH FROM BUS 200599 TO BUS 200595 CKT 1 /* 26ERIE W 345 26WAYNE 345 END

Contingency Name	Contingency Definition
ATSI-P1-2-CEI-345-700T	CONTINGENCY 'ATSI-P1-2-CEI-345-700T' /* PN/ATSI ERIE WEST - ASHTABULA - PERRY 345KV DISCONNECT BRANCH FROM BUS 239036 TO BUS 238547 CKT 1 /* 02PERRY 345 02AT 345 DISCONNECT BRANCH FROM BUS 238547 TO BUS 239082 CKT 1 /* 02AT 345 02S8-ATT 345 DISCONNECT BRANCH FROM BUS 239082 TO BUS 238544 CKT 8 /* 02S8-ATT 345 02ASH_3 138 DISCONNECT BRANCH FROM BUS 238547 TO BUS 200599 CKT 1 /* 02AT 345 26ERIE W 345 END
ATSI-P1-1-CEI-345-711	CONTINGENCY 'ATSI-P1-1-CEI-345-711' /* GEN 02PERRG1 22.0 UNIT 1 REMOVE MACHINE 1 FROM BUS 239035 /* 02PERRG1 22 END
PN-P1-3-PN-230-001T	CONTINGENCY 'PN-P1-3-PN-230-001T' /* CONEMAUGH #1 500/230KV XFMR DISCONNECT BRANCH FROM BUS 200005 TO BUS 200912 CKT 3 /* CONEM-GH 500 26CONEMAGH 230 END

12 Light Load Analysis

Light Load analysis is not required for solar projects.

13 Short Circuit Analysis

The following Breakers are overdutied:

None.

13.1 System Reinforcements - Short Circuit

None.

14 Stability and Reactive Power

To be determined in the Facilities Study Phase.

15 Affected Systems

15.1 NYISO

NYISO Impacts to be determined during later study phases (as applicable).

16 Attachment 1: One Line Diagram

